

Absorption cutoff and stationary singularities for rounded Gaussian random dynamical systems

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Abstract

We study Gaussian random dynamical systems with coordinatewise tanh nonlinearity and nearest-grid rounding. Gaussian symmetry reduces the dynamics to an exact Markov chain for the normalized squared radius. Rounding makes the origin absorbing, and the total variation distance to the absorbing equilibrium equals the survival probability of the absorption time. At fixed width, we identify the critical gain and prove an absorption cutoff with Gaussian profile as the mesh tends to zero. At fixed precision, global contraction yields a large-dimension absorption cutoff, while positive drift produces metastability. In the supercritical regime, we prove a large-dimension cutoff to a nonzero invariant law and show that, at fixed dimension, its mass near the repelling origin has a power-law asymptotic. All six main results are formalized in Lean 4 on top of Mathlib and independently checked against restatements that import only Mathlib.

Keywords: cutoff; absorption; finite precision; Gaussian Markov chains; metastability; renewal tails; random neural networks.

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1. Introduction

We study cutoff for nonlinear Markov chains obtained by rounding random iterated maps to a finite grid. The grid makes exact absorption possible. In the absorbing regimes considered below, the total variation distance to equilibrium equals the survival probability of the absorption time. The cutoff problem therefore reduces to sharp first passage for a random radius.

Random neural-network dynamics provide the motivating example. In real-valued models, the weights are scaled to control signal propagation through depth [GB10, HZRS15, PLR⁺16, SGGSD17]. Finite-precision arithmetic instead confines the numerical state to a finite set [Hig02]. The zero state can then be reached exactly, turning gradual signal decay into absorption.

This phenomenon was observed numerically in a joint work with Karlsson [AK22]. The model iterates random fully connected layers with independent weights uniformly distributed on $[-N^{-1/2}, N^{-1/2}]$:

$$X_{t+1}^{(N)} = \tanh(W_{t+1}^{(N)} X_t^{(N)}), \quad X_t^{(N)} \in [-1, 1]^N, \quad (1.1)$$

where $\tanh$ acts coordinatewise. Rounding makes the origin absorbing. With continuous weights, the unrounded chain started away from zero does not hit the origin at any finite time, so its total variation distance to δ_0 remains one. Numerical experiments at fixed precision showed a sharp transition in depth for dimensions 1 and 2.

For the Gaussian model studied below, we prove a Gaussian-profile absorption cutoff as the mesh tends to zero at fixed width. At fixed precision and growing width, we prove a bounded-window absorption cutoff in the globally contracting regime and exponential persistence in the positive-drift regime. For the unrounded supercritical chain, we prove cutoff to a nonzero invariant law as the width grows and, at fixed width, determine the power-law mass of this law near the repelling origin.

1.1. Gaussian model and rounding. For the remainder of the paper, we replace the bounded weights in the motivating model by independent Gaussian weights with gain parameter $A > 0$:

$$(W_{t,A}^{(N)})_{ij} \sim \mathcal{N}(0, A^2/N), \quad 1 \leq i, j \leq N, \quad t \geq 1. \quad (1.2)$$

For $x \in [-1, 1]^N$, we write $X^{(N)}$ for the resulting unrounded chain:

$$X_0^{(N)} = x, \quad X_{t+1}^{(N)} = \tanh(W_{t+1,A}^{(N)} X_t^{(N)}). \quad (1.3)$$

For a mesh size $\varrho > 0$, let Q_ϱ denote coordinatewise nearest-grid rounding to $\varrho\mathbb{Z}^N$, with ties rounded toward the grid point of smaller absolute value. The rounded chain is

$$Y_0^{(N)} = Q_\varrho x, \quad Y_{t+1}^{(N)} = Q_\varrho \left(\tanh(W_{t+1,A}^{(N)} Y_t^{(N)}) \right), \quad (1.4)$$

and we write $P_{\varrho,A,N}$ for its transition kernel. Since the point 0 is absorbing, define the absorption time as

$$\tau_\varrho^{(N)} := \inf\{t \geq 0 : Y_t^{(N)} = 0\}, \quad (1.5)$$

then the total variation distance to the absorbing equilibrium is exactly

$$\|P_{\varrho,A,N}^t(Q_\varrho x, \cdot) - \delta_0\|_{\text{TV}} = \mathbb{P}(\tau_\varrho^{(N)} > t). \quad (1.6)$$

Thus finite-precision cutoff to δ_0 is an absorption-time question.

Set

$$Z_t^{(N)} := N^{-1} \|X_t^{(N)}\|_2^2$$

and suppose that $Z_t^{(N)} = q$. The preactivation coordinates at the next step are independent centered Gaussian variables with variance $A^2 q$. Hence $Z^{(N)}$ is a Markov chain, and, for independent standard normal variables $G_1, \dots, G_N$, its transition is

$$Z_{t+1}^{(N)} \stackrel{d}{=} \frac{1}{N} \sum_{i=1}^N \tanh^2 \left(A \sqrt{Z_t^{(N)}} G_i \right). \quad (1.7)$$

We denote the transition kernel of $Z^{(N)}$ by $K_{A,N}$. The updates of $X^{(N)}$ and $Y^{(N)}$ are examples of iterated random functions. For invariant laws and convergence under average contraction, see Diaconis and Freedman [DF99]. The corresponding scalar reductions and further notation are given in Section 2.

We first recall the threshold for the unrounded chain as $N \rightarrow \infty$, and then define the three thresholds used for the rounded chain. Let G be a standard normal variable and define

$$V_A(q) := \mathbb{E} \tanh^2(A\sqrt{q}G). \quad (1.8)$$

Since $\tanh u = u + O(u^3)$ near zero, we have $V_A(q) = A^2q + O(q^2)$ for $q > 0$ small. Thus the threshold is $A = 1$ as $N \rightarrow \infty$. Strict concavity of V_A gives a unique nonzero attracting fixed point q_* when $A > 1$.

At fixed N , the transition from q satisfies

$$\sqrt{\frac{Z_{t+1}^{(N)}}{q}} \implies \frac{A}{\sqrt{N}} \chi_N, \quad q \downarrow 0,$$

where χ_N is the Euclidean norm of a standard Gaussian vector in $\mathbb{R}^N$. Set

$$\ell_{A,N} := \frac{A}{\sqrt{N}} \chi_N, \quad \gamma_{A,N} := -\mathbb{E} \log \ell_{A,N}.$$

The equation $\gamma_{A,N} = 0$ has the unique solution

$$A_c(N) = \sqrt{\frac{N}{2}} \exp\left\{-\frac{1}{2}\psi\left(\frac{N}{2}\right)\right\}, \quad (1.9)$$

where ψ is the digamma function. Thus $\gamma_{A,N} = \log(A_c(N)/A)$, and $A < A_c(N)$ is equivalent to $\mathbb{E} \log \ell_{A,N} < 0$. Moreover, $A_c(N) > 1$ for finite N and $A_c(N) \downarrow 1$ as $N \rightarrow \infty$.

For the rounded chain, measure the normalized squared radius in units of ϱ^2 , so that $h = q/\varrho^2$. As $N \rightarrow \infty$ at fixed ϱ , the mean map for h is

$$V_{A,\varrho}(h) := \mathbb{E} \left[Q_1 \left(\varrho^{-1} \tanh(\varrho A \sqrt{h} G) \right)^2 \right]. \quad (1.10)$$

Here Q_1 is nearest-integer rounding, with ties rounded toward the integer of smaller absolute value. Since $|\tanh z| \leq |z|$, we have $V_{A,\varrho}(h) \leq L_A(h)$, where

$$L_A(h) := \mathbb{E} [Q_1(A\sqrt{h}G)^2].$$

Define

$$A_{\text{lat}}^2 := \inf_{\alpha > 0} \frac{\alpha^2}{\mathbb{E} [Q_1(\alpha G)^2]}. \quad (1.11)$$

The condition $A < A_{\text{lat}}$ is equivalent to $L_A(h) < h$ for every $h > 0$. In this case, both L_A and $V_{A,\varrho}$ contract globally toward zero. Set

$$A_{\text{ex}}(\varrho)^2 := \inf_{\alpha > 0} \frac{\alpha^2}{\mathbb{E} \left[Q_1(\varrho^{-1} \tanh(\varrho \alpha G))^2 \right]}. \quad (1.12)$$

Then $A > A_{\text{ex}}(\varrho)$ is equivalent to $V_{A,\varrho}(h) > h$ for some $h > 0$. In this case, the rightmost component of $\{h : V_{A,\varrho}(h) > h\}$ produces metastability.

The three thresholds enter the rounded results as follows. At fixed N , $A < A_c(N)$ gives the cutoff as $\varrho \downarrow 0$. At fixed ϱ and as $N \rightarrow \infty$, $A < A_{\text{lat}}$ gives the absorption cutoff, while $A > A_{\text{ex}}(\varrho)$ gives metastability.

For both absorption cutoffs, the relevant distance is

$$\|P_{\varrho,A,N}^t(Q_\varrho x, \cdot) - \delta_0\|_{\text{TV}}.$$

In the supercritical large-dimension problem, the origin is repelling and $X^{(N)}$ has a nonzero invariant law. The relevant distance is then total variation distance to this law. Here and below, nonzero means that the invariant probability law assigns zero mass to the absorbing state. Mixtures with δ_0 are excluded.

1.2. Cutoff convention. We use the following convention throughout.

Definition 1.1 (Total variation cutoff). Let $\mathcal{I} \subset \mathbb{R}$, and let $r_* \in [-\infty, \infty]$ belong to the closure of $\mathcal{I}$ in the extended real line. For each $r \in \mathcal{I}$, let P_r be a Markov kernel on a measurable space E_r , let π_r be an invariant probability measure, and let $x_r \in E_r$. Set

$$d_r(t) := \|P_r^t(x_r, \cdot) - \pi_r\|_{\text{TV}}, \quad t \in \mathbb{N}_0. \quad (1.13)$$

For $u \in \mathbb{R}$, write $[u]_+ := \max\{0, [u]\}$. Let $t_r \rightarrow \infty$ and $a_r = o(t_r)$ with $a_r > 0$. We say that $(P_r, x_r, \pi_r)_{r \in \mathcal{I}}$ has total variation cutoff at time t_r with window a_r if

$$\lim_{c \rightarrow \infty} \liminf_{r \rightarrow r_*} d_r([t_r - ca_r]_+) = 1, \quad \lim_{c \rightarrow \infty} \limsup_{r \rightarrow r_*} d_r([t_r + ca_r]_+) = 0. \quad (1.14)$$

A function $F : \mathbb{R} \rightarrow [0, 1]$ is a cutoff profile on the scale a_r if, for every $c \in \mathbb{R}$,

$$d_r([t_r + ca_r]_+) \rightarrow F(c), \quad r \rightarrow r_*, \quad (1.15)$$

where $F(c) \rightarrow 1$ as $c \rightarrow -\infty$ and $F(c) \rightarrow 0$ as $c \rightarrow \infty$.

When no window is specified, cutoff at time t_r means that, for every fixed $\delta \in (0, 1)$,

$$d_r([(1 - \delta)t_r]_+) \rightarrow 1, \quad d_r([(1 + \delta)t_r]_+) \rightarrow 0, \quad r \rightarrow r_*. \quad (1.16)$$

Equivalently, there exists $b_r = o(t_r)$ with $b_r > 0$ such that the family has cutoff at time t_r with window b_r .

Definition 1.2 (Mixing time). For $\varepsilon \in (0, 1)$, define

$$t_{\text{mix}}^{(r)}(\varepsilon) := \inf\{t \in \mathbb{N}_0 : d_r(t) \leq \varepsilon\}, \quad (1.17)$$

where the infimum is ∞ if the set is empty.

Remark 1.3. Since d_r is nonincreasing, cutoff with window a_r implies, for every fixed $\varepsilon \in (0, 1)$,

$$t_{\text{mix}}^{(r)}(\varepsilon) = t_r + O_\varepsilon(a_r).$$

Thus window-free cutoff gives $t_{\text{mix}}^{(r)}(\varepsilon) = t_r + o(t_r)$. If $a_r \rightarrow \infty$, the family has profile F , and F is continuous and strictly decreasing through the level ε , then

$$t_{\text{mix}}^{(r)}(\varepsilon) = t_r + c_\varepsilon a_r + o(a_r), \quad F(c_\varepsilon) = \varepsilon.$$

1.3. Main results. We first state the two absorption cutoffs, followed by metastability, the supercritical cutoff, and the behavior of the invariant law near zero. We write Φ_G for the distribution function of a standard Gaussian random variable.

For fixed $N \geq 1$, $0 < A < A_c(N)$, deterministic $x_0 \in \mathbb{R}^N \setminus \{0\}$, and $0 < \varrho < R_0 := \|x_0\|_2$, set

$$L_\varrho := \log \frac{R_0}{\varrho}, \quad \sigma_N^2 := \text{Var}(\log \chi_N) = \frac{1}{4} \psi_1\left(\frac{N}{2}\right),$$

where $\psi_1 = \psi'$ is the trigamma function. For $a \in \mathbb{R}$, define the nonnegative integer

$$t_\varrho(a) := \max\left\{0, \left\lfloor \frac{L_\varrho}{\gamma_{A,N}} + a \frac{\sigma_N}{\gamma_{A,N}^{3/2}} \sqrt{L_\varrho} \right\rfloor\right\}.$$

Theorem 1.4 (Vanishing-mesh cutoff). *Assume $0 < A < A_c(N)$. Then, for every fixed $a \in \mathbb{R}$,*

$$\|P_{\varrho,A,N}^{t_\varrho(a)}(Q_\varrho x_0, \cdot) - \delta_0\|_{\text{TV}} = \mathbb{P}(\tau_\varrho^{(N)} > t_\varrho(a)) \rightarrow \Phi_G(-a), \quad \varrho \downarrow 0. \quad (1.18)$$

In particular, $Y^{(N)}$ has total variation cutoff to δ_0 at time $L_\varrho/\gamma_{A,N}$ with window $\sigma_N \gamma_{A,N}^{-3/2} \sqrt{L_\varrho}$ and Gaussian profile $a \mapsto \Phi_G(-a)$.

For $A > 0$, $\varrho \in (0, 1)$, and deterministic $x_N \in [-1, 1]^N$, define

$$\bar{h}_{0,N} := \frac{1}{N\varrho^2} \|Q_\varrho x_N\|_2^2, \quad \bar{h}_{t+1,N} := V_{A,\varrho}(\bar{h}_{t,N}),$$

and

$$\bar{t}_N := \inf \left\{ t \geq 0 : \bar{h}_{t,N} \leq \frac{1}{\log N} \right\}.$$

Theorem 1.5 (Subcritical dimension cutoff). *Assume $A < A_{\text{lat}}$ and that the rounded initial radii are macroscopic:*

$$\liminf_{N \rightarrow \infty} \bar{h}_{0,N} > 0.$$

Then $\bar{t}_N \rightarrow \infty$ and

$$\|P_{\varrho,A,N}^{\bar{t}_N-1}(Q_\varrho x_N, \cdot) - \delta_0\|_{\text{TV}} \rightarrow 1, \quad \|P_{\varrho,A,N}^{\bar{t}_N+2}(Q_\varrho x_N, \cdot) - \delta_0\|_{\text{TV}} \rightarrow 0.$$

Thus $Y^{(N)}$ has total variation cutoff to δ_0 at the exact deterministic center $\bar{t}_N$, with a bounded window.

Remark 1.6. The macroscopic assumption in Theorem 1.5 is sufficient but not necessary. Continuity and positivity of $V_{A,\varrho}$ imply that it gives $\bar{t}_N \rightarrow \infty$. On the other hand, $A < A_{\text{lat}}$ implies $\bar{h}_{t,N} \leq C_\varrho \theta_A^t$ for some $\theta_A \in (0, 1)$, and hence $\bar{t}_N = O(\log \log N)$. In place of the macroscopic assumption, Theorem 4.5 requires only $\bar{t}_N \rightarrow \infty$ and therefore also permits initial radii that approach zero.

For $\varrho \in (0, 1)$, set

$$M_\varrho := \max_{|u| \leq \varrho^{-1}} Q_1(u)^2, \quad \mathcal{O}_{A,\varrho} := \{h \in (0, M_\varrho) : V_{A,\varrho}(h) > h\},$$

and call its connected components positive-drift components. Write

$$\mathcal{H}_t^{(N)} := \frac{1}{N\varrho^2} \|Y_t^{(N)}\|_2^2.$$

Theorem 1.7 (Metastability). *Assume $A > A_{\text{ex}}(\varrho)$, and let (h_-, h_+) be the rightmost positive-drift component of $\mathcal{O}_{A,\varrho}$. Fix a compact set $B \subset (h_-, h_+]$ and deterministic initial states $Y_0^{(N)}$ satisfying $\mathcal{H}_0^{(N)} \in B$ for every $N \geq 1$. Then there are $T \in \mathbb{N}_0$ and $c > 0$ such that*

$$\mathbb{P}(\tau_\varrho^{(N)} > T + \lfloor \exp(cN) \rfloor) \rightarrow 1.$$

For each fixed N , however, absorption still occurs almost surely for every deterministic choice of $Y_0^{(N)}$.

The persistence of $Y^{(N)}$ describes its behavior before eventual absorption. We next remove rounding and study relaxation as $N \rightarrow \infty$. In the supercritical regime, $X^{(N)}$ approaches a nonzero invariant law, under which $Z^{(N)}$ is concentrated near q_* .

Let $A > 1$, and let q_* and $\mu_A = V'_A(q_*) \in (0, 1)$ be the nonzero attracting fixed point and multiplier of V_A . For $q_0 \in (0, 1] \setminus \{q_*\}$, let $C(q_0) \neq 0$ be determined by

$$V_A^t(q_0) - q_* = C(q_0)\mu_A^t + o(\mu_A^t),$$

and set

$$t_N(q_0) := \frac{\frac{1}{2} \log N + \log |C(q_0)|}{|\log \mu_A|}. \quad (1.19)$$

For $c \in \mathbb{R}$, define the nonnegative integer

$$n_N(c) := \max\{0, \lfloor t_N(q_0) + c \rfloor\}. \quad (1.20)$$

Theorem 1.8 (Supercritical cutoff for $Z^{(N)}$). *Let $x_N \in [-1, 1]^N$ be deterministic and suppose*

$$q_N := N^{-1} \|x_N\|_2^2 \rightarrow q_0.$$

Then, for all sufficiently large N , $K_{A,N}$ has a unique nonzero invariant law $\nu_{A,N}$, and

$$\lim_{c \rightarrow \infty} \liminf_{N \rightarrow \infty} \|K_{A,N}^{n_N(-c)}(q_N, \cdot) - \nu_{A,N}\|_{\text{TV}} = 1, \quad (1.21)$$

while

$$\lim_{c \rightarrow \infty} \limsup_{N \rightarrow \infty} \|K_{A,N}^{n_N(c)}(q_N, \cdot) - \nu_{A,N}\|_{\text{TV}} = 0. \quad (1.22)$$

Corollary 1.9 (Supercritical cutoff for $X^{(N)}$). *Under the assumptions of Theorem 1.8, $X^{(N)}$ has a unique nonzero invariant law $\pi_{A,N}$ for all sufficiently large N , and*

$$\lim_{c \rightarrow \infty} \liminf_{N \rightarrow \infty} \|P_{A,N}^{n_N(-c)}(x_N, \cdot) - \pi_{A,N}\|_{\text{TV}} = 1, \quad (1.23)$$

while

$$\lim_{c \rightarrow \infty} \limsup_{N \rightarrow \infty} \|P_{A,N}^{n_N(c)}(x_N, \cdot) - \pi_{A,N}\|_{\text{TV}} = 0. \quad (1.24)$$

By (1.19), $|C(q_0)|\mu_A^{t_N(q_0)} = N^{-1/2}$. Shifting $t_N(q_0)$ by c multiplies the left-hand side by μ_A^c , so the cutoff window is bounded. Since $n_N(c) = \max\{0, \lfloor t_N(q_0) + c \rfloor\}$, it depends on the fractional part of $t_N(q_0) + c$. Consequently, no nonconstant continuous profile on $\mathbb{R}$, including a Gaussian profile, can arise on this scale. A sharper profile would have to retain the floor in $n_N(c)$ and may require subsequences along which the fractional part of $t_N(q_0)$ converges.

Finally, fix N and consider the nonzero invariant law near the repelling origin. Its mass there has a power-law asymptotic.

Theorem 1.10 (Stationary singularity in the supercritical regime). *Fix $N \geq 1$ and assume $A > A_c(N)$. Let $\pi_{A,N}$ be the nonzero invariant law of $X^{(N)}$. Let $\beta_{A,N} \in (0, N)$ be the unique positive solution of*

$$A^{-\beta} N^{\beta/2} 2^{-\beta/2} \frac{\Gamma((N - \beta)/2)}{\Gamma(N/2)} = 1.$$

Here $\bar{\sigma}_N$ denotes normalized surface measure on $\mathbb{S}^{N-1}$. Then there is a constant $c_{A,N} > 0$ such that, for every Borel set $B \subseteq \mathbb{S}^{N-1}$ satisfying $\bar{\sigma}_N(\partial B) = 0$,

$$s^{-\beta_{A,N}} \pi_{A,N} \left(0 < \|x\|_2 \leq s, \frac{x}{\|x\|_2} \in B \right) \longrightarrow c_{A,N} \bar{\sigma}_N(B) \quad \text{as } s \downarrow 0.$$

In particular,

$$\pi_{A,N}(0 < \|x\|_2 \leq s) \sim c_{A,N} s^{\beta_{A,N}} \quad \text{as } s \downarrow 0.$$

1.4. Previous work and novelty. The closest neural-network literature describes signal propagation through deterministic recursions for variances and correlations. This includes vanishing gradients and variance-based initialization [BSF94, GB10, HZRS15], as well as the order-to-chaos transition in the large-width limit [PLR⁺16, SGGSD17]. We instead study the absorption time created by finite-precision rounding and its concentration as the mesh or width varies.

The identity (1.6) is the standard relation between distance to an absorbing state and the tail of the absorption time. Martínez and Ycart use this relation for families whose initial states leave every finite set. They connect cutoff to concentration of the absorption times and give moment criteria for tree-type birth-and-death chains [MY01]. Here rounding creates the absorbing state, and our estimates use logarithmic-radius random walks and fluid limits.

For general background on cutoff and mixing times, see [Dia96, LP17]. Cutoff criteria have been obtained for L^p semigroups, birth-and-death chains, reversible chains through hitting-time concentration, and chains with nonnegative curvature [CSC08, DSC06, DLP10, BHP17, Sal24]. These results concern ergodic chains or semigroups and do not directly apply to the absorbing rounded dynamics. The local Poincaré criterion of Pedrotti and Salez [PS26] likewise does not apply: it assumes $\mu_t \ll \pi$ for every $t > 0$, whereas here $\pi = \delta_0$ and the law before absorption charges nonzero states.

Related results treat small random perturbations of deterministic dynamics. Barrera and Jara prove total-variation cutoff for small Brownian perturbations under coercivity and growth assumptions [BJ20]. Barrera, Högele, and Pardo obtain total-variation cutoff for a class of Lévy-driven Langevin systems and related Wasserstein results for generalized Ornstein–Uhlenbeck systems [BHP21a, BHP21b]. In joint work with Barrera, we study cutoff and its failure for one-dimensional Langevin–Kolmogorov dynamics by spectral Fokker–Planck methods [AB26]. Here we instead study discrete-time absorption and metastability caused by rounding. Together with Karlsson, we previously observed this cutoff numerically at fixed precision in dimensions one and two [AK22].

The contrast between cutoff and metastable escape also appears in birth-and-death chains: downhill hitting times concentrate, while normalized uphill escape times are asymptotically exponential [BBF09]. Our two behaviors likewise occur in different parameter regimes. The metastability theorem gives exponential persistence, but not a sharp exit-time law or a quasi-stationary limit. See [BdH15, CMSM13, CV23] for general background. The stationary singularity uses Kesten–Goldie renewal theory [Kes73, Gol91, BDM16]. Gaussian isotropy refreshes the angle at each step, reducing the renewal equation to a scalar one. To the best of our knowledge, these absorption cutoffs, our metastability estimate, and the stationary singularity have not previously been proved for this random neural-network model.

1.5. Formalization. All six headline results in Section 1.3 have been formalized in Lean 4 on top of Mathlib. The Lean development is publicly available at <https://github.com/BennyAvelin/AbsorptionCutoff>.¹ The production library contains no `sorry` and declares no custom axiom. A machine-checked audit shows that the seven declarations below use only Mathlib’s three standard foundational axioms (`propext`, `Classical.choice`, `Quot.sound`). The fixed-dimensional absorption clause of the metastability theorem is audited separately.

¹The repository is pinned to Lean v4.32.0 and Mathlib v4.32.0.

Result	Lean declaration
Theorem 1.4	<code>rounded_gaussian_nearest_cutoff</code>
Theorem 1.5	<code>subcritical_dimension_cutoff</code>
Theorem 1.7, persistence	<code>rounded_qualitative_metastability</code>
Theorem 1.7, absorption	<code>rounded_fixed_dimension_absorption</code>
Theorem 1.8	<code>gaussian_process_cutoff</code>
Corollary 1.9	<code>gaussian_vector_cutoff</code>
Theorem 1.10	<code>nd_power_singularity</code>

The Lean comparator checks the six printed results and the separate absorption clause against restatements that import only Mathlib. Each restatement appears in a challenge file with one intentional `sorry`. A matching solution derives it from the development. Challenge files are excluded from the production library. The comparator checks that each pair has the same Lean type, that the solution is accepted by the kernel, and which axioms it uses. It cannot verify that a restatement expresses the intended mathematics.

The paper and Lean code sometimes use different notation or equivalent hypotheses. After unfolding the definitions and equivalences, every printed result follows from its Lean declaration. No formalized result requires an additional hypothesis, and several have stronger conclusions.

Our use of autoformalization is inspired by the recent Lean formalizations of De Giorgi–Nash–Moser theory by Armstrong and Kempe [AK26a], and of coarse-graining theory for elliptic equations by Armstrong and Kuusi [AK26b]. The development was produced by autoformalization. Under the author’s supervision, OpenAI’s ChatGPT and Anthropic’s Claude generated the proofs from the arguments of this paper. The Lean kernel checks every definition, statement, and proof in the production library against Mathlib.

1.6. Proof overview and organization. To prove Theorem 1.4, we couple $Y^{(N)}$ to $X^{(N)}$ until the radius reaches the grid scale ϱ . In logarithmic coordinates, the radius is a perturbed random walk, and a first-passage central limit theorem gives the Gaussian profile. An estimate on the finite grid then turns entrance at this scale into absorption.

For Theorem 1.5, $\mathcal{H}^{(N)}$ concentrates around the iterates of $V_{A,\varrho}$. When $A < A_{\text{lat}}$, the orbit reaches the absorption scale, and absorption occurs within two further steps. For Theorem 1.7, the iterates starting in $B \subset (h_-, h_+]$ approach h_+ , and Hoeffding bounds keep $\mathcal{H}^{(N)}$ near h_+ for exponentially many steps.

For Theorem 1.8, the orbit $V_A^t(q_0)$ reaches an $N^{-1/2}$ -neighborhood of q_* , the scale of the stationary fluctuations. Negative-moment bounds control visits of $Z^{(N)}$ near zero, and concentration localizes the invariant law near q_* . We couple $Z^{(N)}$ to a stationary copy. Synchronous contraction reduces their separation, and a one-step total-variation estimate proves the cutoff. The comparison in Proposition 2.4 gives Corollary 1.9.

To prove Theorem 1.10, we write the invariant equation in log-polar coordinates. Gaussian isotropy refreshes the angular variable in one step, and the radial equation becomes a scalar renewal equation with Cramér root $\beta_{A,N}$.

The common notation and scalar estimates are in Section 2. Sections 3 and 4 prove the two absorption cutoffs, and Section 5 treats metastability. The supercritical cutoff and stationary singularity are proved in Sections 6 and 7. The auxiliary estimates are in Appendices A and B.

2. Gaussian chains and scalar estimates

We recall the vector chains and derive their scalar reductions. We then collect the thresholds, properties of the mean maps, scalar estimates, transition identities, and facts about invariant

laws used later. Additional probabilistic estimates are proved in Appendix A.

We write $\|\cdot\|_2$ for the Euclidean norm on vectors. Fix $A > 0$, $N \geq 1$, and $x \in \mathbb{R}^N$. Throughout the paper the Gaussian weights are

$$(W_{t,A}^{(N)})_{ij} \sim \mathcal{N}(0, A^2/N), \quad 1 \leq i, j \leq N, \quad t \geq 1, \quad (2.1)$$

independently over all indices. The unrounded vector chain is

$$X_0^{(N)} = x, \quad X_{t+1}^{(N)} = \tanh(W_{t+1,A}^{(N)} X_t^{(N)}). \quad (2.2)$$

For $\varrho > 0$, let $Q_\varrho : \mathbb{R}^N \rightarrow \varrho\mathbb{Z}^N$ be coordinatewise nearest-grid rounding, with ties rounded toward the grid point of smaller absolute value. The rounded vector chain is

$$Y_0^{(N)} = Q_\varrho x, \quad Y_{t+1}^{(N)} = Q_\varrho \left(\tanh(W_{t+1,A}^{(N)} Y_t^{(N)}) \right). \quad (2.3)$$

We denote the transition kernel of $Y^{(N)}$ by $P_{\varrho,A,N}$ and its absorption time by

$$\tau_\varrho^{(N)} := \inf\{t \geq 0 : Y_t^{(N)} = 0\}. \quad (2.4)$$

The point 0 is absorbing, and hence δ_0 is invariant. With the convention

$$\|\mu - \nu\|_{\text{TV}} := \sup_{B \in \mathcal{B}(\mathbb{R}^N)} |\mu(B) - \nu(B)|,$$

we have the exact identity

$$\|P_{\varrho,A,N}^t(Q_\varrho x, \cdot) - \delta_0\|_{\text{TV}} = \mathbb{P}(\tau_\varrho^{(N)} > t). \quad (2.5)$$

Thus cutoff to δ_0 is equivalent to concentration of the absorption time.

2.1. Scalar reductions. *The unrounded chain.* By Gaussian rotational invariance, the conditional law of the next preactivation vector depends on the present state only through its Euclidean norm. For $u \in \mathbb{R}^N$, set

$$r_N(u) := \frac{1}{N} \|u\|_2^2.$$

Define

$$R_t^{(N)} := \|X_t^{(N)}\|_2, \quad Z_t^{(N)} := r_N(X_t^{(N)}).$$

Since $R_t^{(N)} = \sqrt{N Z_t^{(N)}}$, both processes are Markov. We call $R^{(N)}$ the unrounded radius chain and $Z^{(N)}$ the unrounded squared-radius chain. Conditionally on $X_t^{(N)}$, the vector $W_{t+1,A}^{(N)} X_t^{(N)}$ has the same law as $(A R_t^{(N)} / \sqrt{N}) G_{t+1}$, where G_{t+1} is a standard Gaussian vector in $\mathbb{R}^N$. Equivalently, given $Z_t^{(N)} = q$,

$$Z_{t+1}^{(N)} \stackrel{d}{=} \frac{1}{N} \sum_{i=1}^N \tanh^2(A \sqrt{q} G_{t+1,i}). \quad (2.6)$$

For $t \geq 1$, the chain $Z^{(N)}$ takes values in $[0, 1]$.

For $g = (g_1, \dots, g_N) \in \mathbb{R}^N$, define

$$F_{A,N}(q, g) := \frac{1}{N} \sum_{i=1}^N \tanh^2(A \sqrt{q} g_i), \quad q \geq 0. \quad (2.7)$$

Let $K_{A,N}$ be the transition kernel of $Z^{(N)}$ on $[0, 1]$, given by

$$K_{A,N}(q, B) = \int_{\mathbb{R}^N} \mathbf{1}_{\{F_{A,N}(q,g) \in B\}} \mathcal{G}_N(dg), \quad (2.8)$$

where $\mathcal{G}_N$ is standard Gaussian measure on $\mathbb{R}^N$. Thus, with $G_{t+1} \sim \mathcal{N}(0, I_N)$,

$$Z_{t+1}^{(N)} = F_{A,N}(Z_t^{(N)}, G_{t+1}). \quad (2.9)$$

For fixed g , the map $q \mapsto F_{A,N}(q, g)$ is nondecreasing.

The conditional mean map of $Z^{(N)}$ is

$$V_A(q) := \mathbb{E} \tanh^2(A\sqrt{q} G), \quad G \sim \mathcal{N}(0, 1). \quad (2.10)$$

For fixed q , the conditional moments are

$$\mathbb{E}[Z_{t+1}^{(N)} \mid Z_t^{(N)} = q] = V_A(q), \quad \text{Var}(Z_{t+1}^{(N)} \mid Z_t^{(N)} = q) = \frac{1}{N} \text{Var}(\tanh^2(A\sqrt{q} G)). \quad (2.11)$$

Moreover, near zero, V_A expands as

$$V_A(q) = A^2 q - 2A^4 q^2 + O(q^3), \quad q \downarrow 0. \quad (2.12)$$

With respect to the natural filtration of $Z^{(N)}$, define the centered one-step fluctuation by

$$\zeta_{t+1}^{(N)} = \frac{1}{N} \sum_{i=1}^N \left[\tanh^2(A\sqrt{Z_t^{(N)}} G_{t+1,i}) - V_A(Z_t^{(N)}) \right], \quad (2.13)$$

so

$$Z_{t+1}^{(N)} = V_A(Z_t^{(N)}) + \zeta_{t+1}^{(N)}. \quad (2.14)$$

Conditionally on $\mathcal{F}_t$, the summands in (2.13) are independent and centered, and each lies in an interval of length at most one. Hence

$$\mathbb{E}[\zeta_{t+1}^{(N)} \mid \mathcal{F}_t] = 0, \quad \text{Var}(\zeta_{t+1}^{(N)} \mid \mathcal{F}_t) \leq \frac{1}{4N}, \quad (2.15)$$

and Hoeffding's inequality gives, for every $u > 0$,

$$\mathbb{P}(|\zeta_{t+1}^{(N)}| > u \mid \mathcal{F}_t) \leq 2 \exp(-2Nu^2). \quad (2.16)$$

The rounded chain. Let $Q_1 : \mathbb{R} \rightarrow \mathbb{Z}$ be nearest-integer rounding with the same tie convention as Q_ϱ . Then

$$Q_\varrho(y)_i = \varrho Q_1(y_i/\varrho), \quad Q_1(u) = 0 \iff |u| \leq \frac{1}{2}, \quad |Q_1(u) - u| \leq \frac{1}{2}.$$

Define

$$\mathcal{H}_t^{(N)} := \frac{1}{N\varrho^2} \|Y_t^{(N)}\|_2^2. \quad (2.17)$$

For $h \geq 0$ and $g \in \mathbb{R}$, put

$$U_{A,\varrho}(h, g) := Q_1\left(\varrho^{-1} \tanh(\varrho A\sqrt{h} g)\right)^2. \quad (2.18)$$

Conditionally on $\mathcal{H}_t^{(N)} = h$, the Gaussian representation and (2.18) give

$$\mathcal{H}_{t+1}^{(N)} \stackrel{d}{=} \frac{1}{N} \sum_{i=1}^N U_{A,\varrho}(h, G_{t+1,i}), \quad (2.19)$$

where the variables $G_{t+1,i}$ are independent standard Gaussians. Thus $\mathcal{H}^{(N)}$ is Markov, and its absorption time satisfies

$$\tau_\varrho^{(N)} = \inf\{t \in \mathbb{N}_0 : \mathcal{H}_t^{(N)} = 0\}. \quad (2.20)$$

Its mean map is

$$V_{A,\varrho}(h) := \mathbb{E} \left[Q_1 \left(\varrho^{-1} \tanh(\varrho A \sqrt{h} G) \right)^2 \right], \quad h \geq 0. \quad (2.21)$$

With respect to the natural filtration of $\mathcal{H}^{(N)}$, define

$$\hat{\zeta}_{t+1}^{(N)} := \mathcal{H}_{t+1}^{(N)} - V_{A,\varrho}(\mathcal{H}_t^{(N)}). \quad (2.22)$$

If $B_\varrho := \max_{|u| \leq \varrho^{-1}} Q_1(u)^2$, then $0 \leq U_{A,\varrho}(h, G) \leq B_\varrho$, and hence

$$\mathbb{E}[\hat{\zeta}_{t+1}^{(N)} \mid \mathcal{F}_t] = 0, \quad \mathbb{E}[(\hat{\zeta}_{t+1}^{(N)})^2 \mid \mathcal{F}_t] \leq \frac{B_\varrho^2}{4N}. \quad (2.23)$$

Moreover, Hoeffding's inequality yields

$$\mathbb{P} \left(\left| \hat{\zeta}_{t+1}^{(N)} \right| > u \mid \mathcal{F}_t \right) \leq 2 \exp \left(-\frac{2Nu^2}{B_\varrho^2} \right), \quad u > 0. \quad (2.24)$$

2.2. Thresholds and mean maps. The large-dimension threshold is $A = 1$. Recall that χ_N denotes the Euclidean norm of a standard Gaussian vector in $\mathbb{R}^N$. At fixed width, set

$$\ell_{A,N} := \frac{A}{\sqrt{N}} \chi_N. \quad (2.25)$$

Define

$$\gamma_{A,N} := -\mathbb{E} \log \ell_{A,N}.$$

The unique zero of $\gamma_{A,N}$ is

$$A = A_c(N) = \sqrt{\frac{N}{2}} \exp \left(-\frac{1}{2} \psi \left(\frac{N}{2} \right) \right), \quad (2.26)$$

where ψ is the digamma function. Thus $A < A_c(N)$ is equivalent to $\gamma_{A,N} > 0$, and

$$\gamma_{A,N} = \log \frac{A_c(N)}{A}, \quad \text{Var}(\log \ell_{A,N}) = \text{Var}(\log \chi_N) = \frac{1}{4} \psi_1 \left(\frac{N}{2} \right). \quad (2.27)$$

Here $\psi_1 = \psi'$ is the trigamma function. Moreover,

$$\frac{F_{A,N}(q, G)}{q} \rightarrow A^2 \frac{\chi_N^2}{N} \quad \text{as } q \downarrow 0 \quad (2.28)$$

almost surely. In particular, $A_c(N) > 1$ for finite N and $A_c(N) \downarrow 1$ as $N \rightarrow \infty$.

Define

$$L_A(h) := \mathbb{E}[Q_1(A\sqrt{h}G)^2], \quad (2.29)$$

and

$$m(\alpha) := \mathbb{E}[Q_1(\alpha G)^2], \quad A_{\text{lat}}^2 := \inf_{\alpha > 0} \frac{\alpha^2}{m(\alpha)}. \quad (2.30)$$

Thus $L_A(h) = m(A\sqrt{h})$. The condition $A < A_{\text{lat}}$ is equivalent to $L_A(h) < h$ for every $h > 0$.

We use the following fixed-point and linearization facts for V_A .

Lemma 2.1. *The map V_A is strictly increasing on $[0, 1]$ and strictly concave on $[0, 1]$, with $V_A(0) = 0$, $V'_A(0+) = A^2$, and $V_A(1) < 1$. If $A \leq 1$, then 0 is its only fixed point. If $A > 1$, then it has a unique nonzero fixed point $q_* \in (0, 1)$ satisfying*

$$q_* = V_A(q_*), \quad \mu_A := V'_A(q_*) \in (0, 1). \quad (2.31)$$

Every orbit of V_A started from $(0, 1]$ converges to q_ . If $q_0 \in (0, 1] \setminus \{q_*\}$, then there is a continuous function C in a neighborhood of q_0 , with $C(q_0) \neq 0$, such that*

$$V_A^t(q) - q_* = C(q)\mu_A^t + o(\mu_A^t) \quad (2.32)$$

locally uniformly for q near q_0 .

Proof. Fix $a > 0$ and put $f_a(q) := \tanh^2(a\sqrt{q})$ for $q > 0$. With $x = a\sqrt{q}$,

$$f'_a(q) = a^2 \frac{\tanh x}{x} \operatorname{sech}^2 x.$$

The functions $x \mapsto \tanh x/x$ and $x \mapsto \operatorname{sech}^2 x$ are strictly decreasing on $(0, \infty)$, hence f'_a is strictly decreasing and f_a is strictly concave. Since $\tanh^2(A\sqrt{q}G) = f_{A|G|}(q)$ and $\mathbb{P}(|G| > 0) = 1$, integration gives strict concavity of V_A . The same formula gives strict monotonicity, and the expansion in (2.12) gives $V'_A(0+) = A^2$. Finally $V_A(1) < 1$ since $\tanh^2(AG) < 1$ almost surely.

Concavity gives no positive fixed point when $A \leq 1$ and exactly one when $A > 1$. At the positive fixed point, concavity gives

$$0 < V'_A(q_*) < \frac{V_A(q_*)}{q_*} = 1.$$

If $A > 1$, then $V_A(q) > q$ for $0 < q < q_*$ and $V_A(q) < q$ for $q_* < q \leq 1$. Since V_A is increasing, the orbit is monotone and bounded on each side of q_* , and therefore converges to q_* .

The asymptotic (2.32) follows from the one-dimensional Koenigs linearization theorem [CG93, Ch. II, Sec. 2, Theorem 2.1]. The Taylor expansion at the attracting fixed point is

$$V_A(q) - q_* = \mu_A(q - q_*) + O((q - q_*)^2)$$

near q_* . After a finite time the orbit is in a neighborhood on which $|V_A(q) - q_*| \leq \rho|q - q_*|$ for some $\rho < 1$. For all later times before the orbit reaches q_* , write

$$\frac{V_A^{t+1}(q) - q_*}{\mu_A^{t+1}} = \frac{V_A^t(q) - q_*}{\mu_A^t} \frac{V_A(V_A^t(q)) - q_*}{\mu_A(V_A^t(q) - q_*)}.$$

The last factor is $1 + O(|V_A^t(q) - q_*|)$, uniformly for q in a small neighborhood of any fixed $q_0 \neq q_*$. Since the orbit is eventually contained in a contraction neighborhood of q_* , the series

$$\sum_{t=0}^{\infty} |V_A^t(q) - q_*|$$

is locally uniformly summable. Hence the logarithms of the product factors are locally uniformly summable, the Koenigs product converges locally uniformly, and we may define

$$C(q) = (V_A^s(q) - q_*)\mu_A^{-s} \prod_{j=s}^{\infty} \frac{V_A(V_A^j(q)) - V_A(q_*)}{\mu_A(V_A^j(q) - q_*)}$$

for any entrance time s after which the orbit stays in the contraction neighborhood. This formula is independent of the chosen sufficiently large s , locally uniform in q , and continuous. It is nonzero at q_0 because every factor is positive and finite and the orbit approaches q_* monotonically from one side without crossing it. This proves (2.32). $\square$

2.3. Common scalar estimates. Let $T_N \in \mathbb{N}$, let $(\mathcal{F}_t)_{0 \leq t \leq T_N}$ be a filtration, and let $(\mathcal{R}_{t,N})_{0 \leq t \leq T_N}$ be adapted. Let τ_N be a stopping time. Suppose that, on $\{t < \tau_N\}$ and for $0 \leq t < T_N$,

$$\mathcal{R}_{t+1,N} = A_{t,N} \mathcal{R}_{t,N} + \eta_{t+1,N}, \quad (2.33)$$

where $A_{t,N}$ is $\mathcal{F}_t$ -measurable, $|A_{t,N}| \leq \alpha_{t,N}$ for deterministic $\alpha_{t,N} \geq 0$, and

$$\mathbb{E}[\eta_{t+1,N} \mid \mathcal{F}_t] = 0, \quad \mathbb{E}[\eta_{t+1,N}^2 \mid \mathcal{F}_t] \leq \frac{C\sigma_{t,N}^2}{N} \quad (2.34)$$

for a constant $C < \infty$ and deterministic $\sigma_{t,N} \geq 0$. Define

$$K_N := 1 \vee \max_{0 \leq s < t \leq T_N} \left\{ \prod_{u=s}^{t-1} (1 \vee \alpha_{u,N}), \sigma_{s,N} \prod_{u=s+1}^{t-1} (1 \vee \alpha_{u,N}) \right\}. \quad (2.35)$$

The following stopped second-moment estimate controls deviations from a deterministic orbit up to a prescribed stopping time.

Lemma 2.2. *Under the preceding assumptions, set $\bar{\mathcal{R}}_{t,N} := \mathcal{R}_{t \wedge \tau_N, N}$. Then*

$$\mathbb{E} \bar{\mathcal{R}}_{T_N, N}^2 \leq CK_N^2 \left(\mathbb{E} \mathcal{R}_{0,N}^2 + \frac{T_N}{N} \right). \quad (2.36)$$

In addition, for every $t \leq T_N$,

$$\mathbb{E}[\mathcal{R}_{t,N}^2 \mathbf{1}_{\{\tau_N > t\}}] \leq \left(\prod_{u=0}^{t-1} \alpha_{u,N}^2 \right) \mathbb{E} \mathcal{R}_{0,N}^2 + \frac{C}{N} \sum_{s=0}^{t-1} \sigma_{s,N}^2 \prod_{u=s+1}^{t-1} \alpha_{u,N}^2, \quad (2.37)$$

with empty products equal to one and empty sums equal to zero. In particular, if $\tau_N = \inf\{t : |\mathcal{R}_{t,N}| > \delta\}$ for some $\delta > 0$, then

$$\mathbb{P}(\tau_N \leq T_N) \leq \frac{CK_N^2}{\delta^2} \left(\mathbb{E} \mathcal{R}_{0,N}^2 + \frac{T_N}{N} \right). \quad (2.38)$$

If $\mathbb{E} \log M_1 < 0$, the affine recursion below is contractive [BDM16]. The next lemma quantifies its entrance into a bounded set.

Lemma 2.3. *Let $(M_j)_{j \geq 1}$ be independent and identically distributed positive random variables with $\mathbb{E} \log M_1 < 0$, and assume $\log(M_1 + \varepsilon)$ has exponential moments in a neighborhood of the origin for each fixed $\varepsilon > 0$. For every $b < \infty$, there exist $K_* < \infty$ and $c_1, c_2, c_3 > 0$, depending only on b and the law of M_1 , such that the affine recursion*

$$U_{t+1} = M_{t+1} U_t + b, \quad U_0 = K,$$

satisfies, uniformly for $K \geq 2$,

$$\mathbb{P}(\inf\{t : U_t \leq K_*\} > c_1 \log K + r) \leq c_2 e^{-c_3 r}, \quad r \geq 0. \quad (2.39)$$

See Appendices A.1 and A.2 for the proofs.

2.4. Transition and reconstruction kernels. For $q \in [0, 1]$, let $J_{A,N}(q, \cdot)$ be the law of $(\tanh(A\sqrt{q} G_1), \dots, \tanh(A\sqrt{q} G_N))$, where $G \sim \mathcal{N}(0, I_N)$. For a probability measure η on $[0, 1]$, we write $\eta J_{A,N}$ for the probability measure on $[-1, 1]^N$ defined by

$$\eta J_{A,N}(B) := \int_0^1 J_{A,N}(q, B) \eta(dq).$$

The next proposition transfers invariant laws and total variation estimates between $K_{A,N}$ and $P_{A,N}$.

Proposition 2.4. *Let $P_{A,N}$ be the transition kernel of $X^{(N)}$ in (2.2). Then, for every bounded measurable $f : [-1, 1]^N \rightarrow \mathbb{R}$,*

$$P_{A,N}f(x) = \int_{[-1,1]^N} f(z) J_{A,N}(r_N(x), dz). \quad (2.40)$$

Consequently, for every deterministic $x \in [-1, 1]^N$, with $q = r_N(x)$, and every $t \geq 1$,

$$P_{A,N}^t(x, \cdot) = K_{A,N}^{t-1}(q, \cdot) J_{A,N}, \quad (r_N)_\# P_{A,N}^t(x, \cdot) = K_{A,N}^t(q, \cdot). \quad (2.41)$$

If $\nu_{A,N}$ is invariant for $K_{A,N}$, then $\pi_{A,N} := \nu_{A,N} J_{A,N}$, that is,

$$\pi_{A,N} = \int_0^1 J_{A,N}(q, \cdot) \nu_{A,N}(dq) \quad (2.42)$$

is an invariant law for $P_{A,N}$. Conversely, if $\pi_{A,N}$ is invariant for $P_{A,N}$ and $\nu_{A,N} := (r_N)_\# \pi_{A,N}$, then $\nu_{A,N}$ is invariant for $K_{A,N}$ and $\pi_{A,N}$ is given by (2.42).

For this pair of invariant laws and $t \geq 1$,

$$\|K_{A,N}^t(q, \cdot) - \nu_{A,N}\|_{\text{TV}} \leq \|P_{A,N}^t(x, \cdot) - \pi_{A,N}\|_{\text{TV}} \leq \|K_{A,N}^{t-1}(q, \cdot) - \nu_{A,N}\|_{\text{TV}}. \quad (2.43)$$

Proof. For a fixed vector x , the random vector $W_A^{(N)}x$ has law $A\sqrt{r_N(x)}G$. Applying tanh coordinatewise gives (2.40). Since $(r_N)_\# J_{A,N}(q, \cdot) = K_{A,N}(q, \cdot)$, induction gives (2.41). The same identities also give the statement about invariant laws. Indeed, if $\nu_{A,N} K_{A,N} = \nu_{A,N}$, then

$$(\nu_{A,N} J_{A,N}) P_{A,N} = (\nu_{A,N} K_{A,N}) J_{A,N} = \nu_{A,N} J_{A,N}.$$

Conversely, if $\pi_{A,N} P_{A,N} = \pi_{A,N}$ and $\nu_{A,N} = (r_N)_\# \pi_{A,N}$, then

$$\pi_{A,N} = \pi_{A,N} P_{A,N} = \nu_{A,N} J_{A,N}$$

and, after applying $(r_N)_\#$, we obtain $\nu_{A,N} K_{A,N} = \nu_{A,N}$.

The lower bound in (2.43) is the contraction of total variation under the measurable map r_N . For the upper bound, use (2.41) and (2.42). The reconstruction kernel $J_{A,N}$ contracts total variation, and hence

$$\|K_{A,N}^{t-1}(q, \cdot) J_{A,N} - \nu_{A,N} J_{A,N}\|_{\text{TV}} \leq \|K_{A,N}^{t-1}(q, \cdot) - \nu_{A,N}\|_{\text{TV}}.$$

□

2.5. Invariant laws. For later use, we write the invariance relation for $K_{A,N}$ on $(0, 1]$ in integral form:

$$\int_0^1 \phi(q) \nu_{A,N}(dq) = \int_0^1 \int_{\mathbb{R}^N} \phi(F_{A,N}(q, g)) \mathcal{G}_N(dg) \nu_{A,N}(dq) \quad (2.44)$$

for every bounded measurable $\phi : [0, 1] \rightarrow \mathbb{R}$.

For $A > 0$, $N \geq 1$, $q \in [0, 1]$, and $T \geq 1$, define

$$\bar{\nu}_T^q := \frac{1}{T} \sum_{t=0}^{T-1} K_{A,N}^t(q, \cdot). \quad (2.45)$$

Proposition 2.5. *The family $(\bar{\nu}_T^q)_{T \geq 1}$ is relatively compact for weak convergence. Every subsequential limit $\bar{\nu}$ is invariant for $K_{A,N}$ and satisfies the stationary equation (2.44). We denote by $\mathcal{I}_{A,N}(q)$ the nonempty set of all such limits.*

If $\bar{\nu} \in \mathcal{I}_{A,N}(q)$ and $\bar{\nu} \neq \delta_0$, then

$$\nu^+ := \frac{\bar{\nu}|_{(0,1]}}{1 - \bar{\nu}(\{0\})}$$

is invariant for $K_{A,N}$ on $(0, 1]$. We write $\mathcal{I}_{A,N}^+(q)$ for the collection of all such nonzero components. For every $\nu^+ \in \mathcal{I}_{A,N}^+(q)$, the measure

$$\pi^+ := \int_0^1 J_{A,N}(s, \cdot) \nu^+(ds) \quad (2.46)$$

is an invariant law for $X^{(N)}$. Equivalently,

$$\int_{[-1,1]^N} \varphi(y) \pi^+(dy) = \int_{[-1,1]^N} \int_{\mathbb{R}^N} \varphi(\tanh(A\sqrt{r_N(x)}g)) \mathcal{G}_N(dg) \pi^+(dx) \quad (2.47)$$

for every bounded measurable $\varphi : [-1, 1]^N \rightarrow \mathbb{R}$.

For $A > A_c(N)$, a negative-moment Foster–Lyapunov estimate shows that Cesàro limits started from positive radii do not charge the origin.

Proposition 2.6. *Let $N \geq 1$ and assume $A > A_c(N)$. Then, for every $q \in (0, 1]$, $\mathcal{I}_{A,N}^+(q)$ is nonempty. In particular, $K_{A,N}$ admits at least one invariant probability on $(0, 1]$, and $X^{(N)}$ admits a corresponding nonzero invariant law.*

The proofs of Propositions 2.5 and 2.6 are in Appendix A.3.

Symbol	Meaning and first definition
$W_{t,A}^{(N)}$	Gaussian weight matrix, (2.1).
$\mathcal{G}_N$	standard Gaussian measure on $\mathbb{R}^N$, (2.8).
$X_t^{(N)}, Y_t^{(N)}$	unrounded and rounded vector chains, (2.2) and (2.3).
Q_ϱ, Q_1	rounding at meshes ϱ and 1, as defined in (2.3) and (2.17).
$\tau_\varrho^{(N)}$	absorption time of the rounded vector chain, (2.4).
$Z_t^{(N)}, R_t^{(N)}$	scalar states of $X^{(N)}$, introduced before (2.6).
$\mathcal{H}_t^{(N)}, U_{A,\varrho}$	scalar state of $Y^{(N)}$ and its one-coordinate update, (2.17) and (2.18).
$F_{A,N}, K_{A,N}$	random map and transition kernel of $Z^{(N)}$, (2.7) and (2.8).
$P_{A,N}, P_{\varrho,A,N}$	unrounded and rounded vector transition kernels, defined before (2.4) and (2.40).
$J_{A,N}$	reconstruction kernel from $Z^{(N)}$ to $X^{(N)}$, introduced before Proposition 2.4.
V_A	mean map of $Z^{(N)}$, (2.10).
$A_c(N), A_c$	fixed-width threshold, with $A_c = A_c(N)$ when N is fixed, as defined in (2.26).
$\gamma_{A,N}$	negative mean of $\log \ell_{A,N}$, (2.25) and (2.27).
χ_N	Euclidean norm of a standard Gaussian vector in $\mathbb{R}^N$, (2.25).
L_A	lattice comparison map, (2.29).
$V_{A,\varrho}$	mean map of $\mathcal{H}^{(N)}$, (2.21).
A_{lat}	threshold for $L_A(h) < h$ on $(0, \infty)$, (2.30).
$q_*, \mu_A, C(q_0)$	fixed point and coefficients in the asymptotic for $V_A^t(q_0)$, (2.31) and (2.32).
$\nu_{A,N}, \pi_{A,N}$	nonzero invariant laws of $Z^{(N)}$ and $X^{(N)}$, (2.42) and (2.44).

Table 1: Frequently used notation.

2.6. Notation guide.

3. Subcritical cutoff as the mesh vanishes

We prove total variation cutoff to the absorbing state for $Y^{(N)}$ as the mesh vanishes at fixed width. Throughout the section $N \geq 1$ and $0 < A < A_c(N)$ are fixed, and $x_0 \in \mathbb{R}^N \setminus \{0\}$ is deterministic. We write

$$R_t := \|X_t^{(N)}\|_2, \quad \hat{R}_t := \|Y_t^{(N)}\|_2,$$

where $X^{(N)}$ starts from x_0 and $Y^{(N)}$ starts from $Q_\varrho x_0$. Generic constants c, C, c_N, C_N may change from line to line. A subscript N indicates dependence only on the fixed width unless further dependence is displayed.

We write $-\log R_t$ as a positive-drift random walk plus an almost surely finite correction. A synchronous coupling transfers its first-passage asymptotics to $\hat{R}_t$ down to the grid scale. A final estimate then gives absorption.

3.1. Unrounded first-passage asymptotics. Let $(\chi_{N,j})_{j \geq 1}$ be independent copies of χ_N . The fixed-width subcritical condition is $\gamma_{A,N} > 0$. Put

$$\xi_j := -\log\left(\frac{A}{\sqrt{N}}\chi_{N,j}\right), \quad \sigma_N^2 := \text{Var}(\xi_1). \quad (3.1)$$

By (2.27),

$$\mathbb{E}\xi_1 = \gamma_{A,N} = \log \frac{A_c(N)}{A}, \quad \sigma_N^2 = \text{Var}(\log \chi_N) = \frac{1}{4}\psi_1\left(\frac{N}{2}\right), \quad (3.2)$$

where ψ_1 is the trigamma function. For $0 < \varrho < R_0$, the cutoff scale is

$$L_\varrho := \log \frac{R_0}{\varrho}, \quad t_\varrho(a) := \max\left\{0, \left\lfloor \frac{L_\varrho}{\gamma_{A,N}} + a \frac{\sigma_N}{\gamma_{A,N}^{3/2}} \sqrt{L_\varrho} \right\rfloor\right\}. \quad (3.3)$$

Here and below Φ_G denotes the distribution function of a standard Gaussian random variable. The nonlinear correction to the logarithmic radius is summable.

Lemma 3.1. *There are an increasing adapted sequence $(E_t)_{t \geq 0}$ and an event $\Omega_{\log}$ of probability one such that, on $\Omega_{\log}$, $R_t > 0$ for every $t \geq 0$,*

$$-\log R_t = -\log R_0 + \sum_{j=1}^t \xi_j + E_t \quad \text{for every } t \geq 0 \quad (3.4)$$

and $E_t \uparrow E_\infty < \infty$. Consequently, $(E_t)_{t \geq 0}$ is tight.

Proof. For each fixed t , $X^{(N)}$ satisfies $R_t > 0$ almost surely. Since the time set is countable, the event

$$\Omega_{\text{nz}} := \bigcap_{t \geq 0} \{R_t > 0\}$$

has probability one. For each $j \geq 1$, define, on $\{R_{j-1} > 0\}$,

$$G^{(j)} := \frac{\sqrt{N}}{AR_{j-1}} W_{j,A}^{(N)} X_{j-1}^{(N)}.$$

Extend $G^{(j)}$ arbitrarily across the null event $\{R_{j-1} = 0\}$. Conditionally on the past, $G^{(j)}$ is standard Gaussian and its law does not depend on the past. Thus $(G^{(j)})_{j \geq 1}$ are independent standard Gaussian vectors. Taking $\chi_{N,j} := \|G^{(j)}\|_2$, the radius update is

$$R_j = \left\| \tanh\left(\frac{A}{\sqrt{N}} G^{(j)} R_{j-1}\right) \right\|_2.$$

Set $M_j := (A/\sqrt{N})\chi_{N,j}$. Since $|\tanh z| \leq |z|$ coordinatewise,

$$R_j \leq M_j R_{j-1} \quad \text{for every } j \geq 1. \quad (3.5)$$

Define

$$\eta_j := \log \frac{M_j R_{j-1}}{R_j} \geq 0, \quad E_t := \sum_{j=1}^t \eta_j, \quad \xi_j := -\log M_j,$$

which gives (3.4) simultaneously for every $t \geq 0$ on Ω_{nz} .

To prove finiteness of the error series, let $u \in \mathbb{R}$, with the value at zero defined by continuity. Then

$$0 \leq \log \frac{|u|}{|\tanh u|} \leq C u^2.$$

Applying this coordinatewise, if $u = (u_1, \dots, u_N)$ and $v_i = \tanh(u_i)$, with ratios at $u_i = 0$ understood by continuity, then

$$\|u\|_2^2 = \sum_{i=1}^N v_i^2 \frac{u_i^2}{v_i^2} \leq \max_{1 \leq i \leq N} \frac{u_i^2}{\tanh^2(u_i)} \|v\|_2^2,$$

and hence $\log(\|u\|_2/\|v\|_2) \leq C\|u\|_2^2$. With $u = (A/\sqrt{N})G^{(j)}R_{j-1}$ and $v = \tanh(u)$, this yields

$$0 \leq \eta_j \leq C M_j^2 R_{j-1}^2. \quad (3.6)$$

Since $A < A_c(N)$, the variables $\log M_j$ have negative mean. By the strong law of large numbers, for some $\kappa > 0$ and all sufficiently large j ,

$$\prod_{i=1}^j M_i \leq e^{-\kappa j}$$

almost surely. Combining this with (3.5) gives $R_j \leq R_0 e^{-\kappa j}$ eventually almost surely. On the other hand, $\chi_{N,j}^2 \leq e^{\kappa j}$ eventually almost surely by the Gaussian tail bound and Borel–Cantelli. Therefore the tail of the error series is almost surely bounded by

$$C \sum_{j \text{ large}} M_j^2 R_{j-1}^2 \leq C \frac{A^2}{N} R_0^2 \sum_{j \text{ large}} e^{-2\kappa(j-1)} \chi_{N,j}^2 \leq C \frac{A^2}{N} R_0^2 \sum_{j \text{ large}} e^{-\kappa(j-2)} < \infty.$$

The finitely many remaining terms are finite almost surely. Intersecting Ω_{nz} with the probability-one events furnished by the strong law and Borel–Cantelli gives an event $\Omega_{\log}$ on which all the preceding identities hold simultaneously for every time and $E_t \uparrow E_\infty < \infty$. $\square$

For a positive threshold ε_ϱ , define

$$L_{\varepsilon_\varrho} := \log \frac{R_0}{\varepsilon_\varrho}, \quad \theta_{\varepsilon_\varrho}^{\text{rad}} := \inf\{t \geq 0 : R_t \leq \varepsilon_\varrho\}.$$

The same profile holds for moving thresholds.

Proposition 3.2. *Let $\varepsilon_\varrho \downarrow 0$ satisfy*

$$L_{\varepsilon_\varrho} = L_\varrho + o(\sqrt{L_\varrho}).$$

Then, for every fixed $a \in \mathbb{R}$,

$$\mathbb{P}\left(\theta_{\varepsilon_\varrho}^{\text{rad}} > t_\varrho(a)\right) \rightarrow \Phi_G(-a), \quad \varrho \downarrow 0. \quad (3.7)$$

The same limit holds if $t_\varrho(a)$ is replaced by $\max\{0, \lfloor t_\varrho(a) + u_\varrho \rfloor\}$ for any deterministic $u_\varrho = o(\sqrt{L_\varrho})$.

Proof. On the probability-one event supplied by Lemma 3.1, simultaneously for every $t \geq 0$,

$$-\log R_t = -\log R_0 + S_t + E_t, \quad S_t := \sum_{j=1}^t \xi_j,$$

where $0 \leq E_t \leq E_\infty < \infty$. Since $R_t > 0$ for every t on this event, $R_t \leq \varepsilon_\varrho$ is equivalent to $S_t + E_t \geq L_{\varepsilon_\varrho}$. Hence, pathwise,

$$\theta_{\varepsilon_\varrho}^{\text{rad}} = \inf\{t \geq 0 : S_t + E_t \geq L_{\varepsilon_\varrho}\}.$$

Let $H_u = \inf\{t \geq 0 : S_t \geq u\}$, and let $\nu(u) = \inf\{t \geq 0 : S_t > u\}$ be the strict first-passage time used in [Gut09]. The law of ξ_1 is absolutely continuous, and hence so is the law of S_t for every $t \geq 1$. Thus, for every fixed $u > 0$,

$$\mathbb{P}(S_t = u \text{ for some } t \geq 0) = 0,$$

and therefore $H_u = \nu(u)$ almost surely. The increments have mean $\gamma_{A,N} > 0$ and variance $\sigma_N^2 < \infty$ by (3.2). Hence [Gut09, Theorem 3.5.1], with $\mu = \gamma_{A,N}$ and $\sigma = \sigma_N$, gives

$$\frac{H_u - u/\gamma_{A,N}}{\sigma_N \gamma_{A,N}^{-3/2} \sqrt{u}} \Rightarrow N(0, 1) \quad \text{as } u \rightarrow \infty.$$

The same limit holds with H_u replaced by H_{u+b_u} whenever $b_u = o(\sqrt{u})$, because the resulting changes in the centering and normalization are negligible.

Fix $B < \infty$. On $\{E_\infty \leq B\}$, the entrance time satisfies the pathwise bounds

$$H_{L_{\varepsilon_\varrho}-B} \leq \theta_{\varepsilon_\varrho}^{\text{rad}} \leq H_{L_{\varepsilon_\varrho}}. \quad (3.8)$$

Therefore, for $t = t_\varrho(a)$,

$$\mathbb{P}(H_{L_{\varepsilon_\varrho}-B} > t) - \mathbb{P}(E_\infty > B) \leq \mathbb{P}(\theta_{\varepsilon_\varrho}^{\text{rad}} > t) \leq \mathbb{P}(H_{L_{\varepsilon_\varrho}} > t) + \mathbb{P}(E_\infty > B).$$

Applying the first-passage limit with $u = L_\varrho$, both hitting probabilities in the last display converge to $\Phi_G(-a)$, since $L_{\varepsilon_\varrho} - L_\varrho = o(\sqrt{L_\varrho})$ and $B = o(\sqrt{L_\varrho})$. Thus

$$\Phi_G(-a) - \mathbb{P}(E_\infty > B) \leq \liminf_{\varrho \downarrow 0} \mathbb{P}(\theta_{\varepsilon_\varrho}^{\text{rad}} > t_\varrho(a))$$

and

$$\limsup_{\varrho \downarrow 0} \mathbb{P}(\theta_{\varepsilon_\varrho}^{\text{rad}} > t_\varrho(a)) \leq \Phi_G(-a) + \mathbb{P}(E_\infty > B).$$

Sending $B \rightarrow \infty$ proves the displayed limit. For the final assertion, fix $\delta > 0$. Since

$$\frac{u_\varrho}{\sigma_N \gamma_{A,N}^{-3/2} \sqrt{L_\varrho}} \rightarrow 0,$$

monotonicity places the perturbed survival probability, for all sufficiently small ϱ , between those at $t_\varrho(a - \delta)$ and $t_\varrho(a + \delta)$, up to a uniformly bounded time shift. The first part and continuity of $a \mapsto \Phi_G(-a)$ give the result after $\delta \downarrow 0$. $\square$

3.2. Rounding comparison and absorption. We couple $Y^{(N)}$ and $X^{(N)}$ with the same Gaussian matrices. The following estimate bounds the accumulated rounding error by the grid size times a polylogarithmic factor on the cutoff time scale.

Lemma 3.3. *Assume $0 < A < A_c(N)$, and let $x_0 \in \mathbb{R}^N$ be deterministic. Run the chain $X^{(N)}$ from $X_0^{(N)} = x_0$ and $Y^{(N)}$ from $Y_0^{(N)} = Q_\varrho(x_0)$ with the same Gaussian matrices. There is $C_N < \infty$, depending only on N , such that the following holds. For $0 < C_0 < \infty$, define*

$$T_\varrho := \lfloor C_0 \log(1/\varrho) \rfloor.$$

There exists $p = p(A, N, C_0)$ satisfying

$$0 \leq p \leq C_N(1 + C_0) \left(1 + \frac{1}{\gamma_{A,N}} \right), \quad (3.9)$$

and

$$\mathbb{P} \left(\max_{0 \leq t \leq T_\varrho} \|Y_t^{(N)} - X_t^{(N)}\|_2 > \varrho (\log(1/\varrho))^p \right) \rightarrow 0 \quad (3.10)$$

as $\varrho \downarrow 0$.

Proof. Step 1: Let $D_t := Y_t^{(N)} - X_t^{(N)}$ and $d_t := \|D_t\|_2$. For the common Gaussian matrix at time $t + 1$, set

$$\mathcal{T}_{t+1}(z) := \tanh(\mathbf{W}_{t+1,A}^{(N)} z).$$

The update rules (2.2) and (2.3) give

$$d_{t+1} \leq \|\mathcal{T}_{t+1}(Y_t^{(N)}) - \mathcal{T}_{t+1}(X_t^{(N)})\|_2 + \|Q_\varrho(\mathcal{T}_{t+1}(Y_t^{(N)})) - \mathcal{T}_{t+1}(Y_t^{(N)})\|_2.$$

The second term is at most $c_N \varrho$. By the one-Lipschitz property of $\tanh$, the first term is at most

$$\|\mathbf{W}_{t+1,A}^{(N)} Y_t^{(N)} - \mathbf{W}_{t+1,A}^{(N)} X_t^{(N)}\|_2 = \|\mathbf{W}_{t+1,A}^{(N)} D_t\|_2.$$

Thus

$$d_{t+1} \leq \|\mathbf{W}_{t+1,A}^{(N)} D_t\|_2 + c_N \varrho. \quad (3.11)$$

Let $\mathcal{F}_t := \sigma(\mathbf{W}_{1,A}^{(N)}, \dots, \mathbf{W}_{t,A}^{(N)})$, so that D_t is $\mathcal{F}_t$ -measurable. Fix a reference unit vector $e_1 \in \mathbb{R}^N$ and define the $\mathcal{F}_t$ -measurable unit vector and the multiplier

$$\theta_t := \frac{D_t}{\|D_t\|_2} \mathbf{1}_{\{D_t \neq 0\}} + e_1 \mathbf{1}_{\{D_t = 0\}}, \quad M_{t+1} := \|\mathbf{W}_{t+1,A}^{(N)} \theta_t\|_2.$$

Conditionally on $\mathcal{F}_t$, the vector θ_t is a fixed unit vector, so the Gaussian weights give

$$\mathbf{W}_{t+1,A}^{(N)} \theta_t \sim \mathcal{N}(0, (A^2/N) I_N),$$

independently of $\mathcal{F}_t$. Thus $(M_j)_{j \geq 1}$ are independent copies of $\ell_{A,N}$ from (2.25). In particular, $\mathbb{E} \log M_j = -\gamma_{A,N} < 0$. Moreover $\|\mathbf{W}_{t+1,A}^{(N)} D_t\|_2 = M_{t+1} d_t$ holds identically, since both sides vanish on $\{D_t = 0\}$, so (3.11) gives the pathwise bound $d_{t+1} \leq M_{t+1} d_t + c_N \varrho$. Since $d_0 \leq c_N \varrho$ and each $M_{t+1} \geq 0$, induction on t yields $d_t \leq U_t$ for all t , where $(U_t)_{t \geq 0}$ is defined by

$$U_{t+1} = M_{t+1} U_t + c_N \varrho, \quad U_0 = c_N \varrho. \quad (3.12)$$

Step 2: Since $M_1 = (A/\sqrt{N}) \chi_N$ and χ_N^2 is chi-squared with N degrees of freedom, the Gamma moment formula $\mathbb{E}(\chi_N^2)^{\alpha/2} = 2^{\alpha/2} \Gamma((N + \alpha)/2) / \Gamma(N/2)$ gives, for every $\alpha \geq 0$,

$$\log \mathbb{E} M_1^\alpha = \alpha \log \frac{A}{\sqrt{N}} + \frac{\alpha}{2} \log 2 + \log \Gamma\left(\frac{N + \alpha}{2}\right) - \log \Gamma\left(\frac{N}{2}\right). \quad (3.13)$$

In particular all positive moments are finite, and differentiating (3.13) twice in α gives

$$\frac{d^2}{d\alpha^2} \log \mathbb{E} M_1^\alpha = \frac{1}{4} \psi_1 \left(\frac{N + \alpha}{2} \right),$$

with ψ_1 the trigamma function. As ψ_1 is positive and decreasing on $(0, \infty)$, this second derivative is bounded on $[0, 1]$ by $K := \frac{1}{4} \psi_1(N/2)$. Put

$$\Lambda(\alpha) := \log \mathbb{E} M_1^\alpha, \quad \alpha_\gamma := \min \left\{ 1, \frac{\gamma_{A,N}}{2K} \right\}.$$

Since $\Lambda(0) = 0$ and $\Lambda'(0) = \mathbb{E} \log M_1 = -\gamma_{A,N}$, Taylor's theorem yields

$$\mathbb{E} M_1^{\alpha_\gamma} \leq \exp \left(-\frac{\alpha_\gamma \gamma_{A,N}}{2} \right) =: q < 1, \quad \alpha_\gamma^{-1} \leq C_N \left(1 + \gamma_{A,N}^{-1} \right).$$

Let $L_\varrho := \log(1/\varrho)$. Markov's inequality and a union bound therefore give, for every $p_0 > 0$,

$$\mathbb{P} \left(\max_{0 \leq s \leq t \leq T_\varrho} \prod_{j=s+1}^t M_j > L_\varrho^{p_0} \right) \leq L_\varrho^{-\alpha_\gamma p_0} \sum_{0 \leq s \leq t \leq T_\varrho} q^{t-s} \leq \frac{T_\varrho + 1}{1 - q} L_\varrho^{-\alpha_\gamma p_0}.$$

We may choose p_0 so that $\alpha_\gamma p_0 > 2$ and, after increasing C_N ,

$$p_0 \leq C_N(1 + C_0) \left(1 + \gamma_{A,N}^{-1} \right).$$

Since $T_\varrho = O(L_\varrho)$, this probability tends to zero. Thus

$$\mathbb{P} \left(\max_{0 \leq s \leq t \leq T_\varrho} \prod_{j=s+1}^t M_j > L_\varrho^{p_0} \right) \rightarrow 0. \quad (3.14)$$

On the complementary event, the explicit solution of (3.12) gives, for every $t \leq T_\varrho$,

$$U_t \leq c_N \varrho \sum_{s=0}^t \prod_{j=s+1}^t M_j \leq C_N \varrho T_\varrho (\log(1/\varrho))^{p_0} \leq \varrho (\log(1/\varrho))^{p_0 + C_N(1+C_0)}$$

for all sufficiently small ϱ . Set $p := p_0 + \lceil C_N(1 + C_0) \rceil$. After increasing C_N , this exponent satisfies (3.9). Since $d_t \leq U_t$, this proves (3.10). $\square$

Once $Y^{(N)}$ enters a ball of radius $K\varrho$, the number of additional steps to absorption is $O(\log K)$, with an exponentially decaying upper tail. The constants below may depend on (A, N) , but not on ϱ , K , or $Y_0^{(N)}$.

Lemma 3.4. *Let $0 < A < A_c(N)$. There are constants $c > 0$ and $C < \infty$ such that, uniformly over $\varrho \in (0, 1)$, $K \geq 2$, and deterministic $Y_0^{(N)} = y \in \varrho \mathbb{Z}^N$ satisfying $\|y\|_2 \leq K\varrho$,*

$$\mathbb{P}_y \left(\tau_\varrho^{(N)} > C \log K + r \right) \leq C e^{-cr}, \quad r \geq 0. \quad (3.15)$$

Here $\mathbb{P}_y$ is the law of $Y^{(N)}$ started from y .

Proof. Step 1: Let

$$K_t := \frac{\widehat{R}_t}{\varrho}.$$

Since $\|Q_\varrho(z) - z\|_2 \leq c_N \varrho$ and $\tanh$ is one-Lipschitz coordinatewise, $\hat{R}_{t+1} \leq \|W_{t+1,A}^{(N)} Y_t^{(N)}\|_2 + c_N \varrho$. Let $\mathcal{F}_t := \sigma(W_{1,A}^{(N)}, \dots, W_{t,A}^{(N)})$ and fix a reference unit vector $e_1 \in \mathbb{R}^N$. Exactly as in the proof of Lemma 3.3, set

$$\theta_t := \frac{Y_t^{(N)}}{\|Y_t^{(N)}\|_2} \mathbb{1}_{\{Y_t^{(N)} \neq 0\}} + e_1 \mathbb{1}_{\{Y_t^{(N)} = 0\}}, \quad M_{t+1} := \|W_{t+1,A}^{(N)} \theta_t\|_2.$$

As in the proof of Lemma 3.3, the variables M_j are independent copies of $\ell_{A,N} = (A/\sqrt{N})\chi_N$. Moreover, $\|W_{t+1,A}^{(N)} Y_t^{(N)}\|_2 = M_{t+1} \hat{R}_t$ identically, with both sides zero when $Y_t^{(N)} = 0$. Hence, pathwise,

$$K_{t+1} \leq M_{t+1} K_t + c_N. \quad (3.16)$$

Define

$$U_0 = K, \quad U_{t+1} = M_{t+1} U_t + c_N.$$

We have $\mathbb{E} \log M_1 = -\gamma_{A,N} < 0$. For every $\varepsilon > 0$, $\log(M_1 + \varepsilon)$ has exponential moments near the origin because M_1 has all positive moments by (3.13). Thus the hypotheses of Lemma 2.3 hold with $b = c_N$. Since $K_0 \leq K = U_0$, (3.16) gives $K_t \leq U_t$. Hence there are $K_* = K_*(A, N) < \infty$ and $c_1, c_2, c_3 > 0$ such that, with $\kappa := \inf\{t \geq 0 : K_t \leq K_*\}$,

$$\mathbb{P}(\kappa > c_1 \log K + r) \leq c_2 e^{-c_3 r}, \quad r \geq 0. \quad (3.17)$$

Step 2: Once $K_t \leq K_*$, $Y^{(N)}$ has a uniform positive probability of absorption in the next step. For a matrix B , write $\|B\|_{\ell^2 \rightarrow \ell^2} := \sup_{\|u\|_2=1} \|Bu\|_2$. Since $W_{t+1,A}^{(N)}$ has a nondegenerate Gaussian law, there is $p_* = p_*(A, N, K_*) > 0$ such that

$$\mathbb{P}\left(\|W_{t+1,A}^{(N)}\|_{\ell^2 \rightarrow \ell^2} \leq \frac{1}{2K_*}\right) \geq p_*.$$

On this small operator norm event, if $\hat{R}_t \leq K_* \varrho$, then

$$\|\tanh(W_{t+1,A}^{(N)} Y_t^{(N)})\|_\infty \leq \|W_{t+1,A}^{(N)} Y_t^{(N)}\|_2 \leq \frac{\varrho}{2},$$

and hence $Y_{t+1}^{(N)} = 0$. Thus each visit to $\{K_t \leq K_*\}$ is followed by absorption with conditional probability at least p_* . To control a failed attempt, start with $K_0 \leq K_*$ and define

$$\kappa^+ := \inf\{t \geq 1 : K_t \leq K_*\}.$$

After one step, $Y^{(N)}$ satisfies $K_1 \leq M_1 K_* + c_N$, where M_1 has law $(A/\sqrt{N})\chi_N$. Hence (3.17), conditionally applied from time one with initial radius $\max\{2, K_1\}$, gives

$$\mathbb{P}(\kappa^+ > 1 + c_1 \log(\max\{2, K_1\}) + r \mid \mathcal{F}_1) \leq c_2 e^{-c_3 r}, \quad r \geq 0.$$

Choose $s_0 \in (0, c_3)$. The layer-cake formula and $K_1 \leq M_1 K_* + c_N$ give

$$B := \sup_{\varrho \in (0,1)} \sup_{K_0 \leq K_*} \mathbb{E} e^{s_0 \kappa^+} \leq C \mathbb{E}(\max\{2, M_1 K_* + c_N\})^{c_1 s_0} < \infty,$$

since χ_N has moments of every positive order.

The one-step absorption event above is independent of the starting point in the fixed ball, and therefore

$$\sup_{\varrho \in (0,1)} \sup_{K_0 \leq K_*} \mathbb{P}(Y_1^{(N)} \neq 0) \leq 1 - p_*.$$

For $s \in (0, s_0)$, Hölder's inequality consequently gives

$$q_s := \sup_{\varrho \in (0,1)} \sup_{K_0 \leq K_*} \mathbb{E} \left[e^{s\kappa^+} \mathbf{1}_{\{Y_1^{(N)} \neq 0\}} \right] \leq B^{s/s_0} (1 - p_*)^{1-s/s_0}.$$

The right-hand side converges to $1 - p_*$ as $s \downarrow 0$. We may thus fix $s \in (0, s_0)$ sufficiently small that $q_s < 1$.

Step 3: For $n \geq 1$, set

$$F_{s,n} := \sup_{\varrho \in (0,1)} \sup_{K_0 \leq K_*} \mathbb{E} e^{s(\min\{\tau_\varrho^{(N)}, n\})} < \infty.$$

On $\{Y_1^{(N)} = 0\}$ we have $\tau_\varrho^{(N)} = 1$. On its complement, the strong Markov property at κ^+ , together with $K_{\kappa^+} \leq K_*$, gives

$$F_{s,n} \leq e^s + q_s F_{s,n}.$$

Consequently $F_{s,n} \leq e^s/(1 - q_s)$ uniformly in n . Monotone convergence now shows that

$$\sup_{\varrho \in (0,1)} \sup_{K_0 \leq K_*} \mathbb{E} e^{s\tau_\varrho^{(N)}} \leq \frac{e^s}{1 - q_s} < \infty.$$

Markov's inequality gives $C_0 < \infty$ and $c_0 > 0$ such that, uniformly in ϱ and deterministic starts with $K_0 \leq K_*$,

$$\mathbb{P}(\tau_\varrho^{(N)} > u) \leq C_0 e^{-c_0 u}, \quad u \geq 0. \quad (3.18)$$

Combining (3.17) with (3.18) and the Markov property at κ gives, for $r \geq 0$,

$$\mathbb{P}(\tau_\varrho^{(N)} > c_1 \log K + r) \leq c_2 e^{-c_3 r/2} + C_0 e^{-c_0 r/2} \leq (c_2 + C_0) e^{-\frac{1}{2} \min\{c_3, c_0\} r}.$$

After increasing the prefactor and the coefficient of $\log K$, this is (3.15). $\square$

3.3. Proof of the cutoff theorem.

Proof of Theorem 1.4. Fix $a \in \mathbb{R}$ and write $\ell_\varrho := \log(1/\varrho)$ and $t_\varrho := t_\varrho(a)$. By (2.5), it is enough to prove $\mathbb{P}(\tau_\varrho^{(N)} > t_\varrho) \rightarrow \Phi_G(-a)$.

Step 1: Since $L_\varrho = \ell_\varrho + \log R_0$ and a is fixed, (3.3) gives $t_\varrho = O(\ell_\varrho)$. Choose $C_0 < \infty$ such that, for all sufficiently small ϱ , $0 \leq t_\varrho \leq T_\varrho := \lfloor C_0 \ell_\varrho \rfloor$. Let $p = p(A, N, C_0)$ be the exponent from Lemma 3.3, put $d_\varrho := \varrho \ell_\varrho^p$ and $M_\varrho := \ell_\varrho^{p+3}$. For the synchronous coupling, define

$$\mathcal{G}_\varrho := \left\{ \max_{0 \leq s \leq T_\varrho} \|Y_s^{(N)} - X_s^{(N)}\|_2 \leq d_\varrho \right\}, \quad \mathbb{P}(\mathcal{G}_\varrho^c) = o(1), \quad (3.19)$$

where the estimate follows from (3.10).

We compare absorption with the unrounded entrance times $\Theta_\varrho^- := \theta_{d_\varrho}^{\text{rad}}$ and $\Theta_\varrho^+ := \theta_{M_\varrho}^{\text{rad}}$. Since both $\log(R_0/d_\varrho)$ and $\log(R_0/(M_\varrho \varrho))$ satisfy the hypotheses of Proposition 3.2,

$$\mathbb{P}(\Theta_\varrho^- > t_\varrho) \rightarrow \Phi_G(-a), \quad \mathbb{P}(\Theta_\varrho^+ > t_\varrho) \rightarrow \Phi_G(-a) \quad (3.20)$$

as $\varrho \downarrow 0$.

Step 2: The lower bound. On $\{\tau_\varrho^{(N)} \leq t_\varrho\} \cap \mathcal{G}_\varrho$, absorption gives $Y_{\tau_\varrho^{(N)}}^{(N)} = 0$, while $\tau_\varrho^{(N)} \leq t_\varrho \leq T_\varrho$. Hence $R_{\tau_\varrho^{(N)}} = \|X_{\tau_\varrho^{(N)}}^{(N)} - Y_{\tau_\varrho^{(N)}}^{(N)}\|_2 \leq d_\varrho$ and thus $\Theta_\varrho^- \leq \tau_\varrho^{(N)}$. Consequently,

$$\{\Theta_\varrho^- > t_\varrho\} \cap \mathcal{G}_\varrho \subseteq \{\tau_\varrho^{(N)} > t_\varrho\}, \quad \mathbb{P}(\tau_\varrho^{(N)} > t_\varrho) \geq \mathbb{P}(\Theta_\varrho^- > t_\varrho) - o(1), \quad (3.21)$$

where we used (3.19) in the last estimate.

Step 3: The upper bound. Let $c_{\text{abs}} > 0$ and $C_{\text{abs}} < \infty$ be constants for which (3.15) holds. Choose $B > C_{\text{abs}}(p+3)$ and set $h_\varrho := \lceil B \log \ell_\varrho \rceil$, $K_\varrho := 2M_\varrho$, and $r_\varrho := h_\varrho - C_{\text{abs}} \log K_\varrho$. Since $\log K_\varrho = \log 2 + (p+3) \log \ell_\varrho$, we have $r_\varrho \rightarrow \infty$, while $h_\varrho = o(\sqrt{L_\varrho})$ and $t_\varrho - h_\varrho \geq 0$ for all sufficiently small ϱ .

Write $\mathcal{E}_\varrho := \{\Theta_\varrho^+ \leq t_\varrho - h_\varrho\} \cap \mathcal{G}_\varrho$. On $\mathcal{E}_\varrho$, we have $\Theta_\varrho^+ \leq T_\varrho$, and therefore

$$\hat{R}_{\Theta_\varrho^+} = \|Y_{\Theta_\varrho^+}^{(N)}\|_2 \leq \|X_{\Theta_\varrho^+}^{(N)}\|_2 + \|Y_{\Theta_\varrho^+}^{(N)} - X_{\Theta_\varrho^+}^{(N)}\|_2 \leq M_\varrho \varrho + d_\varrho \leq K_\varrho \varrho$$

for all sufficiently small ϱ . Define the $\mathcal{F}_{\Theta_\varrho^+}$ -measurable event $\mathcal{A}_\varrho := \{\Theta_\varrho^+ \leq t_\varrho - h_\varrho, \hat{R}_{\Theta_\varrho^+} \leq K_\varrho \varrho\}$. The preceding bound gives $\mathcal{E}_\varrho \subseteq \mathcal{A}_\varrho$. Therefore

$$\begin{aligned} \mathbb{P}(\tau_\varrho^{(N)} > t_\varrho, \mathcal{E}_\varrho) &\leq \mathbb{P}(\tau_\varrho^{(N)} > t_\varrho, \mathcal{A}_\varrho) \\ &= \mathbb{E} \left[\mathbf{1}_{\mathcal{A}_\varrho} \mathbb{P}_{Y_{\Theta_\varrho^+}^{(N)}} \left(\tau_\varrho^{(N)} > t_\varrho - \Theta_\varrho^+ \right) \right] \\ &\leq \mathbb{E} \left[\mathbf{1}_{\mathcal{A}_\varrho} \mathbb{P}_{Y_{\Theta_\varrho^+}^{(N)}} \left(\tau_\varrho^{(N)} > h_\varrho \right) \right] \\ &\leq C_{\text{abs}} e^{-c_{\text{abs}} r_\varrho} = o(1). \end{aligned}$$

Here $\mathbb{P}_y$ denotes the law of $Y^{(N)}$ started from y . The equality is the strong Markov property at Θ_ϱ^+ . The next inequality uses $t_\varrho - \Theta_\varrho^+ \geq h_\varrho$ on $\mathcal{A}_\varrho$. The last inequality follows from Lemma 3.4, since the entrance radius is at most $K_\varrho \varrho$ and $h_\varrho = C_{\text{abs}} \log K_\varrho + r_\varrho$.

Hence

$$\begin{aligned} \{\tau_\varrho^{(N)} > t_\varrho\} &\subseteq \{\Theta_\varrho^+ > t_\varrho - h_\varrho\} \cup \mathcal{G}_\varrho^c \cup (\{\tau_\varrho^{(N)} > t_\varrho\} \cap \mathcal{E}_\varrho), \\ \mathbb{P}(\tau_\varrho^{(N)} > t_\varrho) &\leq \mathbb{P}(\Theta_\varrho^+ > t_\varrho - h_\varrho) + \mathbb{P}(\mathcal{G}_\varrho^c) + \mathbb{P}(\tau_\varrho^{(N)} > t_\varrho, \mathcal{E}_\varrho) \\ &\leq \mathbb{P}(\Theta_\varrho^+ > t_\varrho - h_\varrho) + o(1). \end{aligned} \tag{3.22}$$

Step 4: The profile. By the level calculations in Step 1, the conclusion of (3.20) also holds for Θ_ϱ^+ at $t_\varrho - h_\varrho$, because $h_\varrho = o(\sqrt{L_\varrho})$. Combining these limits with (3.21) and (3.22) yields

$$\Phi_G(-a) \leq \liminf_{\varrho \downarrow 0} \mathbb{P}(\tau_\varrho^{(N)} > t_\varrho) \leq \limsup_{\varrho \downarrow 0} \mathbb{P}(\tau_\varrho^{(N)} > t_\varrho) \leq \Phi_G(-a).$$

Thus the absorption tail converges to $\Phi_G(-a)$, and (2.5) gives (1.18).

Since A and N are fixed, $\sigma_N \gamma_{A,N}^{-3/2} \sqrt{L_\varrho} = o(L_\varrho / \gamma_{A,N})$. The profile just proved satisfies $\Phi_G(-a) \rightarrow 1$ as $a \rightarrow -\infty$ and $\Phi_G(-a) \rightarrow 0$ as $a \rightarrow \infty$. Therefore (1.14) holds with center $L_\varrho / \gamma_{A,N}$ and window $\sigma_N \gamma_{A,N}^{-3/2} \sqrt{L_\varrho}$. $\square$

Remark 3.5. As $A \uparrow A_c(N)$,

$$\gamma_{A,N} = \log \frac{A_c(N)}{A} = \frac{A_c(N) - A}{A_c(N)} + O_N((A_c(N) - A)^2).$$

Hence the cutoff center and window satisfy

$$\frac{L_\varrho}{\gamma_{A,N}} \sim \frac{A_c(N) L_\varrho}{A_c(N) - A}, \quad \frac{\sigma_N}{\gamma_{A,N}^{3/2}} \sqrt{L_\varrho} \sim \frac{\sigma_N A_c(N)^{3/2} \sqrt{L_\varrho}}{(A_c(N) - A)^{3/2}}.$$

For fixed C_0 , (3.9) allows the exponent in Lemma 3.3 to be chosen with

$$p = O_{N,C_0}((A_c(N) - A)^{-1}) \quad \text{as } A \uparrow A_c(N).$$

The constants in Lemma 3.4 may also diverge as $A \uparrow A_c(N)$. For fixed $A < A_c(N)$, p appears only in the bound $\varrho(\log(1/\varrho))^p$ for $\max_{0 \leq t \leq T_\varrho} \|Y_t^{(N)} - X_t^{(N)}\|_2$ in (3.10). The absorption constants determine only the $O(\log \log(1/\varrho))$ additional time allowed after the chains are close to the origin. Neither changes the cutoff center or window.

4. Subcritical cutoff as $N \rightarrow \infty$

Fix $\varrho \in (0, 1)$ and let $N \rightarrow \infty$. We track $\mathcal{H}^{(N)}$ along the iterates of $V_{A, \varrho}$ until they reach $(\log N)^{-1}$. The next step makes $\mathcal{H}^{(N)}$ polynomially small with high probability, and one further step gives absorption. The same tracking estimate excludes absorption one step before the deterministic entrance time.

4.1. Deterministic rounded dynamics. Recall the transition of $\mathcal{H}^{(N)}$ and its centered noise decomposition from (2.19) and (2.22). The deterministic orbit and terminal-scale entrance time are

$$\bar{h}_{0,N} := \frac{1}{N\varrho^2} \|Q_\varrho x_N\|_2^2, \quad \bar{h}_{t+1,N} = V_{A,\varrho}(\bar{h}_{t,N}), \quad (4.1)$$

and

$$\bar{t}_N := \inf \left\{ t \in \mathbb{N}_0 : \bar{h}_{t,N} \leq \frac{1}{\log N} \right\}. \quad (4.2)$$

The tie convention in Section 2 gives

$$|Q_1(u)| \leq |Q_1(v)| \quad \text{whenever } |u| \leq |v|. \quad (4.3)$$

The threshold A_{lat} is defined by (2.30). Equivalently, $A < A_{\text{lat}}$ means that $L_A(h) < h$ for every $h > 0$.

Lemma 4.1. *The threshold in (2.30) belongs to $(0, 1)$. If $A < A_{\text{lat}}$, then there exists $\theta_A \in (0, 1)$ such that*

$$L_A(h) \leq \theta_A h \quad \text{for every } h > 0. \quad (4.4)$$

Consequently,

$$V_{A,\varrho}(h) \leq \theta_A h, \quad h \geq 0. \quad (4.5)$$

Proof. For every $\alpha > 0$, $m(\alpha) > 0$, since $\mathbb{P}(|\alpha G| > 1/2) > 0$. The map $\alpha \mapsto m(\alpha)$ is continuous on $(0, \infty)$ by dominated convergence on compact α -intervals: if $\alpha \leq R$, then

$$Q_1(\alpha G)^2 \leq (|\alpha G| + \tfrac{1}{2})^2 \leq (R|G| + \tfrac{1}{2})^2.$$

Since $Q_1(u) = 0$ on $\{|u| \leq 1/2\}$ and $|Q_1(u)| \leq |u| + 1/2 \leq 2|u|$ on $\{|u| > 1/2\}$,

$$0 \leq \frac{m(\alpha)}{\alpha^2} \leq 4\mathbb{E}[G^2 \mathbf{1}_{\{|G| > 1/(2\alpha)\}}] \xrightarrow{\alpha \downarrow 0} 0. \quad (4.6)$$

The bound $|Q_1(u) - u| \leq 1/2$ gives $Q_1(\alpha G)/\alpha \rightarrow G$ pointwise and

$$\left(\frac{|Q_1(\alpha G)|}{\alpha} \right)^2 \leq \left(|G| + \frac{1}{2\alpha} \right)^2 \leq (|G| + 1)^2 \quad \text{for } \alpha \geq \frac{1}{2}.$$

Dominated convergence therefore gives

$$\frac{m(\alpha)}{\alpha^2} = \mathbb{E} \left[\left(\frac{Q_1(\alpha G)}{\alpha} \right)^2 \right] \xrightarrow{\alpha \rightarrow \infty} \mathbb{E}[G^2] = 1. \quad (4.7)$$

It follows from (4.6) that $\alpha^2/m(\alpha) \rightarrow \infty$ as $\alpha \downarrow 0$, while (4.7) gives $\alpha^2/m(\alpha) \rightarrow 1$ as $\alpha \rightarrow \infty$. Since $\alpha^2/m(\alpha)$ is continuous and positive on $(0, \infty)$, these two endpoint limits imply that its infimum is positive: outside a compact subinterval of $(0, \infty)$ the ratio is bounded below, while on that compact subinterval it has a positive minimum. Hence the threshold A_{lat} belongs to $(0, 1]$. The upper bound is strict. Indeed, for $\alpha = 1/2$,

$$Q_1(\alpha G)^2 \geq 1 \quad \text{on } \{|G| > 1\},$$

and hence

$$m(1/2) \geq \mathbb{P}(|G| > 1) > \frac{1}{4}.$$

The final strict inequality can be checked by an elementary Gaussian tail bound. Therefore $A_{\text{lat}}^2 \leq (1/2)^2/m(1/2) < 1$.

By the change of variables $\alpha = A\sqrt{h}$,

$$\frac{L_A(h)}{h} = A^2 \frac{m(A\sqrt{h})}{(A\sqrt{h})^2}, \quad h > 0.$$

Thus $A < A_{\text{lat}}$ implies $L_A(h) < h$ for every $h > 0$. The same endpoint estimates give

$$\frac{L_A(h)}{h} \rightarrow 0 \quad \text{as } h \downarrow 0, \quad \frac{L_A(h)}{h} \rightarrow A^2 < 1 \quad \text{as } h \rightarrow \infty.$$

Since $h \mapsto L_A(h)/h$ is continuous on $(0, \infty)$, choose $0 < r < R < \infty$ so that the ratio is bounded away from one on $(0, r] \cup [R, \infty)$. Compactness gives the same bound on $[r, R]$, and hence (4.4).

The inequality $|\tanh z| \leq |z|$ and (4.3) give, pointwise in G ,

$$\left| Q_1\left(\varrho^{-1} \tanh(\varrho A\sqrt{h} G)\right) \right| \leq \left| Q_1(A\sqrt{h} G) \right|.$$

Taking squares and expectations yields $V_{A,\varrho}(h) \leq L_A(h)$ for $h > 0$. The case $h = 0$ is immediate, since both sides vanish. Combining this with (4.4) proves (4.5). $\square$

Let $\bar{\Phi}(x) := \mathbb{P}(G > x)$ and $\phi(x) := (2\pi)^{-1/2} \exp(-x^2/2)$. For $0 < \varrho < 1$ set

$$b_{\varrho,k} := \varrho^{-1} \operatorname{arctanh}\left(\varrho(k + \tfrac{1}{2})\right), \quad \mathcal{K}_\varrho := \{k \geq 0 : \varrho(k + \tfrac{1}{2}) < 1\}. \quad (4.8)$$

Then

$$V_{A,\varrho}(h) = 2 \sum_{k \in \mathcal{K}_\varrho} (2k+1) \bar{\Phi}\left(\frac{b_{\varrho,k}}{A\sqrt{h}}\right), \quad V'_{A,\varrho}(h) = \sum_{k \in \mathcal{K}_\varrho} (2k+1) \frac{b_{\varrho,k}}{Ah^{3/2}} \phi\left(\frac{b_{\varrho,k}}{A\sqrt{h}}\right). \quad (4.9)$$

Indeed, $Q_1(u)^2 = \sum_{k \geq 0} (2k+1) \mathbb{1}_{\{|u| > k + \frac{1}{2}\}}$ outside Gaussian-null boundary events, and the preimage of the layer $k + \frac{1}{2}$ under $u \mapsto \varrho^{-1} \tanh(\varrho u)$ is $b_{\varrho,k}$. Since $\mathcal{K}_\varrho$ is finite, differentiating term by term is immediate. The small-radius estimate in (4.11) also gives $V'_{A,\varrho}(h) \rightarrow 0$ as $h \downarrow 0$. We use this one-sided extension at the origin.

By (4.9), $V_{A,\varrho}$ is positive and $V'_{A,\varrho}$ is bounded on compact subsets of $(0, \infty)$. The threshold $b_{\varrho,0} > 0$ in (4.8) yields exponential decay in $1/h$ near the origin.

We will use regularity of $V_{A,\varrho}$ both away from and near the origin.

Lemma 4.2. *Fix $\varrho \in (0, 1)$. For every $0 < r < R < \infty$ there are constants $m, L \in (0, \infty)$, depending on A, ϱ, r, R , such that*

$$m \leq V_{A,\varrho}(h), \quad V'_{A,\varrho}(h) \leq L, \quad r \leq h \leq R. \quad (4.10)$$

For every $r \in (0, 1)$ there are constants $c, C \in (0, \infty)$, depending on A, ϱ, r , such that, for $0 < h \leq r$,

$$V_{A,\varrho}(h) \leq C \exp\left(-\frac{c}{h}\right), \quad V'_{A,\varrho}(h) \leq Ch^{-2} \exp\left(-\frac{c}{h}\right). \quad (4.11)$$

Proof. Since $0 < \varrho < 1$, the set $\mathcal{K}_\varrho$ is finite and nonempty. In particular $0 \in \mathcal{K}_\varrho$ and

$$b_{\varrho,0} = \varrho^{-1} \operatorname{arctanh}(\varrho/2) > 0.$$

The representation (4.9) is a finite sum of continuous functions of $h > 0$. Hence $V_{A,\varrho}$ is continuous on $(0, \infty)$. Moreover, for every $h > 0$,

$$V_{A,\varrho}(h) \geq 2\bar{\Phi}\left(\frac{b_{\varrho,0}}{A\sqrt{h}}\right) > 0.$$

It follows that

$$m := \min_{h \in [r, R]} V_{A,\varrho}(h) > 0.$$

Similarly, (4.9) is a finite sum of continuous functions on $[r, R]$. Thus

$$L := \max_{h \in [r, R]} V'_{A,\varrho}(h) < \infty,$$

which proves (4.10).

For the small-radius bounds, $C < \infty$ may change from line to line and depends only on A, ϱ, r . Put $b_0 := b_{\varrho,0}$ and

$$c := \frac{b_0^2}{2A^2}.$$

Since $k \mapsto b_{\varrho,k}$ is increasing on $\mathcal{K}_\varrho$, we have $b_{\varrho,k} \geq b_0$ for every $k \in \mathcal{K}_\varrho$. Using $\bar{\Phi}(x) \leq \exp(-x^2/2)$ for $x \geq 0$ in (4.9), we obtain, for $h > 0$,

$$V_{A,\varrho}(h) \leq 2 \sum_{k \in \mathcal{K}_\varrho} (2k+1) \exp\left(-\frac{b_{\varrho,k}^2}{2A^2h}\right) \leq C \exp\left(-\frac{c}{h}\right).$$

For the derivative, the same lower bound on the thresholds and (4.9) give

$$V'_{A,\varrho}(h) \leq \frac{1}{\sqrt{2\pi}} \sum_{k \in \mathcal{K}_\varrho} (2k+1) \frac{b_{\varrho,k}}{Ah^{3/2}} \exp\left(-\frac{b_{\varrho,k}^2}{2A^2h}\right) \leq Ch^{-3/2} \exp\left(-\frac{c}{h}\right).$$

If $0 < h \leq r < 1$, then $h^{-3/2} \leq h^{-2}$, and the desired derivative estimate follows after increasing C . This proves (4.11). $\square$

4.2. Tracking $\mathcal{H}^{(N)}$. Assume $A < A_{\text{lat}}$, put $a_N = (\log N)^{-1}$, and consider an orbit $h_{t+1,N} = V_{A,\varrho}(h_{t,N})$ with $h_{0,N} \leq C_0$. For $\delta \in (0, 1/4)$, set

$$L_{t,N} := \sup_{\substack{u \geq 0 \\ |u - h_{t,N}| \leq \delta(h_{t,N} + a_N)}} V'_{A,\varrho}(u), \quad \beta_{t,N} := \frac{h_{t,N} + a_N}{h_{t+1,N} + a_N}, \quad \alpha_{t,N} := L_{t,N} \beta_{t,N}.$$

The tracking argument below normalizes $\mathcal{H}_t^{(N)} - \bar{h}_{t,N}$ by $\bar{h}_{t,N} + a_N$. When $h_{t,N} = \bar{h}_{t,N}$, the factors $\alpha_{t,N}$ and $\beta_{t,N}$ control the propagation of this ratio. The following estimate bounds the resulting products.

Lemma 4.3. *There is a constant $C < \infty$, depending only on A, ϱ, δ, C_0 , such that, for all $T_N \in \mathbb{N}_0$ and all sufficiently large N ,*

$$\prod_{u=s}^{t-1} (1 \vee \alpha_{u,N}) + \beta_{s,N} \prod_{u=s+1}^{t-1} (1 \vee \alpha_{u,N}) \leq \exp\{C(\log \log N)^2\} \quad (4.12)$$

for all $0 \leq s < t \leq T_N$.

Proof. Fix $r \in (0, 1/2)$ and set

$$\tau_r := \inf\{t : h_{t,N} \leq r\}, \quad \tau_a := \inf\{t : h_{t,N} \leq a_N\}.$$

Let $I_{t,N}$ denote the interval over which the supremum defining $L_{t,N}$ is taken.

Step 1: The region $h_{t,N} > r$. By (4.5), $h_{t,N} \leq \theta_A^t C_0$, and hence $\tau_r \leq C$. For $t < \tau_r$, we have $h_{t,N} \in [r, C_0]$. Since $a_N \leq r$ for all sufficiently large N and $\delta < 1/4$, it follows that $I_{t,N} \subseteq [r/2, 2(C_0 + r)]$. The bounds in (4.10) therefore give

$$h_{t+1,N} = V_{A,\varrho}(h_{t,N}) \geq m, \quad L_{t,N} \leq L.$$

Thus $\alpha_{t,N}$ and $\beta_{t,N}$ are bounded uniformly in N for $t < \tau_r$.

Step 2: The region $a_N < h_{t,N} \leq r$. The contraction bound also gives $\tau_a \leq C \log(1/a_N) = C \log \log N$. For $\tau_r \leq t < \tau_a$ and $u \in I_{t,N}$,

$$(1 - 2\delta)h_{t,N} \leq u \leq (1 + 2\delta)h_{t,N},$$

because $a_N < h_{t,N}$. In particular, $I_{t,N} \subseteq [0, 2r]$. By (4.11) and the boundedness of $u^{-2} \exp(-c/u)$ on $(0, 2r]$, we have $L_{t,N} \leq C$. Moreover,

$$\beta_{t,N} = \frac{h_{t,N} + a_N}{h_{t+1,N} + a_N} \leq \frac{2r}{a_N} \leq C \log N.$$

Hence $\alpha_{t,N} \leq C \log N$ for $\tau_r \leq t < \tau_a$.

Step 3: The region $h_{t,N} \leq a_N$. For $t \geq \tau_a$, the contraction gives $h_{t,N} \leq a_N$, and $I_{t,N} \subseteq [0, 2a_N]$. Since $u \mapsto u^{-2} \exp(-c/u)$ is increasing in a sufficiently small right-neighborhood of the origin, the derivative bound in (4.11) gives

$$L_{t,N} \leq C a_N^{-2} \exp\left(-\frac{c}{2a_N}\right) = C(\log N)^2 N^{-c/2} \leq N^{-c/3}$$

for all sufficiently large N , after decreasing $c > 0$ if necessary. Moreover, $h_{t+1,N} + a_N \geq a_N$, and therefore

$$\beta_{t,N} = \frac{h_{t,N} + a_N}{h_{t+1,N} + a_N} \leq 2.$$

Consequently, after decreasing c once more, $\alpha_{t,N} = L_{t,N} \beta_{t,N} \leq N^{-c}$.

Thus $1 \vee \alpha_{t,N}$ is uniformly bounded for at most C times, bounded by $C \log N$ for at most $C \log \log N$ times, and equal to 1 for $t \geq \tau_a$. It follows that

$$\prod_{u=s}^{t-1} (1 \vee \alpha_{u,N}) \leq \exp\{C(\log \log N)^2\}.$$

The preceding three steps also give $\beta_{s,N} \leq C \log N$ for every s . This factor is absorbed into the same bound, which proves (4.12). $\square$

Conditional independence and (4.12) show that $\mathcal{H}^{(N)}$ tracks $(\bar{h}_{t,N})_{t \geq 0}$ in relative error. The estimate below is the discrete-time counterpart of the stopped L^2 fluid-limit estimate of Darling and Norris [DN08]. Compare also with the bounded-increment formulation of Warnke [War19].

Proposition 4.4. *Assume $A < A_{\text{lat}}$, fix $\varrho \in (0, 1)$, and let $x_N \in [-1, 1]^N$. Let $\bar{T}_N \in \mathbb{N}_0$ be a sequence such that*

$$\frac{\bar{T}_N}{N} \exp\{C(\log \log N)^2\} \rightarrow 0 \quad \text{for every fixed } C < \infty.$$

Then

$$\sup_{0 \leq t \leq \bar{T}_N} \frac{|\mathcal{H}_t^{(N)} - \bar{h}_{t,N}|}{\bar{h}_{t,N} + (\log N)^{-1}} \xrightarrow{\mathbb{P}} 0. \quad (4.13)$$

Proof. Put $a_N = (\log N)^{-1}$, and let $U_{A,\varrho}$ be the coordinate observable in (2.18). If $\bar{T}_N = 0$, then the assertion is immediate because $\mathcal{H}_0^{(N)} = \bar{h}_{0,N}$. We therefore assume $\bar{T}_N \geq 1$.

Step 1: Bounding $\bar{h}_{0,N}$ and the moments of $U_{A,\varrho}$. Since $x_N \in [-1, 1]^N$ and $|Q_1(u) - u| \leq 1/2$, we have $|Q_\varrho(x_N)_i| \leq 1 + \varrho/2$ for every coordinate. Hence,

$$\bar{h}_{0,N} \leq C_\varrho := \frac{(1 + \varrho/2)^2}{\varrho^2}. \quad (4.14)$$

Furthermore, $U_{A,\varrho}(h, G)$ is uniformly bounded in h , because

$$\varrho^{-1} \tanh(\varrho A \sqrt{h} G) \in [-\varrho^{-1}, \varrho^{-1}].$$

For $0 < h \leq 1$, the definition of $b_{\varrho,0}$ in (4.8) gives

$$\{U_{A,\varrho}(h, G) \neq 0\} \subseteq \{A\sqrt{h}|G| > b_{\varrho,0}\}.$$

The probability of this event is at most $C \exp(-c/h)$. Therefore, for every integer $p \geq 1$,

$$\mathbb{E}U_{A,\varrho}(h, G)^p \leq C_p h^p, \quad h \geq 0. \quad (4.15)$$

Indeed, for $0 < h \leq 1$ we use $e^{-c/h} \leq C_p h^p$, for $h = 0$ the left-hand side vanishes, and for $h \geq 1$ the bound follows from the uniform boundedness of $U_{A,\varrho}$ and $h^p \geq 1$.

Step 2: The conditional variance of $\hat{\zeta}_{t+1}^{(N)}$ on $\{t < \tau_\delta\}$. Fix $\delta \in (0, 1/4)$ and set

$$e_t := \mathcal{H}_t^{(N)} - \bar{h}_{t,N}, \quad \mathcal{R}_{t,N} := \frac{e_t}{\bar{h}_{t,N} + a_N}, \quad \tau_\delta := \inf\{t \geq 0 : |\mathcal{R}_{t,N}| > \delta\}.$$

Let $\mathcal{F}_t$ be the natural filtration. By (2.22) and (2.23), $\mathbb{E}[\hat{\zeta}_{t+1}^{(N)} | \mathcal{F}_t] = 0$. Since $\mathcal{H}_{t+1}^{(N)}$ is an average of N conditionally independent copies of $U_{A,\varrho}(\mathcal{H}_t^{(N)}, G)$, the conditional variance is bounded by

$$\mathbb{E}[(\hat{\zeta}_{t+1}^{(N)})^2 | \mathcal{F}_t] \leq \frac{1}{N} \mathbb{E}U_{A,\varrho}(\mathcal{H}_t^{(N)}, G)^2.$$

On $\{t < \tau_\delta\}$, we have

$$\mathcal{H}_t^{(N)} \leq (1 + \delta)\bar{h}_{t,N} + \delta a_N \leq C(\bar{h}_{t,N} + a_N).$$

Thus (4.15) with $p = 2$ gives, on this event,

$$\mathbb{E}[(\hat{\zeta}_{t+1}^{(N)})^2 | \mathcal{F}_t] \leq \frac{C(\bar{h}_{t,N} + a_N)^2}{N}. \quad (4.16)$$

Step 3: The error recursion. Define the $\mathcal{F}_t$ -measurable difference quotient

$$D_{t,N} := \begin{cases} \frac{V_{A,\varrho}(\mathcal{H}_t^{(N)}) - V_{A,\varrho}(\bar{h}_{t,N})}{\mathcal{H}_t^{(N)} - \bar{h}_{t,N}}, & \mathcal{H}_t^{(N)} \neq \bar{h}_{t,N}, \\ V'_{A,\varrho}(\bar{h}_{t,N}), & \mathcal{H}_t^{(N)} = \bar{h}_{t,N}. \end{cases}$$

On $\{t < \tau_\delta\}$, the interval between $\mathcal{H}_t^{(N)}$ and $\bar{h}_{t,N}$ is contained in the interval defining $L_{t,N}$. The mean-value theorem and the definition of $D_{t,N}$ therefore give $|D_{t,N}| \leq L_{t,N}$.

The definitions of e_t and $\hat{\zeta}_{t+1}^{(N)}$, together with $\bar{h}_{t+1,N} = V_{A,\varrho}(\bar{h}_{t,N})$, give the error recursion

$$e_{t+1} = V_{A,\varrho}(\mathcal{H}_t^{(N)}) - V_{A,\varrho}(\bar{h}_{t,N}) + \hat{\zeta}_{t+1}^{(N)} = D_{t,N}e_t + \hat{\zeta}_{t+1}^{(N)}.$$

Use $\alpha_{t,N}$ and $\beta_{t,N}$ from Lemma 4.3 with $h_{t,N} = \bar{h}_{t,N}$, and set

$$\zeta_{t+1} := \frac{\hat{\zeta}_{t+1}^{(N)}}{\bar{h}_{t,N} + a_N}, \quad A_{t,N} := \mathbb{1}_{\{t < \tau_\delta\}} \frac{D_{t,N}(\bar{h}_{t,N} + a_N)}{\bar{h}_{t+1,N} + a_N}.$$

Dividing the error recursion by $\bar{h}_{t+1,N} + a_N$ shows that, on $\{t < \tau_\delta\}$,

$$\mathcal{R}_{t+1,N} = A_{t,N} \mathcal{R}_{t,N} + \beta_{t,N} \zeta_{t+1}. \quad (4.17)$$

Moreover, $|A_{t,N}| \leq \alpha_{t,N}$: on $\{t < \tau_\delta\}$ this follows from $|D_{t,N}| \leq L_{t,N}$, while $A_{t,N} = 0$ on the complementary event.

The hypotheses of Lemma 4.3 hold with $C_0 = C_\varrho$, because (4.14) gives $\bar{h}_{0,N} \leq C_\varrho$ and $\bar{h}_{t+1,N} = V_{A,\varrho}(\bar{h}_{t,N})$. Let

$$K_N := 1 \vee \max_{0 \leq s < t \leq \bar{T}_N} \left\{ \prod_{u=s}^{t-1} (1 \vee \alpha_{u,N}), \beta_{s,N} \prod_{u=s+1}^{t-1} (1 \vee \alpha_{u,N}) \right\}.$$

By Lemma 4.3, $K_N \leq \exp\{C(\log \log N)^2\}$.

Step 4: $\mathbb{P}(\tau_\delta \leq \bar{T}_N) \rightarrow 0$. Set

$$\eta_{t+1,N} := \mathbb{1}_{\{t < \tau_\delta\}} \beta_{t,N} \zeta_{t+1}, \quad \sigma_{t,N} := \beta_{t,N}.$$

On $\{t < \tau_\delta\}$, (4.17) becomes $\mathcal{R}_{t+1,N} = A_{t,N} \mathcal{R}_{t,N} + \eta_{t+1,N}$. The variables $A_{t,N}$ and $\eta_{t+1,N}$ are respectively $\mathcal{F}_t$ - and $\mathcal{F}_{t+1}$ -measurable. Since $\hat{\zeta}_{t+1}^{(N)}$ is conditionally centered and its denominator in ζ_{t+1} is deterministic,

$$\mathbb{E}[\eta_{t+1,N} \mid \mathcal{F}_t] = 0.$$

(4.16) gives

$$\mathbb{E}[\eta_{t+1,N}^2 \mid \mathcal{F}_t] = \mathbb{1}_{\{t < \tau_\delta\}} \beta_{t,N}^2 \frac{\mathbb{E}[(\hat{\zeta}_{t+1}^{(N)})^2 \mid \mathcal{F}_t]}{(\bar{h}_{t,N} + a_N)^2} \leq \frac{C \beta_{t,N}^2}{N}.$$

Thus the hypotheses of Lemma 2.2 hold with $T_N = \bar{T}_N$, $\tau_N = \tau_\delta$, and the amplification constant K_N defined above. Since $\mathcal{H}_0^{(N)} = \bar{h}_{0,N}$, we have $\mathcal{R}_{0,N} = 0$. By (2.38),

$$\begin{aligned} \mathbb{P}\left(\sup_{0 \leq t \leq \bar{T}_N} |\mathcal{R}_{t,N}| > \delta\right) &= \mathbb{P}(\tau_\delta \leq \bar{T}_N) \\ &\leq \frac{CK_N^2 \bar{T}_N}{\delta^2 N} \leq \frac{C \bar{T}_N}{\delta^2 N} \exp\{C'(\log \log N)^2\} \rightarrow 0. \end{aligned}$$

This holds for every fixed $\delta \in (0, 1/4)$, and hence $\sup_{0 \leq t \leq \bar{T}_N} |\mathcal{R}_{t,N}| \rightarrow 0$ in probability. By the definition of $\mathcal{R}_{t,N}$, this is (4.13). $\square$

4.3. Terminal layer and cutoff.

Theorem 4.5 (Fixed-precision subcritical dimension cutoff). *Assume $A < A_{\text{lat}}$, fix $\varrho \in (0, 1)$, and let $x_N \in [-1, 1]^N$ be deterministic. Suppose that $\bar{t}_N \rightarrow \infty$. Then*

$$\mathbb{P}\left(\tau_\varrho^{(N)} > \bar{t}_N - 1\right) \rightarrow 1, \quad \mathbb{P}\left(\tau_\varrho^{(N)} > \bar{t}_N + 2\right) \rightarrow 0. \quad (4.18)$$

Equivalently, by the total-variation/absorption identity (2.5),

$$\|P_{\varrho, A, N}^{\bar{t}_N - 1}(Q_\varrho x_N, \cdot) - \delta_0\|_{\text{TV}} \rightarrow 1, \quad \|P_{\varrho, A, N}^{\bar{t}_N + 2}(Q_\varrho x_N, \cdot) - \delta_0\|_{\text{TV}} \rightarrow 0. \quad (4.19)$$

Thus $Y^{(N)}$ has total-variation cutoff to δ_0 at $\bar{t}_N$ with a window of order one. In particular, for every fixed $\varepsilon \in (0, 1)$,

$$t_{\text{mix}}^{(N)}(\varepsilon) = \bar{t}_N + O(1).$$

Proof. Step 1: Put $a_N := (\log N)^{-1}$ and $\bar{T}_N := \bar{t}_N + 1$. By Lemma 4.1 and (4.5),

$$\bar{h}_{t,N} \leq C_\varrho \theta_A^t, \quad t \geq 0,$$

for some $\theta_A \in (0, 1)$. The definition of $\bar{t}_N$ therefore gives $\bar{t}_N = O(\log \log N)$, and the same bound holds for $\bar{T}_N$. Since $(\log \log N)^2 = o(\log N)$, there is $C_0 < \infty$ such that, for every fixed $C < \infty$ and all sufficiently large N ,

$$\frac{\bar{T}_N}{N} \exp\{C(\log \log N)^2\} \leq \frac{C_0 \log \log N}{N} \exp\{C(\log \log N)^2\} \rightarrow 0.$$

The assumptions $A < A_{\text{lat}}$, fixed $\varrho \in (0, 1)$, and $x_N \in [-1, 1]^N$ are precisely the remaining hypotheses of Proposition 4.4. Hence (4.13) holds on $[0, \bar{T}_N]$.

Step 2: The lower bound. Since $\bar{t}_N \rightarrow \infty$, we have $\bar{t}_N \geq 1$ for all sufficiently large N . By the definition of $\bar{t}_N$,

$$\bar{h}_{\bar{t}_N-1,N} > a_N.$$

On the event $\{\mathcal{H}_{\bar{t}_N-1}^{(N)} = 0\}$, it follows that

$$\frac{|\mathcal{H}_{\bar{t}_N-1}^{(N)} - \bar{h}_{\bar{t}_N-1,N}|}{\bar{h}_{\bar{t}_N-1,N} + a_N} = \frac{\bar{h}_{\bar{t}_N-1,N}}{\bar{h}_{\bar{t}_N-1,N} + a_N} > \frac{1}{2}.$$

Proposition 4.4 therefore gives $\mathbb{P}(\mathcal{H}_{\bar{t}_N-1}^{(N)} = 0) \rightarrow 0$. By (2.20) and the fact that 0 is absorbing, this is equivalent to $\mathbb{P}(\tau_\varrho^{(N)} > \bar{t}_N - 1) \rightarrow 1$.

Step 3: Define

$$b_\varrho := \varrho^{-1} \operatorname{arctanh}(\varrho/2). \quad (4.20)$$

By the definition of Q_1 , conditionally on $\mathcal{H}_t^{(N)} = h$, $Y_{t+1}^{(N)} = 0$ precisely when

$$\left| \varrho^{-1} \tanh(\varrho A \sqrt{h} G_{t+1,i}) \right| \leq \frac{1}{2}, \quad 1 \leq i \leq N.$$

Since $\tanh$ is increasing, this event is equivalently

$$\max_{1 \leq i \leq N} |A \sqrt{h} G_{t+1,i}| \leq b_\varrho. \quad (4.21)$$

Step 4: Set

$$s_N := \bar{t}_N + 1 = \bar{T}_N.$$

By the definition of $\bar{t}_N$, we have $\bar{h}_{\bar{t}_N,N} \leq a_N$. The small-radius estimate in Lemma 4.2 therefore yields, for some constants $c, C > 0$ and all sufficiently large N ,

$$\bar{h}_{s_N,N} = V_{A,\varrho}(\bar{h}_{\bar{t}_N,N}) \leq C \exp\left(-\frac{c}{a_N}\right) = CN^{-c}.$$

Here the inequality is immediate if $\bar{h}_{\bar{t}_N,N} = 0$. Consequently, $\bar{h}_{s_N,N} \log N \rightarrow 0$ and

$$(\bar{h}_{s_N,N} + a_N) \log N \rightarrow 1.$$

Since $s_N = \bar{T}_N$, the uniform concentration estimate (4.13) applies at time s_N . Writing the exact radius as its deterministic value plus the normalized tracking error, we obtain

$$\begin{aligned} \mathcal{H}_{s_N}^{(N)} \log N &= \bar{h}_{s_N, N} \log N \\ &+ \frac{\mathcal{H}_{s_N}^{(N)} - \bar{h}_{s_N, N}}{\bar{h}_{s_N, N} + a_N} (\bar{h}_{s_N, N} + a_N) \log N \xrightarrow{\mathbb{P}} 0. \end{aligned}$$

Step 5: Final absorption. Fix

$$0 < \eta < \frac{b_\varrho^2}{2A^2}$$

and define

$$M_{s_N+1, N} := \max_{1 \leq i \leq N} |G_{s_N+1, i}|,$$

where G_{s_N+1} is the standard Gaussian vector used in the update from s_N to $s_N + 1$. By (4.21), the event $\{\tau_\varrho^{(N)} > s_N + 1\}$ requires $A\sqrt{\mathcal{H}_{s_N}^{(N)}} M_{s_N+1, N} > b_\varrho$. Consequently,

$$\{\tau_\varrho^{(N)} > s_N + 1\} \cap \left\{ \mathcal{H}_{s_N}^{(N)} \leq \frac{\eta}{\log N} \right\} \subseteq \left\{ M_{s_N+1, N} > \frac{b_\varrho}{A} \sqrt{\frac{\log N}{\eta}} \right\}.$$

Combining the convergence $\mathcal{H}_{s_N}^{(N)} \log N \rightarrow 0$ established in Step 4 with Gaussian tails and a union bound over the N coordinates gives

$$\begin{aligned} \mathbb{P}(\tau_\varrho^{(N)} > \bar{t}_N + 2) &\leq \mathbb{P}\left(\mathcal{H}_{s_N}^{(N)} \log N > \eta\right) + \mathbb{P}\left(M_{s_N+1, N} > \frac{b_\varrho}{A} \sqrt{\frac{\log N}{\eta}}\right) \\ &\leq \mathbb{P}\left(\mathcal{H}_{s_N}^{(N)} \log N > \eta\right) + 2N^{1-b_\varrho^2/(2A^2\eta)} \rightarrow 0. \end{aligned}$$

Hence (4.18) holds. The total variation formulation (4.19) follows from the absorption identity (2.5). The mixing-time estimate follows from the monotonicity of total variation distance. $\square$

Remark 4.6. Put

$$b_\varrho := \varrho^{-1} \operatorname{arctanh}(\varrho/2), \quad \kappa_{A, \varrho} := \frac{b_\varrho^2}{2A^2}.$$

At radius $h = (\log N)^{-1}$, the expected number of coordinates not rounded to zero is

$$2N\mathbb{P}\left(G > \frac{b_\varrho \sqrt{\log N}}{A}\right) \asymp \frac{N^{1-\kappa_{A, \varrho}}}{\sqrt{\log N}}.$$

Thus, when $\kappa_{A, \varrho} < 1$, entrance below $(\log N)^{-1}$ does not uniformly imply absorption in the next step. One further deterministic iterate is $o((\log N)^{-1})$, and the following update absorbs with probability tending to one. The upper time improves from $\bar{t}_N + 2$ to $\bar{t}_N + 1$ if

$$\limsup_{N \rightarrow \infty} (\log N) \bar{h}_{\bar{t}_N, N} < \kappa_{A, \varrho}.$$

5. Metastability at fixed precision

Fix $A > 0$ and $\varrho \in (0, 1)$. We identify the positive-drift components of $V_{A, \varrho}$. When the rightmost component (h_-, h_+) exists, $\mathcal{H}^{(N)}$ enters a neighborhood of h_+ in bounded time and remains there for $\exp(cN)$ steps. For each fixed N , the finite-state chain $Y^{(N)}$ nevertheless absorbs almost surely.

5.1. Positive-drift components. Let $\mathcal{H}_t^{(N)}$ be the chain from (2.17). Its mean map is $V_{A,\varrho}$ from (2.21). Throughout this section we write

$$V := V_{A,\varrho}.$$

Let

$$M_\varrho := \max_{|u| \leq \varrho^{-1}} Q_1(u)^2 < \infty. \quad (5.1)$$

For $\alpha > 0$, define

$$m_\varrho(\alpha) := \mathbb{E} \left[Q_1(\varrho^{-1} \tanh(\varrho \alpha G))^2 \right]. \quad (5.2)$$

The rounding monotonicity in (4.3) implies that V is nondecreasing on $[0, \infty)$. In the layer notation of (4.9),

$$m_\varrho(\alpha) = 2 \sum_{k \in \mathcal{K}_\varrho} (2k+1) \bar{\Phi} \left(\frac{b_{\varrho,k}}{\alpha} \right). \quad (5.3)$$

Recall the threshold introduced in (1.12):

$$A_{\text{ex}}(\varrho)^2 := \inf_{\alpha > 0} \frac{\alpha^2}{m_\varrho(\alpha)}. \quad (5.4)$$

Define the positive-drift set

$$\mathcal{O}_{A,\varrho} := \{h \in (0, M_\varrho) : V_{A,\varrho}(h) > h\}. \quad (5.5)$$

We call the connected components of $\mathcal{O}_{A,\varrho}$ positive-drift components. The rightmost component produces metastability near its attracting right endpoint. The corresponding vector states form a grid shell.

We first identify the fixed points that bound the positive-drift components.

Proposition 5.1. *The threshold $A_{\text{ex}}(\varrho)$ belongs to $(0, \infty)$. If $A > A_{\text{ex}}(\varrho)$, then $V_{A,\varrho}$ has at least two positive fixed points in $(0, M_\varrho)$. There exist $0 < \alpha_- < \alpha_+$ such that*

$$A^2 m_\varrho(\alpha_\pm) = \alpha_\pm^2, \quad (5.6)$$

and $\hat{h}_\pm := \alpha_\pm^2 / A^2$ are fixed points of $V_{A,\varrho}$.

Moreover, $\mathcal{O}_{A,\varrho}$ is nonempty and has finitely many connected components. If (h_-, h_+) is its rightmost positive-drift component, then

$$0 < h_- < h_+ < M_\varrho, \quad V(h_-) = h_-, \quad V(h_+) = h_+, \quad (5.7)$$

and $V(h) > h$ on (h_-, h_+) . The fixed point h_+ is isolated, and there exists $\eta_0 > 0$ such that

$$h_+ - \eta_0 > h_-, \quad h_+ + \eta_0 < M_\varrho, \quad V(h) < h \quad \text{for } h \in (h_+, h_+ + \eta_0]. \quad (5.8)$$

Proof. Step 1: $A_{\text{ex}}(\varrho) \in (0, \infty)$. By the definitions in (4.8), the set $\mathcal{K}_\varrho$ is finite and nonempty, and each threshold $b_{\varrho,k}$ is positive. Hence (5.3) shows that m_ϱ is continuous and positive on $(0, \infty)$.

To control the quotient in (5.4), put $b_{\varrho,0} = \varrho^{-1} \operatorname{arctanh}(\varrho/2)$. The random variable in (5.2) can be nonzero only when $|G| > b_{\varrho,0}/\alpha$, and its square is bounded by M_ϱ . Hence

$$m_\varrho(\alpha) \leq 2M_\varrho \bar{\Phi} \left(\frac{b_{\varrho,0}}{\alpha} \right), \quad \alpha > 0,$$

and the Gaussian tail bound implies $m_\varrho(\alpha)/\alpha^2 \rightarrow 0$ as $\alpha \downarrow 0$. The bound $m_\varrho(\alpha) \leq M_\varrho$ also gives $m_\varrho(\alpha)/\alpha^2 \rightarrow 0$ as $\alpha \rightarrow \infty$. Therefore

$$\frac{\alpha^2}{m_\varrho(\alpha)} \longrightarrow \infty \quad \text{as } \alpha \downarrow 0 \quad \text{and as } \alpha \rightarrow \infty. \quad (5.9)$$

Since the quotient is continuous and strictly positive on $(0, \infty)$, (5.9) implies that the infimum in (5.4) is attained on a compact subinterval of $(0, \infty)$ and is positive and finite. This proves $A_{\text{ex}}(\varrho) \in (0, \infty)$.

Step 2: $V_{A,\varrho}$ has two fixed points in $(0, M_\varrho)$. Assume now that $A > A_{\text{ex}}(\varrho)$. Then there is $\alpha_0 > 0$ such that $A^2 m_\varrho(\alpha_0) - \alpha_0^2 > 0$. By (5.9), $A^2 m_\varrho(\alpha) - \alpha^2 < 0$ for all sufficiently small and all sufficiently large α . The intermediate value theorem therefore gives two roots $0 < \alpha_- < \alpha_+$ of $A^2 m_\varrho(\alpha) = \alpha^2$.

Since

$$V_{A,\varrho}(h) = m_\varrho(A\sqrt{h}),$$

the points $\hat{h}_\pm = \alpha_\pm^2/A^2$ satisfy

$$V_{A,\varrho}(\hat{h}_\pm) = m_\varrho(\alpha_\pm) = \frac{\alpha_\pm^2}{A^2} = \hat{h}_\pm.$$

They are positive because $\alpha_\pm > 0$. Moreover,

$$\hat{h}_\pm = V_{A,\varrho}(\hat{h}_\pm) = \mathbb{E}\left[Q_1(\varrho^{-1} \tanh(\varrho \alpha_\pm G))^2\right],$$

where the integrand is bounded by M_ϱ and vanishes when $|G| \leq b_{\varrho,0}/\alpha_\pm$. Thus $0 < \hat{h}_\pm < M_\varrho$.

Step 3: $\mathcal{O}_{A,\varrho}$ is nonempty and has finitely many components. Set $D(h) := V(h) - h$. The point $h_0 = \alpha_0^2/A^2$ satisfies $D(h_0) > 0$. Since $V(h_0) \leq M_\varrho$, we also have $h_0 < M_\varrho$, and hence $\mathcal{O}_{A,\varrho}$ is nonempty.

Substituting $\alpha = A\sqrt{h}$ into (5.9) gives

$$\frac{V(h)}{h} = A^2 \frac{m_\varrho(A\sqrt{h})}{(A\sqrt{h})^2} \longrightarrow 0 \quad \text{as } h \downarrow 0.$$

Thus $D(h) < 0$ for all sufficiently small $h > 0$. Moreover, $V(M_\varrho) < M_\varrho$, because the integrand in its definition is bounded by M_ϱ and vanishes with positive probability. By continuity, there is $\delta > 0$ such that

$$D(h) < 0 \quad \text{for } h \in (0, \delta] \cup [M_\varrho - \delta, M_\varrho].$$

By (5.3), V and hence D are real analytic on $(0, \infty)$. The function D is not identically zero because it is negative near the origin. Its zeros are therefore isolated, so it has only finitely many zeros in $[\delta, M_\varrho - \delta]$. Every component of $\mathcal{O}_{A,\varrho}$ is contained in this interval and has endpoints in this zero set. Consequently, $\mathcal{O}_{A,\varrho}$ has finitely many components.

Step 4: $V(h) < h$ immediately to the right of h_+ . Let (h_-, h_+) be the rightmost component. Since D is continuous, $D(h_-) = D(h_+) = 0$, while $D(h) > 0$ for $h \in (h_-, h_+)$. This proves (5.7).

The zero h_+ is isolated, so D has a constant nonzero sign immediately to its right. This sign cannot be positive, since that would produce a component of $\mathcal{O}_{A,\varrho}$ to the right of (h_-, h_+) . We may therefore choose $\eta_0 > 0$ such that $h_+ - \eta_0 > h_-$, $h_+ + \eta_0 < M_\varrho$, and $V(h) < h$ for every $h \in (h_+, h_+ + \eta_0]$. This proves (5.8). $\square$

5.2. Entrance and exponential persistence. Conditionally on $\mathcal{H}_t^{(N)} = h$, the random variables in (2.19) are independent and take values in $[0, M_\varrho]$. Hence Hoeffding's inequality gives, for every $u > 0$,

$$\mathbb{P}\left(\left|\mathcal{H}_{t+1}^{(N)} - V(\mathcal{H}_t^{(N)})\right| > u \mid \mathcal{H}_t^{(N)}\right) \leq 2 \exp\left(-\frac{2Nu^2}{M_\varrho^2}\right). \quad (5.10)$$

Theorem 5.2 (Metastability from the rightmost positive-drift component). *Assume that $A > A_{\text{ex}}(\varrho)$, and let (h_-, h_+) be the rightmost positive-drift component of $\mathcal{O}_{A,\varrho}$ in (5.5). Let $\eta_0 > 0$ be as in (5.8), and let $B \subset (h_-, h_+]$ be compact. For every $0 < \eta < \eta_0$, there are $T_\eta \in \mathbb{N}_0$ and*

constants $c_0, c_1, C \in (0, \infty)$, depending only on A, ϱ, B, η , with $c_1 < c_0$, such that the following holds. If $Y_0^{(N)}$ is deterministic and $\mathcal{H}_0^{(N)} \in B$, then

$$\mathbb{P}\left(\left|\mathcal{H}_{T_\eta}^{(N)} - h_+\right| > \frac{\eta}{2}\right) \leq C \exp(-c_0 N), \quad (5.11)$$

and, for every $T \in \mathbb{N}_0$,

$$\mathbb{P}\left(\max_{0 \leq s \leq T} \left|\mathcal{H}_{T_\eta+s}^{(N)} - h_+\right| > \eta\right) \leq C(1+T) \exp(-c_0 N). \quad (5.12)$$

In particular,

$$\mathbb{P}\left(\tau_\varrho^{(N)} > T_\eta + \lfloor \exp(c_1 N) \rfloor\right) \rightarrow 1. \quad (5.13)$$

For each fixed N and each deterministic choice of $Y_0^{(N)}$ in the finite rounded state space, the absorption time $\tau_\varrho^{(N)}$ is nevertheless finite almost surely.

Proof. Step 1: $V^t(h) \rightarrow h_+$ uniformly for $h \in B$. By (4.3), the map V is nondecreasing. Since (h_-, h_+) is a component of $\mathcal{O}_{A, \varrho}$,

$$h < V(h) \leq h_+ \quad \text{for } h \in (h_-, h_+),$$

and $V(h_+) = h_+$. Thus $V^t(h)$ is nondecreasing in t and bounded above by h_+ for every $h \in (h_-, h_+]$. Its limit is a fixed point of V , and there is no fixed point in (h_-, h_+) . Hence $V^t(h) \rightarrow h_+$.

Since B is compact in $(h_-, h_+]$, choose $b > h_-$ such that $B \subset [b, h_+]$. On the compact interval $[b, h_+ - \eta/4]$, when it is nonempty, the continuous function $V(h) - h$ has a positive minimum. Thus every orbit from B enters the invariant interval $[h_+ - \eta/4, h_+]$ after a uniformly bounded number of steps. Hence there exists $T_\eta \in \mathbb{N}$ such that

$$V^{T_\eta}(B) \subset [h_+ - \eta/4, h_+ + \eta/4]. \quad (5.14)$$

Step 2: $\mathcal{H}_{T_\eta}^{(N)}$ lies within $\eta/2$ of h_+ with exponentially high probability. Define the modulus of continuity

$$\omega(r) := \sup\{|V(u) - V(v)| : u, v \in [0, M_\varrho], |u - v| \leq r\}.$$

Then $\omega(r) \rightarrow 0$ as $r \downarrow 0$. Set $q_s := V^s(\mathcal{H}_0^{(N)})$ and $e_s := |\mathcal{H}_s^{(N)} - q_s|$. Since $e_0 = 0$,

$$e_{s+1} \leq |\mathcal{H}_{s+1}^{(N)} - V(\mathcal{H}_s^{(N)})| + \omega(e_s).$$

For $\delta > 0$, define $E_0(\delta) := 0$ and $E_{s+1}(\delta) := \delta + \omega(E_s(\delta))$. For each fixed s , $E_s(\delta) \rightarrow 0$ as $\delta \downarrow 0$. Since T_η is fixed, we may choose $\delta_0 > 0$ such that $\max_{0 \leq s \leq T_\eta} E_s(\delta_0) \leq \eta/4$.

Suppose that $\mathcal{H}_0^{(N)} \in B$ and

$$\left|\mathcal{H}_{s+1}^{(N)} - V(\mathcal{H}_s^{(N)})\right| \leq \delta_0 \quad \text{for } 0 \leq s < T_\eta.$$

Then induction gives $e_s \leq E_s(\delta_0) \leq \eta/4$ for $s \leq T_\eta$. Hence (5.14) yields

$$\left|\mathcal{H}_{T_\eta}^{(N)} - h_+\right| \leq e_{T_\eta} + |q_{T_\eta} - h_+| \leq \frac{\eta}{4} + \frac{\eta}{4} = \frac{\eta}{2}.$$

A union bound and (5.10) give

$$\mathbb{P}\left(\left|\mathcal{H}_{T_\eta}^{(N)} - h_+\right| > \frac{\eta}{2}\right) \leq 2T_\eta \exp\left(-\frac{2N\delta_0^2}{M_\varrho^2}\right).$$

Thus (5.11) holds with constants $C_{\text{ent}}, c_{\text{ent}} > 0$ depending only on A, ϱ, B, η .

Step 3: The exit bound and survival up to $\exp(c_1 N)$. Put

$$J_\eta := [h_+ - \eta, h_+ + \eta].$$

For $h \in [h_+ - \eta, h_+)$, we have $h < V(h) \leq h_+$. By (5.8) and the monotonicity of V ,

$$h_+ \leq V(h) < h \quad \text{for } h \in (h_+, h_+ + \eta].$$

Consequently, $V(J_\eta) \subset J_\eta^\circ$. Since J_η is compact, there exists $d_\eta > 0$ such that

$$\text{dist}(V(J_\eta), J_\eta^c) \geq d_\eta.$$

Therefore, on the event $\mathcal{H}_t^{(N)} \in J_\eta$, the additional event

$$\left| \mathcal{H}_{t+1}^{(N)} - V(\mathcal{H}_t^{(N)}) \right| \leq d_\eta/2$$

implies $\mathcal{H}_{t+1}^{(N)} \in J_\eta$. Applying (5.10) with $u = d_\eta/2$, and taking a union bound over $T_\eta, \dots, T_\eta + T - 1$, gives

$$\begin{aligned} & \mathbb{P} \left(\max_{0 \leq s \leq T} \left| \mathcal{H}_{T_\eta+s}^{(N)} - h_+ \right| > \eta \right) \\ & \leq C_{\text{ent}} e^{-c_{\text{ent}} N} + 2T \exp \left(-\frac{N d_\eta^2}{2M_\varrho^2} \right) \leq C(1+T)e^{-c_0 N}, \end{aligned}$$

where $c_0 := \min\{c_{\text{ent}}, d_\eta^2/(2M_\varrho^2)\} > 0$. After enlarging C , this proves (5.11) and (5.12) with the same constants C, c_0 .

Since $J_\eta \subset (0, M_\varrho)$, the complementary event in (5.12) prevents absorption. Choose $c_1 \in (0, c_0)$. Taking $T = \lfloor \exp(c_1 N) \rfloor$, the right-hand side of (5.12) is bounded by

$$2C \exp(-(c_0 - c_1)N) \rightarrow 0.$$

This is (5.13).

Step 4: $\tau_\varrho^{(N)} < \infty$ almost surely for fixed N . Let

$$\mathcal{S}_N := (\varrho\mathbb{Z})^N \cap [-1 - \varrho/2, 1 + \varrho/2]^N.$$

The chain $Y^{(N)}$ is supported on the finite set $\mathcal{S}_N$. If $y \in \mathcal{S}_N \setminus \{0\}$, then, conditionally on $Y_t^{(N)} = y$, the coordinates of $W_{t+1,A}^{(N)} y$ are independent nondegenerate Gaussian variables. Thus each coordinate has positive probability of satisfying $Q_\varrho(\tanh((W_{t+1,A}^{(N)} y)_i)) = 0$. Independence therefore gives

$$\mathbb{P}(Y_{t+1}^{(N)} = 0 \mid Y_t^{(N)} = y) > 0.$$

Taking the minimum over $y \in \mathcal{S}_N \setminus \{0\}$ gives a constant $p_N > 0$. The Markov property now gives $\mathbb{P}(\tau_\varrho^{(N)} > m) \leq (1 - p_N)^m \rightarrow 0$, which proves the claim. $\square$

6. Supercritical cutoff as $N \rightarrow \infty$

Fix $A > 1$ and let $N \rightarrow \infty$. We use $Z^{(N)}$, $K_{A,N}$, and V_A from Section 2.1. We first prove uniqueness of the nonzero invariant laws and identify the cutoff scale from the decay of $V_A^t(q_0) - q_*$. Negative moments and concentration then localize $\nu_{A,N}$ near q_* . Before the cutoff scale, the deterministic bias separates the law of $Z^{(N)}$ from $\nu_{A,N}$. After that scale, synchronous coupling makes the separation from a stationary copy of order $N^{-1/2}$, and one further transition converts this bound into convergence in total variation. Generic constants depend only on the fixed parameters in the surrounding statement.

6.1. Transition density and uniqueness of invariant laws. For $q > 0$, set $Y_q := \tanh^2(A\sqrt{q}G)$. For $y \in (0, 1)$, the change of variables $u = \operatorname{arctanh}(\sqrt{y})$ gives the strictly positive density

$$f_q(y) = \frac{1}{A\sqrt{2\pi q}} \frac{1}{\sqrt{y}(1-y)} \exp\left(-\frac{\operatorname{arctanh}(\sqrt{y})^2}{2A^2q}\right). \quad (6.1)$$

The N -fold convolution of f_q , followed by scaling by N^{-1} , gives a strictly positive transition density for $K_{A,N}$ on $(0, 1)$. This yields uniqueness of $\nu_{A,N}$. The kernel $J_{A,N}$ then yields uniqueness of $\pi_{A,N}$.

Proposition 6.1. *Fix $A > 0$ and $N \geq 1$. If $K_{A,N}$ admits a nonzero invariant probability $\nu_{A,N}$ on $(0, 1]$, then this probability is unique and has a strictly positive density on $(0, 1)$.*

Proof. Step 1: The transition density is strictly positive. The density of $\sum_{i=1}^N Y_{i,q}$ is the convolution f_q^{*N} . It is strictly positive on $(0, N)$: for $s \in (0, N)$, choose $y_1, \dots, y_N \in (0, 1)$ with $\sum_i y_i = s$ and integrate over a small neighborhood of $(y_1, \dots, y_{N-1})$ on which all coordinates, including the remaining coordinate, stay in $(0, 1)$. After scaling by N^{-1} , the transition kernel $K_{A,N}(q, \cdot)$ therefore has a density $p_{A,N}(q, r)$ that is strictly positive for every $r \in (0, 1)$.

Step 2: The invariant law is unique and has a positive density. The positivity of $p_{A,N}$ makes $Z^{(N)}$ on $(0, 1]$ Lebesgue-irreducible. By Nummelin [Num84], it therefore admits at most one invariant probability. If $\nu K_{A,N} = \nu$, then invariance and Fubini's theorem give

$$\nu(dr) = \left(\int_{(0,1]} p_{A,N}(q, r) \nu(dq) \right) dr.$$

The density in parentheses is strictly positive for every $r \in (0, 1)$. □

Remark 6.2. On $[0, 1]$, the point 0 is absorbing. Hence, whenever the nonzero invariant law exists, the invariant probabilities on $[0, 1]$ are exactly

$$\alpha \delta_0 + (1 - \alpha) \nu_{A,N}, \quad 0 \leq \alpha \leq 1. \quad (6.2)$$

Indeed, any invariant law has a fixed mass α at $\{0\}$, and its normalized restriction to $(0, 1]$, if nonzero, is invariant for the restricted chain.

Corollary 6.3 (Uniqueness of the nonzero vector invariant law). *Fix $A > 0$ and $N \geq 1$ for which $K_{A,N}$ admits a nonzero invariant law $\nu_{A,N}$ on $(0, 1]$. Then $P_{A,N}$ has a unique invariant probability $\pi_{A,N}$ on $[-1, 1]^N$ with $\pi_{A,N}(\{0\}) = 0$, namely*

$$\pi_{A,N} = \nu_{A,N} J_{A,N}. \quad (6.3)$$

Proof. Existence. By Proposition 2.4, $\nu_{A,N} J_{A,N}$ is invariant for $P_{A,N}$. It has no mass at 0 because $J_{A,N}(q, \cdot)$ is absolutely continuous for every $q > 0$.

Uniqueness. Let π be invariant for $P_{A,N}$ with $\pi(\{0\}) = 0$. By Proposition 2.4, $\nu := (r_N)_\# \pi$ is invariant for $K_{A,N}$, and since $r_N(x) = 0$ if and only if $x = 0$, we have $\nu(\{0\}) = 0$. By Proposition 6.1, $\nu = \nu_{A,N}$. Invariance and (2.40) give

$$\pi = \pi P_{A,N} = ((r_N)_\# \pi) J_{A,N} = \nu_{A,N} J_{A,N}.$$

□

6.2. Deterministic bias and cutoff scale. For $A > 1$, Lemma 2.1 gives the unique nonzero fixed point q_* and the multiplier $\mu_A = V'_A(q_*) \in (0, 1)$.

Let $q_0 \in (0, 1]$ with $q_0 \neq q_*$, and let $C(q_0)$ be the nonzero constant in (2.32). By (2.32), the deterministic bias decays like $C(q_0) \mu_A^t$. The cutoff scale is therefore the time at which this bias reaches the $N^{-1/2}$ fluctuation scale, and the resulting center and integer times are given by (1.19) and (1.20). The proof of Theorem 1.8 is given in Section 6.5.

6.3. Localization of $\nu_{A,N}$ near q_* . We prove exponential localization of $\nu_{A,N}$ in a stable interval around q_* . Negative moments control the mass near zero, while deterministic entrance and concentration control the remaining radii. The estimates are proved in Appendix B.

Assumption 6.4 (Macroscopic initial radius). *We fix $A > 1$ and let q_* and μ_A be the unique nonzero fixed point and multiplier from Lemma 2.1. We also fix $q_0 \in (0, 1]$ with $q_0 \neq q_*$. Every deterministic initial-radius sequence used below is assumed to converge to this q_0 .*

Throughout the remainder of this subsection, Assumption 6.4 is in force.

For a deterministic initial-radius sequence $q_N \rightarrow q_0$, put

$$q_{0,N} := q_N, \quad q_{t+1,N} := V_A(q_{t,N}). \quad (6.4)$$

The deterministic orbit must first enter a region where the mean map is contractive. We record that entrance estimate here.

Lemma 6.5. *Let Assumption 6.4 hold, i.e. $q_N \rightarrow q_0$. There are an integer s_* , a compact interval $I_* \subset (0, 1)$, constants $\delta_* > 0$ and $\kappa < 1$, and $N_0 < \infty$ such that, for all $N \geq N_0$,*

$$q_{t,N} \in I_* \quad \text{for every } t \geq s_*, \quad \inf_{t \geq s_*} \text{dist}(q_{t,N}, \partial I_*) \geq \delta_*, \quad (6.5)$$

and

$$\sup_{q \in I_*} |V'_A(q)| \leq \kappa. \quad (6.6)$$

Proof. Since q_* is stable, choose a compact interval $I_* \subset (0, 1)$ containing q_* in its interior such that $\sup_{I_*} |V'_A| < 1$. Since q_0 lies in the basin of q_* , the orbit $V_A^t(q_0)$ enters the interior of I_* and remains in a smaller compact subinterval of I_* after a finite time. By continuity of V_A^t for each fixed t , the same is true uniformly for all q_N sufficiently close to q_0 , after increasing the entrance time if necessary. This proves (6.5), while the derivative bound follows from the choice of I_* . $\square$

The next estimates control invariant mass near zero. They use negative moments of order γN , first for $\ell_{A,N}$ and then uniformly for small radii.

Remark 6.6. For every positive random variable X , Markov's inequality gives $\mathbb{P}(X \leq r) \leq r^p \mathbb{E} X^{-p}$, so negative moments provide small-ball bounds. Near $q = 0$,

$$\frac{F_{A,N}(q, G)}{q} \approx A^2 \frac{\chi_N^2}{N}.$$

Here χ_N is the Euclidean norm of a standard Gaussian vector in $\mathbb{R}^N$, as in (2.25). To obtain an exponentially small bound on $\nu_{A,N}((0, r])$, we take $p = \gamma N$. If $Z_N \sim \chi_N^2$ and $\gamma \in (0, 1/2)$, direct integration gives

$$\mathbb{E} \left[\left(A^2 \frac{Z_N}{N} \right)^{-\gamma N} \right] = A^{-2\gamma N} \left(\frac{N}{2} \right)^{\gamma N} \frac{\Gamma(N/2 - \gamma N)}{\Gamma(N/2)}.$$

The moment is finite precisely when $\gamma < 1/2$, and Stirling's formula yields

$$\frac{1}{N} \log \mathbb{E} \left[\left(A^2 \frac{Z_N}{N} \right)^{-\gamma N} \right] \rightarrow \Lambda_A(\gamma) := -2\gamma \log A + \gamma + \left(\frac{1}{2} - \gamma \right) \log(1 - 2\gamma).$$

Since $\Lambda_A(0) = 0$ and $\Lambda'_A(0) = -2 \log A < 0$ for $A > 1$, this exponent is negative for all sufficiently small $\gamma > 0$. The truncated estimate below makes this linearized calculation uniform over a neighborhood of the origin.

Lemma 6.7. *Let $A > 1$. There are constants $\gamma \in (0, 1/2)$, $\kappa > 0$, $L \geq 1$, and $\varepsilon \in (0, 1)$ such that, for all sufficiently large N ,*

$$\mathbb{E} \left[\left((1 - \varepsilon) A^2 \frac{1}{N} \sum_{i=1}^N \min(G_i^2, L^2) \right)^{-\gamma N} \right] \leq e^{-\kappa N}. \quad (6.7)$$

For sufficiently small R_0 , the ratio $F_{A,N}(q, G)/q$ dominates the truncated quadratic average in Lemma 6.7, uniformly for $0 < q \leq R_0$. Since $x \mapsto x^{-\gamma N}$ is decreasing, the lemma then gives the required negative-moment contraction near zero.

Proposition 6.8. *Let Assumption 6.4 hold, and let I_* be the compact interval selected in Lemma 6.5. There are $R_0 \in (0, \inf I_*)$, $\gamma \in (0, 1/2)$, and $\kappa > 0$ such that, for all sufficiently large N ,*

$$\sup_{0 < q \leq R_0} \mathbb{E} \left[\left(\frac{F_{A,N}(q, G)}{q} \right)^{-\gamma N} \right] \leq e^{-\kappa N}. \quad (6.8)$$

Away from zero, there is $c_{A,R_0} > 0$ such that $\tanh^2(A\sqrt{q}g) \geq c_{A,R_0} \min(g^2, 1)$ for $q \in [R_0, 1]$. After averaging over the coordinates, this reduces the estimate to a negative-moment bound for a truncated Gaussian square average.

Lemma 6.9. *Let $R_0 \in (0, 1)$ and $\gamma \in (0, 1/2)$. There is $b < \infty$ such that, for every $N \geq 1$,*

$$\sup_{q \in [R_0, 1]} \mathbb{E}[F_{A,N}(q, G)^{-\gamma N}] \leq e^{bN}. \quad (6.9)$$

See Appendix B.1 for the proofs of Lemmas 6.7 and 6.9 and Proposition 6.8.

By Propositions 2.6 and 6.1, and the fact that $A_c(N) \downarrow 1$, $K_{A,N}$ has a unique nonzero invariant law for all sufficiently large N . We denote it by $\nu_{A,N}$.

Remark 6.10. For fixed $A > 1$ and all sufficiently large N ,

$$\mathbb{E} \left[\left(A^2 \frac{\chi_N^2}{N} \right)^{-1} \right] = A^{-2} \frac{N}{N-2} < 1,$$

so the choice $V(q) = q^{-1}$ in Proposition 2.6 gives a Foster–Lyapunov estimate. Consequently, every Cesàro limit of the laws of $Z^{(N)}$ considered there assigns no mass to $\{0\}$. To prove Proposition 6.12, we need $\nu_{A,N}((0, r]) \leq Ce^{-cN}$ for some fixed $r > 0$. We therefore set $V_N(q) = q^{-\gamma N}$. A bound of the form $\int V_N d\nu_{A,N} \leq Ce^{bN}$ gives

$$\nu_{A,N}((0, r]) \leq r^{\gamma N} \int V_N d\nu_{A,N} \leq C \exp\{N(b + \gamma \log r)\}.$$

Choosing $r < e^{-b/\gamma}$ makes the exponent negative.

Combining the near-zero contraction estimate with the bound on $[R_0, 1]$ yields the invariant-law estimate below.

Proposition 6.11. *Let Assumption 6.4 hold. Then, for all sufficiently large N , the unique nonzero invariant law $\nu_{A,N}$ of $K_{A,N}$ satisfies*

$$\int_0^1 q^{-\gamma N} \nu_{A,N}(dq) \leq \frac{e^{bN}}{1 - e^{-\kappa N}}, \quad (6.10)$$

where R_0, γ, κ are as in Proposition 6.8, and b is the constant in Lemma 6.9 for this choice of R_0, γ .

To prove the next proposition, choose $r < R_0$ so that the preceding negative-moment estimate gives $\nu_{A,N}((0, r]) \leq Ce^{-cN}$. Uniform deterministic entrance from $[r, 1]$ into I_* , together with a fixed-time concentration estimate, then gives, for some fixed m , $\sup_{q \in [r, 1]} K_{A,N}^m(q, I_*^c) \leq Ce^{-cN}$. Invariance combines these two bounds.

Proposition 6.12. *Let Assumption 6.4 hold. Let $\nu_{A,N}$ be the unique nonzero invariant law of $K_{A,N}$. Then there are constants $C, c > 0$ such that*

$$\nu_{A,N}(I_*^c) \leq Ce^{-cN} \quad (6.11)$$

for all sufficiently large N . In particular,

$$\nu_{A,N}(I_*^c) \leq \frac{C}{N} \quad (6.12)$$

after enlarging C .

The proofs of Propositions 6.11 and 6.12 are in Appendices B.1 and B.2.

6.4. Concentration and coupling at the cutoff scale. Once two radii lie in I_* , the following estimate converts their separation into total variation after one step. We then combine it with concentration and synchronous coupling.

Lemma 6.13. *Let $I \subseteq (0, 1)$. There is a constant $C_I < \infty$ such that, for every $q, q' \in I$,*

$$\|K_{A,N}(q, \cdot) - K_{A,N}(q', \cdot)\|_{\text{TV}} \leq C_I \sqrt{N} |q - q'|. \quad (6.13)$$

The stationary chain satisfies the analogous concentration estimate.

Proposition 6.14. *Let Assumption 6.4 hold, and let $\nu_{A,N}$ be the unique nonzero invariant law of $K_{A,N}$. Then*

$$\int_0^1 (q - q_*)^2 \nu_{A,N}(dq) \leq \frac{C}{N}. \quad (6.14)$$

The next lemma controls $Z^{(N)}$ around $(q_{t,N})_{t \geq 0}$.

Lemma 6.15. *Let Assumption 6.4 hold. Then, for every $0 < C_0 < \infty$ there are constants $C, c_0 > 0$ such that, with $T_N := \lfloor C_0 \log N \rfloor$ and*

$$\tau_*^{(N)} := \inf\{t \geq s_* : Z_t^{(N)} \notin I_*\},$$

one has, for all sufficiently large N ,

$$\mathbb{P}(\tau_*^{(N)} \leq T_N) \leq CT_N \exp(-c_0 N), \quad (6.15)$$

and

$$\sup_{0 \leq t \leq T_N} \mathbb{E} \left[\left(Z_t^{(N)} - q_{t,N} \right)^2 \right] \leq \frac{C}{N}. \quad (6.16)$$

See Appendix B.2 for the proofs.

We use this lemma with a horizon constant chosen before the cutoff offset. Fix

$$C_0 > \frac{1}{2|\log \mu_A|}. \quad (6.17)$$

For every fixed $c \in \mathbb{R}$, the definitions of $t_N(q_0)$ and $n_N(c)$ give

$$\frac{n_N(c)}{\log N} \rightarrow \frac{1}{2|\log \mu_A|}.$$

Hence, by (6.17),

$$0 \leq n_N(c) \leq C_0 \log N \quad (6.18)$$

for all sufficiently large N . Thus the constants supplied by Lemma 6.15 for this choice of C_0 are independent of c . Only the lower bound on N may depend on the fixed value of c .

The next lemma controls the synchronous coupling at time $(n_N(c) - 1)_+$, one step before $n_N(c)$. The expected separation of the two radii is $O(\mu_A^c N^{-1/2})$. Applying Lemma 6.13 at the next transition then gives a total-variation error of order μ_A^c .

Lemma 6.16. *Let Assumption 6.4 hold. Let $q_N \rightarrow q_0$ with $q_N > 0$, let $\nu_{A,N}$ be the unique nonzero invariant law of $K_{A,N}$, and let $(Z_t^{(N)}, \tilde{Z}_t^{(N)})$ be the synchronous coupling driven by the same Gaussian vectors, with $Z_0^{(N)} = q_N$ and $\tilde{Z}_0^{(N)} \sim \nu_{A,N}$. Then there is a constant $C < \infty$ such that, for every $c \geq 0$,*

$$\limsup_{N \rightarrow \infty} \sqrt{N} \mathbb{E} \left| Z_{(n_N(c)-1)_+}^{(N)} - \tilde{Z}_{(n_N(c)-1)_+}^{(N)} \right| \leq C \mu_A^c. \quad (6.19)$$

See Appendix B.3 for the proof.

6.5. Proof of the supercritical cutoff theorem. The lower bound separates the evolving and stationary laws before the deterministic cutoff scale. The coupling estimate gives the upper bound after that scale.

Proof of Theorem 1.8. Step 1: Setup. The theorem assumptions give Assumption 6.4. Moreover, $q_N > 0$ for all sufficiently large N because $q_N \rightarrow q_0 > 0$. Since $A > 1$ and $A_c(N) \downarrow 1$, the existence of a nonzero invariant law for all sufficiently large N follows from Proposition 2.6. Uniqueness follows from Proposition 6.1. We denote this law by $\nu_{A,N}$.

Step 2: The lower bound. Fix $c > 0$ and put $t_N^- := n_N(-c)$, with n_N defined in (1.20). By (6.18), this time is positive and bounded by $C_0 \log N$ for all large N , with the same C_0 for every fixed c . Let $d_N := |q_{t_N^-, N} - q_*|$. Since $q_0 \neq q_*$, the orbit $V_A^t(q_0)$ never crosses q_* . Hence $d_N > 0$ for all large N . We separate the dynamic law and the invariant law by the half-line

$$H_N := \left\{ q \in [0, 1] : \text{sgn}(q_{t_N^-, N} - q_*)(q - q_*) \geq \frac{1}{2} d_N \right\}. \quad (6.20)$$

Write $\varepsilon_N := \text{sgn}(q_{t_N^-, N} - q_*) \in \{-1, 1\}$. Then $\varepsilon_N(q_{t_N^-, N} - q_*) = d_N$. If $Z_{t_N^-}^{(N)} \notin H_N$, then, by the definition of H_N ,

$$\varepsilon_N(Z_{t_N^-}^{(N)} - q_*) < \frac{1}{2} d_N.$$

Subtracting this inequality from the preceding identity gives

$$\varepsilon_N(q_{t_N^-, N} - Z_{t_N^-}^{(N)}) = \varepsilon_N(q_{t_N^-, N} - q_*) - \varepsilon_N(Z_{t_N^-}^{(N)} - q_*) > \frac{1}{2} d_N.$$

In particular,

$$\{Z_{t_N^-}^{(N)} \notin H_N\} \subseteq \left\{ \left| Z_{t_N^-}^{(N)} - q_{t_N^-, N} \right| \geq \frac{1}{2} d_N \right\}.$$

Chebyshev's inequality and Lemma 6.15 give

$$\begin{aligned} K_{A,N}^{t_N^-}(q_N, H_N^c) &= \mathbb{P}_{q_N}(Z_{t_N^-}^{(N)} \notin H_N) \\ &\leq \mathbb{P}_{q_N} \left(\left| Z_{t_N^-}^{(N)} - q_{t_N^-, N} \right| \geq \frac{1}{2} d_N \right) \\ &\leq \frac{4}{d_N^2} \mathbb{E} \left[\left(Z_{t_N^-}^{(N)} - q_{t_N^-, N} \right)^2 \right] \leq \frac{C}{N d_N^2}. \end{aligned} \quad (6.21)$$

In the last step we used (6.16), since $t_N^- \leq C_0 \log N$. Similarly, if $q \in H_N$, then $|q - q_*| \geq d_N/2$. Chebyshev's inequality and Proposition 6.14 therefore give

$$\nu_{A,N}(H_N) \leq \frac{4}{d_N^2} \int_0^1 (q - q_*)^2 \nu_{A,N}(dq) \leq \frac{C}{Nd_N^2}. \quad (6.22)$$

Since total variation is the supremum over measurable sets,

$$\begin{aligned} \|K_{A,N}^{t_N^-}(q_N, \cdot) - \nu_{A,N}\|_{\text{TV}} &\geq K_{A,N}^{t_N^-}(q_N, H_N) - \nu_{A,N}(H_N) \\ &\geq 1 - K_{A,N}^{t_N^-}(q_N, H_N^c) - \nu_{A,N}(H_N). \end{aligned}$$

The locally uniform asymptotic (2.32), applied at $t_N^- = n_N(-c)$, gives

$$d_N = |C(q_N)| \mu_A^{t_N^-}(1 + o(1)).$$

Moreover, $C(q_N) \rightarrow C(q_0) \neq 0$ and the definition of the cutoff center gives $\sqrt{N} |C(q_0)| \mu_A^{t_N(q_0)} = 1$. Since $t_N^- - (t_N(q_0) - c) \in [-1, 0]$, the integer-time rounding is uniformly bounded. Hence there is $a_0 > 0$, independent of c , such that

$$\liminf_{N \rightarrow \infty} \sqrt{N} d_N \geq a_0 \mu_A^{-c}.$$

Combining this bound with (6.21) and (6.22) yields

$$\liminf_{N \rightarrow \infty} \|K_{A,N}^{t_N^-}(q_N, \cdot) - \nu_{A,N}\|_{\text{TV}} \geq 1 - C \mu_A^{2c}.$$

Here C is independent of c . Sending $c \rightarrow \infty$ proves (1.21).

Step 3: The upper bound. Fix $c \geq 0$ and put $t_N^+ := (n_N(c) - 1)_+$. Let $(Z_t^{(N)}, \tilde{Z}_t^{(N)})$ be the synchronous coupling from Lemma 6.16, with $Z_0^{(N)} = q_N$ and $\tilde{Z}_0^{(N)} \sim \nu_{A,N}$. By invariance, $\tilde{Z}_{t_N^+}^{(N)} \sim \nu_{A,N}$. The Markov property and convexity of total variation imply the standard coupling inequality

$$\begin{aligned} \|K_{A,N}^{t_N^++1}(q_N, \cdot) - \nu_{A,N}\|_{\text{TV}} &= \|\mathbb{E}[K_{A,N}(Z_{t_N^+}^{(N)}, \cdot) - K_{A,N}(\tilde{Z}_{t_N^+}^{(N)}, \cdot)]\|_{\text{TV}} \\ &\leq \mathbb{E} \|K_{A,N}(Z_{t_N^+}^{(N)}, \cdot) - K_{A,N}(\tilde{Z}_{t_N^+}^{(N)}, \cdot)\|_{\text{TV}}. \end{aligned} \quad (6.23)$$

Set

$$E_N := \{Z_{t_N^+}^{(N)} \in I_*, \tilde{Z}_{t_N^+}^{(N)} \in I_*\}.$$

We claim that $\mathbb{P}(E_N^c) = o_N(1)$. Indeed, (6.18) gives $t_N^+ \leq C_0 \log N$ for all large N , with C_0 independent of c , and $t_N^+ \geq s_*$. Thus Lemma 6.15 gives

$$\mathbb{P}(Z_{t_N^+}^{(N)} \notin I_*) \leq C(\log N) e^{-c_0 N} = o_N(1),$$

while stationarity of $\tilde{Z}^{(N)}$ and Proposition 6.12 give

$$\mathbb{P}(\tilde{Z}_{t_N^+}^{(N)} \notin I_*) = \nu_{A,N}(I_*^c) \leq \frac{C}{N}.$$

These two bounds give $\mathbb{P}(E_N^c) = o_N(1)$.

On E_N , the one-step total variation estimate Lemma 6.13, applied with $I = I_*$, gives

$$\|K_{A,N}(Z_{t_N^+}^{(N)}, \cdot) - K_{A,N}(\tilde{Z}_{t_N^+}^{(N)}, \cdot)\|_{\text{TV}} \leq C\sqrt{N} |Z_{t_N^+}^{(N)} - \tilde{Z}_{t_N^+}^{(N)}|.$$

On E_N^c we only use the trivial bound by one. Hence (6.23) yields

$$\|K_{A,N}^{t_N^++1}(q_N, \cdot) - \nu_{A,N}\|_{\text{TV}} \leq C\sqrt{N} \mathbb{E} \left| Z_{t_N^+}^{(N)} - \tilde{Z}_{t_N^+}^{(N)} \right| + \mathbb{P}(E_N^c)$$

For all sufficiently large N , one has $n_N(c) \geq 1$ and hence $t_N^++1 = n_N(c)$. Since $t_N^+ = (n_N(c) - 1)_+$, Lemma 6.16 gives

$$\limsup_{N \rightarrow \infty} \|K_{A,N}^{n_N(c)}(q_N, \cdot) - \nu_{A,N}\|_{\text{TV}} \leq C\mu_A^c.$$

Here C is independent of c . Sending $c \rightarrow \infty$ proves (1.22). $\square$

Proof of Corollary 1.9. The lower bound follows from Proposition 2.4, the left inequality in (2.43), and Theorem 1.8. The right inequality in (2.43) gives

$$\|P_{A,N}^{n_N(c)}(x_N, \cdot) - \pi_{A,N}\|_{\text{TV}} \leq \|K_{A,N}^{(n_N(c)-1)_+}(q_N, \cdot) - \nu_{A,N}\|_{\text{TV}}.$$

For all sufficiently large N , one has $(n_N(c) - 1)_+ = n_N(c - 1)$. Thus (1.22) proves the upper bound after sending $c \rightarrow \infty$. By Corollary 6.3, $\pi_{A,N} = \nu_{A,N} J_{A,N}$ is the unique $P_{A,N}$ -invariant probability that assigns no mass to $\{0\}$. $\square$

7. The supercritical stationary law at fixed N

Fix N and assume $A > A_c(N)$. We study the invariant law $\pi_{A,N}$ with $\pi_{A,N}(\{0\}) = 0$. We first derive its stationary density equation and rewrite it in log-polar coordinates. Near the origin, the nonlinear update is a quadratic perturbation of its linearization. Gaussian isotropy refreshes the angular variable in one step, so exponential tilting at $\beta_{A,N}$ reduces the linearized equation to a scalar renewal equation. We then show that the nonlinear correction gives a directly Riemann integrable forcing and apply the renewal theorem to prove the directional and radial power laws in Theorem 1.10. This is the implicit-renewal method for stationary random recursions [Kes73], [Gol91], and [BDM16].

We use the Gaussian matrix $W_A^{(N)}$ from (2.1), without a time index. The chain $X^{(N)}$ and its transition kernel $P_{A,N}$ are given in (2.2) and Proposition 2.4. Below, W denotes an independent copy of $W_A^{(N)}$. We use $\ell_{A,N}$ from (2.25). Set

$$\mathcal{K}_N := [-1, 1]^N.$$

The origin is absorbing, so δ_0 is always invariant. We write $\pi_{A,N}$ for the nonzero invariant law, which exists under the present assumption by Proposition 2.6. Its uniqueness follows from Corollary 6.3.

7.1. The stationary density equation. For $x \neq 0$, the coordinatewise change of variables $z_i = \text{arctanh}(y_i)$ shows that one Gaussian step has a product density. We record the resulting stationary equation.

Proposition 7.1. *Let $\pi_{A,N}$ be an invariant probability for $P_{A,N}$ on $\mathcal{K}_N$ with $\pi_{A,N}(\{0\}) = 0$. Then, for every bounded measurable $\varphi : \mathcal{K}_N \rightarrow \mathbb{R}$,*

$$\int_{\mathcal{K}_N} \varphi(y) \pi_{A,N}(\text{d}y) = \int_{\mathcal{K}_N} \int_{\mathbb{R}^N} \varphi(\tanh(A\|x\|_2 g / \sqrt{N})) \mathcal{G}_N(\text{d}g) \pi_{A,N}(\text{d}x). \quad (7.1)$$

Moreover, $\pi_{A,N}(\partial[-1, 1]^N) = 0$ and $\pi_{A,N}$ is absolutely continuous on $(-1, 1)^N$. If $f_{A,N}$ denotes its density, then, for $x \neq 0$ and $y \in (-1, 1)^N$,

$$\kappa_{A,N}(x, y) = \prod_{i=1}^N \frac{\sqrt{N}}{A\|x\|_2 \sqrt{2\pi}} \frac{1}{1 - y_i^2} \exp\left(-\frac{N \text{arctanh}(y_i)^2}{2A^2\|x\|_2^2}\right) \quad (7.2)$$

is the transition density from x to y , and

$$f_{A,N}(y) = \int_{\mathcal{K}_N \setminus \{0\}} \kappa_{A,N}(x, y) \pi_{A,N}(\mathrm{d}x). \quad (7.3)$$

If $\pi_{A,N}(\mathrm{d}x) = f_{A,N}(x) \mathrm{d}x$, this becomes

$$f_{A,N}(y) = \int_{\mathcal{K}_N \setminus \{0\}} \kappa_{A,N}(x, y) f_{A,N}(x) \mathrm{d}x. \quad (7.4)$$

Proof. Step 1: The transition density from $x \neq 0$. Fix $x \neq 0$. By (2.1), the rows of $\mathbf{W}_A^{(N)}$ are independent, and the coordinates of $\mathbf{W}_A^{(N)}x$ are independent centered Gaussians with variance $A^2\|x\|_2^2/N$. The change of variables $y_i = \tanh z_i$, or $z_i = \operatorname{arctanh}(y_i)$, gives the product density (7.2). Moreover, $\tanh z \in (-1, 1)$ for every finite z , so this transition law gives no mass to $\partial[-1, 1]^N$.

Step 2: The density of the invariant law. The weak equation (7.1) is the invariance relation $\pi_{A,N} = \pi_{A,N} P_{A,N}$. Since $\pi_{A,N}(\{0\}) = 0$, the boundary property from Step 1 and invariance give $\pi_{A,N}(\partial[-1, 1]^N) = 0$. Integrating the transition density (7.2) against $\pi_{A,N}$ gives (7.3) and proves absolute continuity on $(-1, 1)^N$. Substituting $\pi_{A,N}(\mathrm{d}x) = f_{A,N}(x) \mathrm{d}x$ gives (7.4). $\square$

7.2. Log-polar coordinates. To analyze the invariant law near 0, we use log-polar coordinates. Let $\mathbb{S}^{N-1}$ be the Euclidean unit sphere and write $\hat{z} = z/\|z\|_2$ for $z \neq 0$. For $x \neq 0$, set

$$r = \|x\|_2, \quad \theta = \frac{x}{\|x\|_2}, \quad y = -\log r.$$

Since $r = e^{-y}$, the limit $r \downarrow 0$ corresponds to $y \rightarrow \infty$, and the factor r^β becomes $e^{-\beta y}$.

Let $\tilde{\pi}_{A,N}$ be the image of $\pi_{A,N}$ under $x \mapsto (Y, \Theta) = (-\log\|x\|_2, x/\|x\|_2)$. Since $0 < \|x\|_2 \leq \sqrt{N}$ on $\mathcal{K}_N \setminus \{0\}$, the variable Y takes values in $[-\frac{1}{2} \log N, \infty)$. For $r > 0$, $\theta \in \mathbb{S}^{N-1}$, and $W\theta \neq 0$, define

$$\eta(r, \theta, W) := \log \frac{r\|W\theta\|_2}{\|\tanh(rW\theta)\|_2}, \quad (7.5)$$

$$\Theta_+(r, \theta, W) := \frac{\tanh(rW\theta)}{\|\tanh(rW\theta)\|_2}, \quad \Theta_0(\theta, W) := \widehat{W\theta}. \quad (7.6)$$

Here η is the correction to the logarithmic radius caused by the nonlinearity, Θ_+ is the direction after the nonlinear update, and Θ_0 is the direction after the corresponding linear update. Since $W\theta \neq 0$ almost surely, the stationary equation (7.1) becomes in these coordinates

$$\int \varphi(y, \theta) \tilde{\pi}_{A,N}(\mathrm{d}y, \mathrm{d}\theta) = \int \mathbb{E} \left[\varphi \left(y - \log\|W\theta\|_2 + \eta(e^{-y}, \theta, W), \Theta_+(e^{-y}, \theta, W) \right) \right] \tilde{\pi}_{A,N}(\mathrm{d}y, \mathrm{d}\theta) \quad (7.7)$$

for every bounded measurable φ .

7.3. The linear renewal problem. In (7.7), set $\eta = 0$ and replace Θ_+ by Θ_0 . This gives the linearized chain

$$(Y, \Theta) \mapsto (Y + \Delta, \Theta_0), \quad \Delta = -\log\|\mathbf{W}_A^{(N)}\Theta\|_2. \quad (7.8)$$

For a general matrix ensemble, Θ is a Markov chain and Δ is an additive functional of it [BL85]. Renewal theory for this chain is developed in [Kes73] and further discussed in [BDM16]. For Gaussian W , the law of $W\Theta$ does not depend on Θ . The angle is therefore refreshed at each step.

Let $\bar{\sigma}_N$ denote normalized surface measure on $\mathbb{S}^{N-1}$. We equip $C(\mathbb{S}^{N-1})$ with the supremum norm. Define the linear log-polar transition kernel by

$$P_{A,N}^{\text{lin}}(\theta, \cdot, \cdot) := \text{Law}\left(-\log\|\mathbf{W}_A^{(N)}\theta\|_2, \widehat{\mathbf{W}_A^{(N)}\theta}\right).$$

For $0 \leq \beta < N$, define the transfer operator on $C(\mathbb{S}^{N-1})$ by

$$\mathcal{L}_{A,N,\beta}\varphi(\theta) := \int e^{\beta z} \varphi(\theta') P_{A,N}^{\text{lin}}(\theta, dz, d\theta') = \mathbb{E}\left[\|\mathbf{W}_A^{(N)}\theta\|_2^{-\beta} \varphi(\widehat{\mathbf{W}_A^{(N)}\theta})\right]. \quad (7.9)$$

The Gaussian transfer operator can be computed explicitly.

Lemma 7.2. *For every $0 \leq \beta < N$, the operator $\mathcal{L}_{A,N,\beta}$ is bounded on $C(\mathbb{S}^{N-1})$ and satisfies*

$$\mathcal{L}_{A,N,\beta}\varphi(\theta) = \mathcal{M}_{A,N}(\beta) \int_{\mathbb{S}^{N-1}} \varphi d\bar{\sigma}_N, \quad \theta \in \mathbb{S}^{N-1}, \quad (7.10)$$

where

$$\mathcal{M}_{A,N}(\beta) := \mathbb{E}\|\mathbf{W}_A^{(N)}\theta\|_2^{-\beta} = \left(\frac{\sqrt{N}}{A}\right)^\beta 2^{-\beta/2} \frac{\Gamma((N-\beta)/2)}{\Gamma(N/2)}. \quad (7.11)$$

Consequently, its spectral radius $r_{A,N}(\beta)$ is given by

$$r_{A,N}(\beta) = \mathcal{M}_{A,N}(\beta) = A^{-\beta} \mathcal{M}_{1,N}(\beta). \quad (7.12)$$

Proof. Fix $\theta \in \mathbb{S}^{N-1}$. By (2.1), the vector $\mathbf{W}_A^{(N)}\theta$ has law $(A/\sqrt{N})G$, where $G \sim \mathcal{N}(0, I_N)$. Hence

$$\mathbb{E}\|\mathbf{W}_A^{(N)}\theta\|_2^{-\beta} = \left(\frac{\sqrt{N}}{A}\right)^\beta \mathbb{E}\|G\|_2^{-\beta}.$$

The last expectation is finite exactly for $\beta < N$, since the standard Gaussian density is bounded and the singularity of $\|z\|_2^{-\beta}$ at the origin is integrable in dimension N precisely in this range. The displayed Gamma-function formula is the standard moment formula for $\chi_N = \|G\|_2$.

The same isotropy also shows that $\widehat{\mathbf{W}_A^{(N)}\theta}$ has law $\bar{\sigma}_N$ and is independent of $\|\mathbf{W}_A^{(N)}\theta\|_2$. Therefore (7.10) holds. A rank-one positive operator of the form $\mathcal{M}_{A,N}(\beta)\bar{\sigma}_N(\varphi)\mathbf{1}$ has spectral radius $\mathcal{M}_{A,N}(\beta)$. $\square$

The Cramér exponent is the unique positive solution of the criticality equation for this operator.

Lemma 7.3. *Assume $A > A_c(N)$, with $A_c(N)$ defined in (2.26). Then there is a unique $\beta_{A,N} \in (0, N)$ satisfying*

$$r_{A,N}(\beta_{A,N}) = 1. \quad (7.13)$$

Equivalently, $\beta_{A,N}$ is the unique positive solution of

$$A^{-\beta} N^{\beta/2} 2^{-\beta/2} \frac{\Gamma((N-\beta)/2)}{\Gamma(N/2)} = 1, \quad 0 < \beta < N. \quad (7.14)$$

Proof. Recall $\ell_{A,N}$ from (2.25), and set

$$F(\beta) := \log \mathbb{E}\ell_{A,N}^{-\beta}.$$

By (2.26), the condition $A > A_c(N)$ is equivalent to $\mathbb{E}\log \ell_{A,N} > 0$. Thus $F(0) = 0$ and $F'(0) = -\mathbb{E}\log \ell_{A,N} < 0$. Moreover F is strictly convex by Hölder's inequality and the nondegeneracy of $\ell_{A,N}$, and $F(\beta) \rightarrow \infty$ as $\beta \uparrow N$, because $\mathbb{E}\chi_N^{-\beta} \uparrow \infty$. Therefore $F(\beta) = 0$ has exactly one solution in $(0, N)$. The explicit equation is just (7.11). $\square$

By Lemma 7.2, tilting at $\beta_{A,N}$ changes only the law of the increment. The updated direction remains independent of the increment and has law $\bar{\sigma}_N$. The renewal equation therefore reduces to a scalar convolution in y . The following lemma applies Feller's whole-line renewal theorem [Fel71] to this convolution and recovers the angular limit as a multiple of $\bar{\sigma}_N$. A related formulation is given in [Gut09].

Lemma 7.4. *Assume $A > A_c(N)$ and let $\beta_{A,N}$ be the exponent from Lemma 7.3. Using $\ell_{A,N}$ from (2.25), define*

$$\int g(z) \hat{\mu}_{A,N}(dz) := \mathbb{E} \left[\ell_{A,N}^{-\beta_{A,N}} g(-\log \ell_{A,N}) \right]. \quad (7.15)$$

Then $\hat{\mu}_{A,N}$ is a nonlattice probability measure. Its drift satisfies

$$\hat{m}_{A,N} := \int_{\mathbb{R}} z \hat{\mu}_{A,N}(dz) \in (0, \infty).$$

Set $\hat{P}_{A,N}(\theta, dz, d\theta') := \bar{\sigma}_N(d\theta') \hat{\mu}_{A,N}(dz)$.

Let $(\mathcal{H}_y)_{y \in \mathbb{R}}$ be a locally bounded family of finite signed measures on $\mathbb{S}^{N-1}$ satisfying the boundary conditions

$$\lim_{y \rightarrow -\infty} \|\mathcal{H}_y\|_{\text{TV}} = 0, \quad \lim_{y \rightarrow \infty} e^{-\beta_{A,N} y} \|\mathcal{H}_y\|_{\text{TV}} = 0. \quad (7.16)$$

Assume also that the forcing $(\Psi_y)_{y \in \mathbb{R}}$ satisfies

$$\sum_{k \in \mathbb{Z}} \sup_{y \in [k, k+1]} \|\Psi_y\|_{\text{TV}} < \infty. \quad (7.17)$$

For every continuous φ , assume that $y \mapsto \Psi_y(\varphi)$ is continuous and that

$$\mathcal{H}_y(\varphi) = \int_{\mathbb{S}^{N-1}} \int \varphi(\theta') \hat{P}_{A,N}(\theta, dz, d\theta') \mathcal{H}_{y-z}(d\theta) + \Psi_y(\varphi). \quad (7.18)$$

Then

$$\mathcal{H}_y(\varphi) \rightarrow \frac{1}{\hat{m}_{A,N}} \left(\int_{\mathbb{R}} \Psi_t(\mathbf{1}) dt \right) \int_{\mathbb{S}^{N-1}} \varphi d\bar{\sigma}_N, \quad y \rightarrow \infty, \quad (7.19)$$

where $\mathbf{1}$ denotes the constant function on $\mathbb{S}^{N-1}$.

Proof. Step 1: Tilted law. The identity $r_{A,N}(\beta_{A,N}) = 1$ says precisely that $\hat{\mu}_{A,N}$ has total mass one. Its law has a density, since χ_N has a density on $(0, \infty)$, and hence it is nonlattice. The drift is finite because $\mathbb{E} \chi_N^{-\beta} |\log \chi_N| < \infty$ for $\beta < N$. It is positive since

$$\hat{m}_{A,N} = \frac{d}{d\beta} \log \mathcal{M}_{A,N}(\beta) \Big|_{\beta=\beta_{A,N}},$$

and this derivative is positive at the unique positive zero $\beta_{A,N}$ of the strictly convex function $\log \mathcal{M}_{A,N}(\beta)$.

Step 2: Scalar renewal representation. Put $h_y := \mathcal{H}_y(\mathbf{1})$. Taking $\varphi = \mathbf{1}$ in (7.18) gives the scalar renewal equation

$$h_y = \int_{\mathbb{R}} h_{y-z} \hat{\mu}_{A,N}(dz) + \Psi_y(\mathbf{1}).$$

Let $S_n = Z_1 + \cdots + Z_n$, where the variables Z_i are independent with law $\hat{\mu}_{A,N}$. Iterating the last display gives

$$h_y = \mathbb{E} h_{y-S_n} + \sum_{k=0}^{n-1} \mathbb{E} \Psi_{y-S_k}(\mathbf{1}).$$

We claim that, for this fixed y , the terminal term tends to zero as $n \rightarrow \infty$. Since $\hat{m}_{A,N} > 0$, we have $S_n \rightarrow \infty$ almost surely. Fix $L < \infty$. On the half-line $(-\infty, L]$, local boundedness together with the first condition in (7.16) gives $\sup_{u \leq L} \|\mathcal{H}_u\|_{\text{TV}} < \infty$. Since $y - S_n \rightarrow -\infty$ almost surely, dominated convergence gives

$$\mathbb{E}[\|\mathcal{H}_{y-S_n}\|_{\text{TV}} \mathbf{1}_{\{y-S_n \leq L\}}] \rightarrow 0.$$

On the complementary event, the second condition in (7.16) gives, for L sufficiently large,

$$\mathbb{E}[\|\mathcal{H}_{y-S_n}\|_{\text{TV}} \mathbf{1}_{\{y-S_n > L\}}] \leq \varepsilon e^{\beta_{A,N} y} \mathbb{E} e^{-\beta_{A,N} S_n} = \varepsilon e^{\beta_{A,N} y}.$$

The equality uses the defining tilt (7.15). Adding the two contributions and then letting $\varepsilon \downarrow 0$ gives

$$\mathbb{E} \|\mathcal{H}_{y-S_n}\|_{\text{TV}} \rightarrow 0.$$

Since $\|\mathbf{1}\|_\infty = 1$, the signed terminal term is controlled by this total variation bound,

$$|\mathbb{E} h_{y-S_n}| = |\mathbb{E} \mathcal{H}_{y-S_n}(\mathbf{1})| \leq \mathbb{E} \|\mathcal{H}_{y-S_n}\|_{\text{TV}} \rightarrow 0.$$

Therefore, after first fixing y and then letting $n \rightarrow \infty$,

$$h_y = \sum_{k=0}^{\infty} \mathbb{E} \Psi_{y-S_k}(\mathbf{1}).$$

Step 3: Scalar renewal limit. Let $y \rightarrow \infty$ and decompose the signed forcing as $\Psi_y(\mathbf{1}) = \Psi_y(\mathbf{1})^+ - \Psi_y(\mathbf{1})^-$. Both parts are nonnegative and directly Riemann integrable by (7.17). The increment law $\hat{\mu}_{A,N}$ is nonarithmetic with mean $\hat{m}_{A,N} \in (0, \infty)$ and charges both half-lines, so the renewal measure $U := \sum_{k \geq 0} \mathcal{L}(S_k)$ obeys the general renewal theorem on the whole line: $U\{I + y\} \rightarrow |I|/\hat{m}_{A,N}$ as $y \rightarrow +\infty$ and $U\{I + y\} \rightarrow 0$ as $y \rightarrow -\infty$ for every finite interval I [Fel71, Sec. XI.9, (9.2)–(9.3)]. Applying the upper- and lower-step-function argument of [Fel71, Sec. XI.1, (1.10)–(1.17)] to these whole-line interval limits extends the conclusion to any directly Riemann integrable integrand in the sense of (7.17). Apply it separately to $\Psi(\mathbf{1})^+$ and $\Psi(\mathbf{1})^-$. The identity $\int_{\mathbb{R}} z(y-s) U(ds) = \sum_{k \geq 0} \mathbb{E} z(y-S_k)$ then gives

$$\sum_{k=0}^{\infty} \mathbb{E} \Psi_{y-S_k}(\mathbf{1})^\pm \rightarrow \frac{1}{\hat{m}_{A,N}} \int_{\mathbb{R}} \Psi_t(\mathbf{1})^\pm dt, \quad y \rightarrow \infty,$$

and subtracting the two limits yields

$$h_y \rightarrow \frac{1}{\hat{m}_{A,N}} \int_{\mathbb{R}} \Psi_t(\mathbf{1}) dt.$$

Step 4: Angular reconstruction. For a general continuous φ , the product form of $\hat{P}_{A,N}$ gives

$$\mathcal{H}_y(\varphi) = \left(\int_{\mathbb{S}^{N-1}} \varphi d\bar{\sigma}_N \right) \int_{\mathbb{R}} h_{y-z} \hat{\mu}_{A,N}(dz) + \Psi_y(\varphi).$$

Using the scalar renewal equation to replace the convolution by $h_y - \Psi_y(\mathbf{1})$, and using direct Riemann integrability to get $\Psi_y(\varphi) \rightarrow 0$ and $\Psi_y(\mathbf{1}) \rightarrow 0$, yields (7.19). $\square$

Remark 7.5. The simplification in Lemma 7.2 is special to the Gaussian ensemble in (2.1). For a non-isotropic matrix ensemble the direction would not be refreshed from $\bar{\sigma}_N$ in one step, and the analogue of Lemma 7.4 would have to be replaced by a genuine Markov renewal theorem for products of random matrices [Kes73] and [BDM16].

7.4. The nonlinear renewal equation. We compare (7.7) with its linearization (7.8). After exponential tilting, their difference is the forcing Ψ^π defined below. Its integrability uses exponential moments of order less than $\beta_{A,N}$.

Lemma 7.6. *Assume $A > A_c(N)$ and let $\pi_{A,N}$ be a nonzero invariant law as in Proposition 7.1. Let (Y, Θ) have law $\tilde{\pi}_{A,N}$. Then, for every $0 \leq \alpha < \beta_{A,N}$,*

$$\mathbb{E} e^{\alpha Y} < \infty. \quad (7.20)$$

Proof. The case $\alpha = 0$ is trivial, so assume $0 < \alpha < \beta_{A,N}$. Let ν be the image of $\pi_{A,N}$ under the squared-radius map $r_N(x) = N^{-1}\|x\|_2^2$, defined before Proposition 2.4. By Proposition 2.4, ν is a nonzero invariant law for $K_{A,N}$.

Step 1: Contraction of $q^{-\alpha/2}$ near zero. Since $\alpha < \beta_{A,N}$ and $\beta_{A,N}$ is the first positive zero of $\log \mathcal{M}_{A,N}$, we have

$$\mathbb{E} \left[\left(\frac{A}{\sqrt{N}} \chi_N \right)^{-\alpha} \right] < 1.$$

Put $p = \alpha/2$ and $V(q) = q^{-p}$. Since

$$\frac{F_{A,N}(q, G)}{q} = \frac{1}{N} \sum_{i=1}^N A^2 G_i^2 \left(\frac{\tanh(A\sqrt{q} G_i)}{A\sqrt{q} G_i} \right)^2 \xrightarrow{q \downarrow 0} \frac{A^2}{N} \chi_N^2 \quad \text{a.s.,}$$

the fixed- N version of the truncation argument in Proposition 6.8 gives

$$\mathbb{E} \left[\left(\frac{F_{A,N}(q, G)}{q} \right)^{-p} \right] \longrightarrow \mathbb{E} \left[\left(\frac{A}{\sqrt{N}} \chi_N \right)^{-\alpha} \right] < 1.$$

Hence we may choose $R_0 > 0$ and $a < 1$ such that, for $0 < q \leq R_0$,

$$K_{A,N} V(q) = q^{-p} \mathbb{E} \left[\left(\frac{F_{A,N}(q, G)}{q} \right)^{-p} \right] \leq a V(q).$$

Step 2: A finite negative moment of ν . Since $\alpha < \beta_{A,N} < N$, the fixed- N Gaussian small-ball estimate used above gives

$$B := \sup_{R_0 \leq q \leq 1} K_{A,N} V(q) < \infty.$$

Therefore

$$K_{A,N} V(q) \leq a V(q) + B, \quad q \in (0, 1].$$

The Cesàro construction in the proof of Proposition 2.6 therefore produces a nonzero invariant law $\bar{\nu}$ with $\int q^{-p} \bar{\nu}(dq) < \infty$. By the uniqueness of the nonzero invariant law in Proposition 6.1, $\bar{\nu} = \nu$.

Step 3: The exponential moment of Y . If $X \sim \pi_{A,N}$, then $r_N(X) \sim \nu$ and

$$\mathbb{E} e^{\alpha Y} = N^{-\alpha/2} \int_{(0,1]} q^{-\alpha/2} \nu(dq) < \infty.$$

This proves (7.20). □

The next lemma bounds the difference between the nonlinear and linearized updates.

Lemma 7.7. *Assume $A > A_c(N)$ and let $\beta_{A,N} \in (0, N)$ be the exponent from Lemma 7.3. Fix $\alpha \in (0, 2)$ and set $h(t) := \min\{1, e^{-Nt}\}$. Then there exists $C < \infty$ such that*

$$\sum_{k \in \mathbb{Z}} \sup_{t \in [k, k+1]} e^{\beta_{A,N} t} h(t) < \infty \quad (7.21)$$

and, for every $0 < r \leq \sqrt{N}$ and every $t \in \mathbb{R}$,

$$\left\| \mathbb{E} \left[\mathbf{1}_{\{\|\tanh(rU)\|_2 < re^{-t}\}} \delta_{\widehat{\tanh(rU)}} \right] - \mathbb{E} \left[\mathbf{1}_{\{\|U\|_2 < e^{-t}\}} \delta_{\widehat{U}} \right] \right\|_{\text{TV}} \leq Cr^\alpha h(t), \quad (7.22)$$

where δ_z denotes the Dirac mass at z , $U = (A/\sqrt{N})G$, and $G \sim \mathcal{N}(0, I_N)$.

Proof. Step 1: Reduction to the densities of U and $T_r U$. Put

$$T_r u := r^{-1} \tanh(ru), \quad u \in \mathbb{R}^N.$$

Then $\widehat{\tanh(rU)} = \widehat{T_r U}$ and $\{\|\tanh(rU)\|_2 < re^{-t}\} = \{\|T_r U\|_2 < e^{-t}\}$. Let p_0 be the density of U . The map T_r is a diffeomorphism from $\mathbb{R}^N$ onto $(-r^{-1}, r^{-1})^N$, and the density of $T_r U$ is

$$p_r(v) = p_0(r^{-1} \operatorname{arctanh}(rv)) \prod_{i=1}^N (1 - r^2 v_i^2)^{-1} \mathbf{1}_{\{|rv_i| < 1, 1 \leq i \leq N\}}. \quad (7.23)$$

Since angular projection and restriction to the ball $\{\|v\|_2 < e^{-t}\}$ cannot increase total variation, the left side of (7.22) is at most

$$\int_{\{\|v\|_2 < e^{-t}\}} |p_r(v) - p_0(v)| dv. \quad (7.24)$$

It remains to prove that, for every $\rho > 0$ and $0 < r \leq \sqrt{N}$,

$$\int_{\{\|v\|_2 < \rho\}} |p_r(v) - p_0(v)| dv \leq Cr^\alpha \min\{1, \rho^N\}. \quad (7.25)$$

Step 2: The density estimate for $1/4 \leq r \leq \sqrt{N}$. The densities p_r are uniformly bounded for $1/4 \leq r \leq \sqrt{N}$. Indeed, by (7.23), it is enough to note that the one-dimensional function

$$s \mapsto \exp\{-c \operatorname{arctanh}(s)^2\} (1 - s^2)^{-1}, \quad |s| < 1,$$

is bounded for every $c > 0$. In the density p_r , the corresponding value is $c = N/(2A^2 r^2)$, which ranges over a compact subset of $(0, \infty)$. Hence

$$\int_{\{\|v\|_2 < \rho\}} |p_r(v) - p_0(v)| dv \leq C\rho^N.$$

The same integral is at most 2 because p_r and p_0 are probability densities. Since $r^\alpha \geq 4^{-\alpha}$ on this range, these two bounds give (7.25).

Step 3: The density estimate for $0 < r \leq 1/4$. We first prove the pointwise bound

$$|p_r(v) - p_0(v)| \leq Cr^2 (1 + \|v\|_2^4) e^{-c\|v\|_2^2}, \quad \|v\|_2 \leq (2r)^{-1}.$$

First suppose $\|v\|_2 \leq r^{-1/2}$. Then $|rv_i| \leq \sqrt{r} \leq 1/2$, and hence

$$r^{-1} \operatorname{arctanh}(rv_i) = v_i + O(r^2 |v_i|^3), \quad -\log(1 - r^2 v_i^2) = O(r^2 v_i^2),$$

uniformly in i . Expanding the logarithm of the density ratio $p_r(v)/p_0(v)$ gives

$$|\log(p_r(v)/p_0(v))| \leq Cr^2 (1 + \|v\|_2^4),$$

whose right side is bounded on this region. Since $p_r = p_0 \exp\{\log(p_r/p_0)\}$ and $p_0(v) \leq Ce^{-c\|v\|_2^2}$, the inequality $|e^x - 1| \leq C|x|$ for bounded x gives the pointwise bound above, after decreasing c . On the remaining annulus $r^{-1/2} < \|v\|_2 \leq (2r)^{-1}$, both $p_0(v)$ and $p_r(v)$ are bounded by $Ce^{-c\|v\|_2^2}$, while $r^2(1 + \|v\|_2^4) \geq 1$. Thus the pointwise bound also holds on this annulus, after decreasing

c. Integrating over $\{\|v\|_2 < \rho, \|v\|_2 \leq (2r)^{-1}\}$, and using that $\int_{\{\|v\|_2 < \rho\}} (1 + \|v\|_2^4) e^{-c\|v\|_2^2} dv \leq C \min\{1, \rho^N\}$, since the integrand is bounded near the origin and integrable over all of $\mathbb{R}^N$, this contributes at most $Cr^2 \min\{1, \rho^N\} \leq Cr^\alpha \min\{1, \rho^N\}$. The part of $\{\|v\|_2 < \rho\}$ where $\|v\|_2 > (2r)^{-1}$ is empty when $\rho < 1$. When $\rho \geq 1$, its contribution is at most

$$\begin{aligned} & \int_{\{\|v\|_2 > (2r)^{-1}\}} (p_0(v) + p_r(v)) dv \\ & \leq \mathbb{P}(\|U\|_2 > (2r)^{-1}) + \mathbb{P}(\|T_r U\|_2 > (2r)^{-1}) \leq Ce^{-c/r^2}, \end{aligned}$$

where the last inequality uses $\|T_r U\|_2 \leq \|U\|_2$ and the Gaussian tail bound. Since $e^{-c/r^2} \leq Cr^\alpha$, this proves (7.25).

Step 4: The kernel bound and summability. Taking $\rho = e^{-t}$ in (7.25), and using $\min\{1, \rho^N\} = h(t)$, proves (7.22). For $t < 0$, $e^{\beta_{A,N}t} h(t) = e^{\beta_{A,N}t}$, while for $t \geq 0$ it equals $e^{-(N-\beta_{A,N})t}$. Since $0 < \beta_{A,N} < N$, both tails are summable, which proves (7.21). $\square$

Definition 7.8 (The forcing Ψ^π). Fix an invariant law $\pi_{A,N}$ as in Proposition 7.1, let $(Y, \Theta) \sim \tilde{\pi}_{A,N}$, and let W be an independent copy of $W_A^{(N)}$. Put

$$\Delta = -\log\|W\Theta\|_2, \quad \Theta_0 = \widehat{W\Theta}, \quad \Theta_+ = \Theta_+(e^{-Y}, \Theta, W), \quad \eta = \eta(e^{-Y}, \Theta, W).$$

For each continuous test function φ on $\mathbb{S}^{N-1}$, define

$$\begin{aligned} \Psi_y^\pi(\varphi) &:= e^{\beta_{A,N}y} \mathbb{E} \left[\varphi(\Theta_+) \mathbb{1}_{\{Y+\Delta+\eta>y\}} \right. \\ &\quad \left. - \varphi(\Theta_0) \mathbb{1}_{\{Y+\Delta>y\}} \right]. \end{aligned} \tag{7.26}$$

This bounded linear functional on $C(\mathbb{S}^{N-1})$ defines a finite signed measure, also denoted by Ψ_y^π . Set

$$c_{\Psi^\pi} := \frac{1}{\widehat{m}_{A,N}} \int_{\mathbb{R}} \Psi_t^\pi(\mathbf{1}) dt, \tag{7.27}$$

where $\widehat{m}_{A,N} := \int_{\mathbb{R}} z \widehat{\mu}_{A,N}(dz)$ is the tilted drift. We say that the nonlinear renewal forcing is admissible if $(\Psi_y^\pi)_{y \in \mathbb{R}}$ satisfies (7.17) and $c_{\Psi^\pi} > 0$.

We next verify that this forcing is admissible.

Proposition 7.9. Assume $A > A_c(N)$ and let $\pi_{A,N}$ be a nonzero invariant law as in Proposition 7.1. Then the forcing defined in Definition 7.8 is admissible.

Proof. Step 1: A summable bound for $\|\Psi_y^\pi\|_{\text{TV}}$. Choose $\alpha \in (0, \min\{2, \beta_{A,N}\})$. Conditional on (Y, Θ) , Gaussian isotropy shows that $W\Theta$ has the same law as $U := (A/\sqrt{N})G$. This law does not depend on (Y, Θ) . Set $r = e^{-Y}$ and $t = y - Y$. Since $Y \geq -\frac{1}{2} \log N$, we have $0 < r \leq \sqrt{N}$. Conditional on (Y, Θ) , the signed measure in (7.26) is therefore the difference estimated in Lemma 7.7. It follows that

$$\|\Psi_y^\pi\|_{\text{TV}} \leq Ce^{\beta_{A,N}y} \mathbb{E}[e^{-\alpha Y} h(y - Y)] = C \mathbb{E}[e^{(\beta_{A,N}-\alpha)Y} H(y - Y)], \tag{7.28}$$

where $h(t) = \min\{1, e^{-Nt}\}$. Put $H(t) := e^{\beta_{A,N}t} h(t)$, so that

$$H(t) = \begin{cases} e^{\beta_{A,N}t}, & t < 0, \\ e^{-(N-\beta_{A,N})t}, & t \geq 0. \end{cases}$$

Since $0 < \beta_{A,N} < N$, both tails decay exponentially and

$$\sup_{a \in \mathbb{R}} \sum_{k \in \mathbb{Z}} \sup_{y \in [k, k+1]} H(y - a) \leq C.$$

Using this bound in (7.28) gives

$$\begin{aligned} \sum_{k \in \mathbb{Z}} \sup_{y \in [k, k+1]} \|\Psi_y^\pi\|_{\text{TV}} &\leq C \mathbb{E} \left[e^{(\beta_{A,N} - \alpha)Y} \sum_{k \in \mathbb{Z}} \sup_{y \in [k, k+1]} H(y - Y) \right] \\ &\leq C \mathbb{E} e^{(\beta_{A,N} - \alpha)Y}. \end{aligned}$$

The last expectation is finite by Lemma 7.6, applied with exponent $\beta_{A,N} - \alpha$. This proves (7.17).

Step 2: Continuity of $y \mapsto \Psi_y^\pi(\varphi)$. Fix a continuous φ . Conditional on (Y, Θ) , the vector $W\Theta$ has the nondegenerate Gaussian law $\mathcal{N}(0, (A^2/N)I_N)$. The map

$$v \mapsto \tanh(e^{-Y}v)$$

is a diffeomorphism from $\mathbb{R}^N$ onto $(-1, 1)^N$. Thus both $W\Theta$ and $\tanh(e^{-Y}W\Theta)$ have conditional densities. Since Euclidean spheres have Lebesgue measure zero, the two variables

$$Y + \Delta = Y - \log\|W\Theta\|_2, \quad Y + \Delta + \eta = -\log\|\tanh(e^{-Y}W\Theta)\|_2$$

have no atoms, conditionally and hence unconditionally. If $y_n \rightarrow y$, the two indicators in (7.26) therefore converge almost surely. The boundedness of φ and dominated convergence show that $\Psi_{y_n}^\pi(\varphi) \rightarrow \Psi_y^\pi(\varphi)$.

Step 3: Positivity of c_{Ψ^π} . Taking $\varphi = \mathbf{1}$ and using $\eta \geq 0$, we obtain

$$\Psi_y^\pi(\mathbf{1}) = e^{\beta_{A,N}y} \mathbb{P}(Y + \Delta \leq y < Y + \Delta + \eta) \geq 0.$$

The bound in Step 1 implies that this function is integrable. Tonelli's theorem gives

$$\int_{\mathbb{R}} \Psi_y^\pi(\mathbf{1}) dy = \frac{1}{\beta_{A,N}} \mathbb{E} \left[e^{\beta_{A,N}(Y+\Delta)} (e^{\beta_{A,N}\eta} - 1) \right].$$

Since $Y < \infty$ and $W\Theta \neq 0$ almost surely, the strict inequality

$$\|\tanh(e^{-Y}W\Theta)\|_2 < e^{-Y}\|W\Theta\|_2 \quad \text{almost surely}$$

implies $\eta > 0$ almost surely. Hence the last expectation is strictly positive. Since $\hat{m}_{A,N} > 0$ by Lemma 7.4, (7.27) gives $c_{\Psi^\pi} > 0$, which proves the proposition. $\square$

Proof of Theorem 1.10. Step 1: Boundary conditions for $\mathcal{H}_y$. Let (Y, Θ) have law $\tilde{\pi}_{A,N}$ and define, for $y \in \mathbb{R}$,

$$\mathcal{H}_y(B) := e^{\beta_{A,N}y} \mathbb{E} [\mathbf{1}_{\{Y > y\}} \mathbf{1}_{\{\Theta \in B\}}]. \quad (7.29)$$

Each $\mathcal{H}_y$ is a nonnegative finite measure on $\mathbb{S}^{N-1}$. Since $\tilde{\pi}_{A,N}$ is the image of $\pi_{A,N}$ under $x \mapsto (-\log\|x\|_2, \hat{x})$, we have

$$\mathcal{H}_y(B) = e^{\beta_{A,N}y} \pi_{A,N}(0 < \|x\|_2 < e^{-y}, \hat{x} \in B). \quad (7.30)$$

By Proposition 7.1, $\pi_{A,N}$ is absolutely continuous on $(-1, 1)^N$ and therefore charges no sphere $\{\|x\|_2 = s\}$. Thus the strict radial inequality in (7.30) may be replaced by $\|x\|_2 \leq e^{-y}$.

Since $Y \geq -\frac{1}{2} \log N$, $\|\mathcal{H}_y\|_{\text{TV}} = e^{\beta_{A,N}y}$ for $y < -\frac{1}{2} \log N$, and hence $\|\mathcal{H}_y\|_{\text{TV}} \rightarrow 0$ as $y \rightarrow -\infty$. Moreover,

$$e^{-\beta_{A,N}y} \|\mathcal{H}_y\|_{\text{TV}} = \mathbb{P}(Y > y) \rightarrow 0, \quad y \rightarrow \infty.$$

The bound $\|\mathcal{H}_y\|_{\text{TV}} \leq e^{\beta_{A,N}y}$ also gives local boundedness. Hence $\mathcal{H}_y$ satisfies (7.16).

Step 2: The renewal equation for $\mathcal{H}_y$. Let W be an independent copy of $W_A^{(N)}$, independent of (Y, Θ) , and use the notation of Definition 7.8. For a continuous test function φ on $\mathbb{S}^{N-1}$, stationarity and (7.7) give

$$\mathcal{H}_y(\varphi) = e^{\beta_{A,N}y} \mathbb{E} [\varphi(\Theta_+) \mathbf{1}_{\{Y + \Delta + \eta > y\}}]. \quad (7.31)$$

Add and subtract the linearized term with $(\Theta_0, Y + \Delta)$. The difference between the two terms is precisely $\Psi_y^\pi(\varphi)$ from (7.26). Since $\hat{P}_{A,N}$ is the $e^{\beta_{A,N}z}$ -tilt of the linear log-polar transition kernel $P_{A,N}^{\text{lin}}$ from (7.9),

$$\begin{aligned} & \int_{\mathbb{S}^{N-1}} \int \varphi(\theta') \hat{P}_{A,N}(\theta, dz, d\theta') \mathcal{H}_{y-z}(d\theta) \\ &= e^{\beta_{A,N}y} \mathbb{E}[\varphi(\Theta_0) \mathbf{1}_{\{Y+\Delta > y\}}]. \end{aligned}$$

Therefore

$$\mathcal{H}_y(\varphi) = \int_{\mathbb{S}^{N-1}} \int \varphi(\theta') \hat{P}_{A,N}(\theta, dz, d\theta') \mathcal{H}_{y-z}(d\theta) + \Psi_y^\pi(\varphi). \quad (7.32)$$

This is (7.18).

Step 3: The limit from the renewal lemma. Step 1 verifies the boundary conditions and local boundedness of $\mathcal{H}_y$, and Step 2 gives the renewal equation. By Proposition 7.9, Ψ^π satisfies (7.17). Continuity of $y \mapsto \Psi_y^\pi(\varphi)$ was also established in its proof. Thus the assumptions of Lemma 7.4 hold. The constant c_{Ψ^π} in that lemma is positive by Proposition 7.9. By the uniqueness in Corollary 6.3, it depends only on (A, N) . Set $c_{A,N} := c_{\Psi^\pi}$. The lemma gives

$$\mathcal{H}_y(\varphi) \rightarrow c_{A,N} \int_{\mathbb{S}^{N-1}} \varphi(\theta) \bar{\sigma}_N(d\theta) \quad \text{as } y \rightarrow \infty. \quad (7.33)$$

Step 4: The directional and radial asymptotics. The convergence in (7.33) means that $\mathcal{H}_y$ converges weakly to $c_{A,N} \bar{\sigma}_N$. By the Portmanteau theorem, $\mathcal{H}_y(B) \rightarrow c_{A,N} \bar{\sigma}_N(B)$ for every Borel set B with $\bar{\sigma}_N(\partial B) = 0$. Taking $y = -\log s$ in (7.30) and using that $\pi_{A,N}$ charges no sphere gives

$$s^{-\beta_{A,N}} \pi_{A,N}(0 < \|x\|_2 \leq s, \hat{x} \in B) = \mathcal{H}_{-\log s}(B) \longrightarrow c_{A,N} \bar{\sigma}_N(B), \quad s \downarrow 0.$$

This is the directional asymptotic. Taking $B = \mathbb{S}^{N-1}$, which has empty boundary, gives the radial asymptotic. $\square$

8. Numerical illustrations

The figures compare the theoretical predictions with numerical calculations. We evaluate V_A and $V_{A,\varrho}$ by Gauss–Hermite quadrature or Poisson-summation formulas. These curves contain no sampling error. The chains $R^{(N)}$, $Z^{(N)}$, and $\mathcal{H}^{(N)}$ are simulated with fixed random seeds. The dashed curves in the supercritical cutoff plot are heuristic finite- N Gaussian approximations included for comparison. The scripts are available from the author upon request.

The map V_A .

As $N \rightarrow \infty$, the iterates of $V_A(q) = \mathbb{E} \tanh^2(A\sqrt{q}G)$ describe the deterministic limit of $Z^{(N)}$. Figure 8.1 shows the transition at $A = 1$ and, in the supercritical regime, convergence to the unique positive fixed point q_* . Its multiplier $\mu_A = V'_A(q_*) \in (0, 1)$ determines the cutoff scale in dimension.

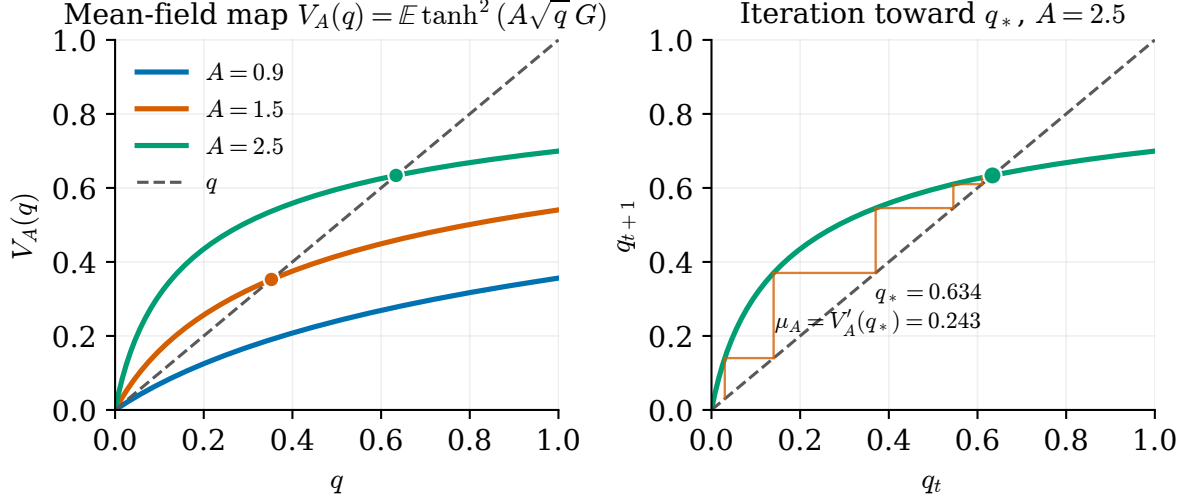


Figure 8.1: The map V_A . Left: V_A and the diagonal for three gains, showing the positive fixed point for $A > 1$. Right: iteration of V_A toward q_* for $A = 2.5$.

Supercritical dimension cutoff. Figure 8.2 shows the empirical total variation distance $\|K_{A,N}^t(q_0, \cdot) - \nu_{A,N}\|_{\text{TV}}$ of $Z^{(N)}$ to its nonzero invariant law, plotted against the centered time $t - t_N(q_0)$ for a range of widths N . The curves stay near one before the cutoff time and drop toward zero afterwards, consistently with Theorem 1.8.

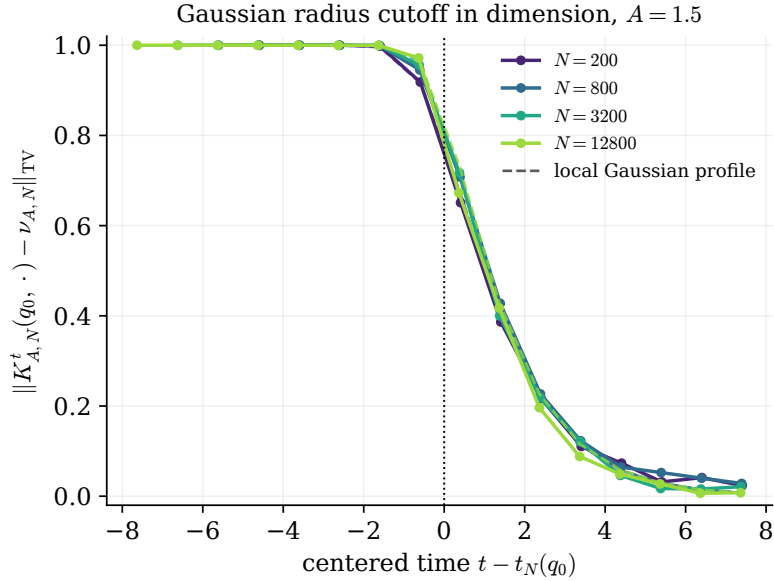


Figure 8.2: Numerical supercritical dimension cutoff for $Z^{(N)}$, with $A = 1.5$ and $q_0 = 0.05$. The solid curves are histogram estimates of the total variation distance to $\nu_{A,N}$ from 4,000 simulated paths. The dashed curves are heuristic local Gaussian approximations.

Invariant law and power-law singularity. In the supercritical regime, $X^{(N)}$ has a nonzero invariant law characterized by Proposition 7.1. Figure 8.3 compares a long run of $R^{(N)}$ with a numerical solution of a discretized invariant equation. For $N = 1$, the transition probabilities are computed from the exact conditional distribution on a mesh refined near zero. The logarithmic lower-tail plot also shows a reference line with the Cramér slope $\beta_{A,N}$ from Theorem 1.10.

Near-origin stationary density, $N = 1$, $A = 2.5$

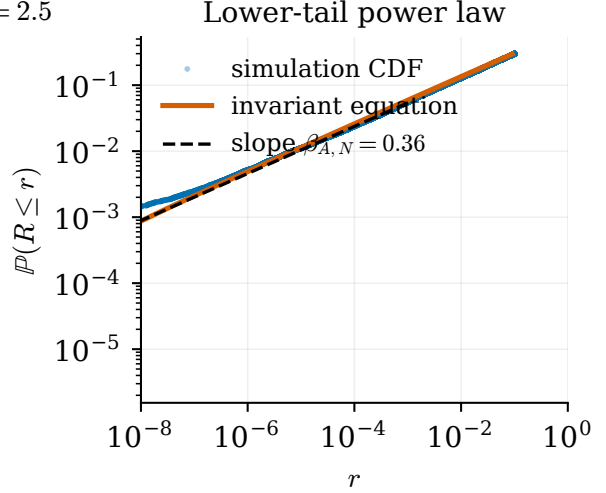
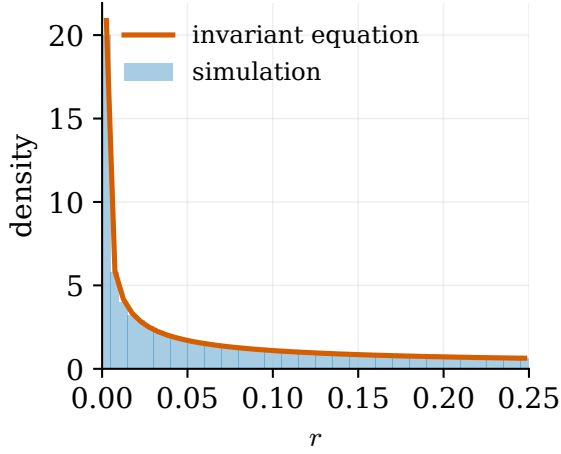


Figure 8.3: Supercritical stationary Euclidean radius for $N = 1$ and $A = 2.5$. Left: the near-origin density from simulation and a numerical solution of the invariant equation. The density increases toward zero because $\beta_{A,1} < 1$. Right: the empirical and numerical lower tails and a reference line of slope $\beta_{A,1}$. The observed slope is consistent with Theorem 1.10.

Vanishing-mesh cutoff at fixed width. At fixed width and subcritical gain, (2.5) identifies the survival probability with the total variation distance to δ_0 . Figure 8.4 plots this probability on the theorem scale, with center $L_\varrho/\gamma_{A,N}$ and window $\sigma_N \gamma_{A,N}^{-3/2} \sqrt{L_\varrho}$. As ϱ decreases, the curves approach the Gaussian first-passage profile $\Phi_G(-a)$ of Theorem 1.4 and Proposition 3.2. For the smallest displayed ϱ , the figure also includes the random-walk approximation, which shows the lattice correction.

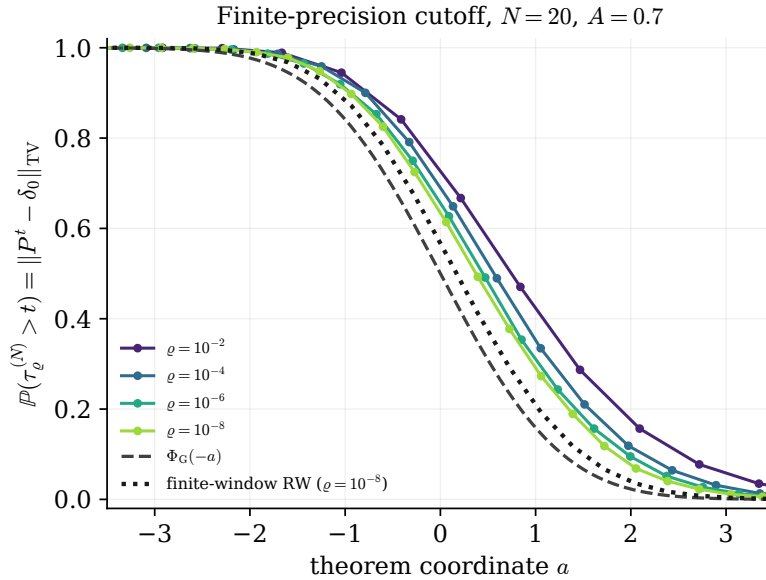


Figure 8.4: Vanishing-mesh cutoff at fixed width, with $N = 20$ and $A = 0.7$. The survival probability, equal to the total variation distance to δ_0 , is plotted using the center and window in Theorem 1.4. The dashed curve is $\Phi_G(-a)$. The dotted curve is the random-walk approximation at $\varrho = 10^{-8}$.

Subcritical cutoff as $N \rightarrow \infty$. Fix the mesh size and let the dimension grow with $A < A_{\text{lat}}$. For the displayed parameters, $A = 0.7 < A_{\text{lat}} \approx 0.8663$ and $\varrho = 0.5$. By Theorem 4.5, the

survival probability remains near one at time $\bar{t}_N - 1$ and is near zero by time $\bar{t}_N + 2$, where $\bar{t}_N$ is the first time at which $(\bar{h}_{t,N})_{t \geq 0}$ enters the scale $(\log N)^{-1}$. Figure 8.5 plots the empirical survival probability against the centered integer time $t - \bar{t}_N$. For this figure, we sample each step of $\mathcal{H}^{(N)}$ exactly.

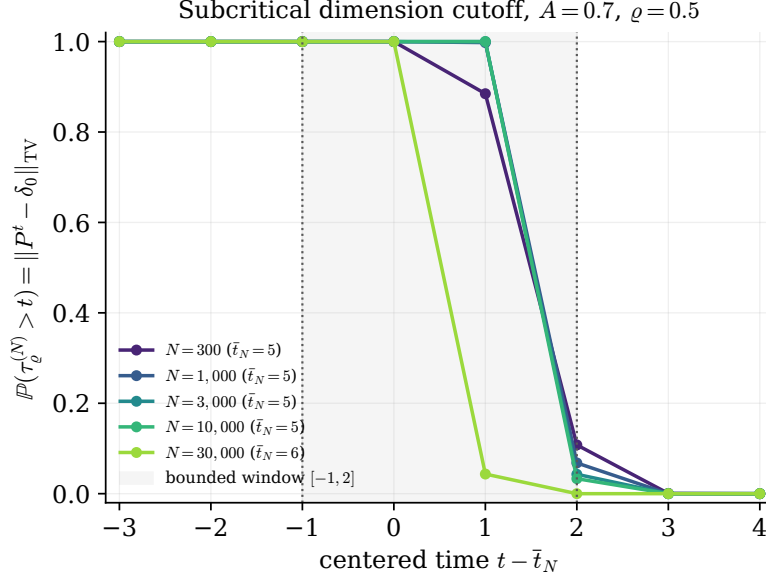


Figure 8.5: Fixed-precision subcritical dimension cutoff ($A = 0.7$, $\varrho = 0.5$, and $x_N = (1, \dots, 1)$, with 40,000 paths per dimension). The empirical survival probability $\mathbb{P}(\tau_\varrho^{(N)} > t) = \|P_{\varrho,A,N}^t(Q_\varrho x_N, \cdot) - \delta_0\|_{\text{TV}}$ is plotted against $t - \bar{t}_N$. The dotted lines mark the lower and upper time bounds in Theorem 4.5, namely $\bar{t}_N - 1$ and $\bar{t}_N + 2$.

Metastability at fixed precision. When the rightmost positive-drift component (h_-, h_+) exists, $\mathcal{H}^{(N)}$ approaches h_+ and remains nearby until a rare fluctuation triggers absorption. See Proposition 5.1. Figure 8.6 shows $V_{A,\varrho}$, sample paths exhibiting the two time scales, and the growth of the absorption time with width.

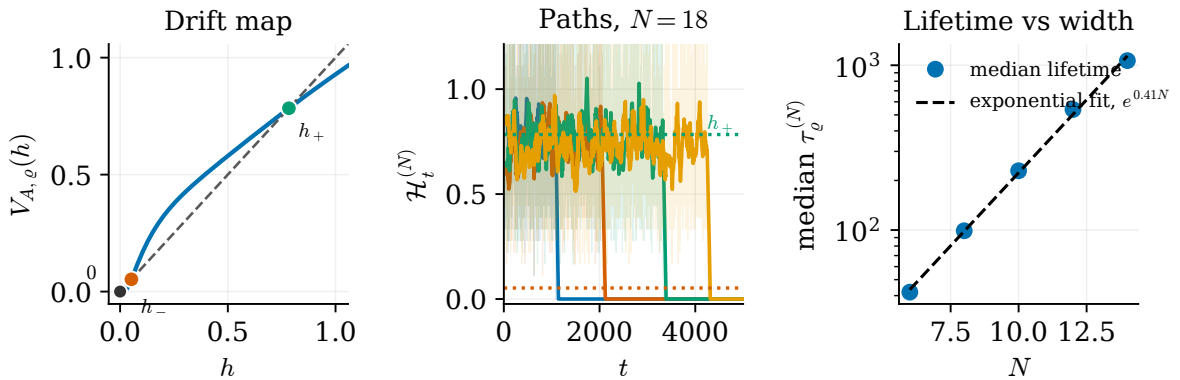


Figure 8.6: Metastability for $A = 1.15$ and $\varrho = 0.5$. Left: the map $V_{A,\varrho}$ with unstable endpoint h_- and stable endpoint h_+ . Middle: representative paths and moving averages with window 60. Right: median absorption times and an exponential fit. The observed growth is consistent with Theorem 5.2.

Critical curves. Figure 8.7 compares $A_c(N)$ from (2.26), which decreases to 1, with A_{lat} and $A_{\text{ex}}(\varrho)$.

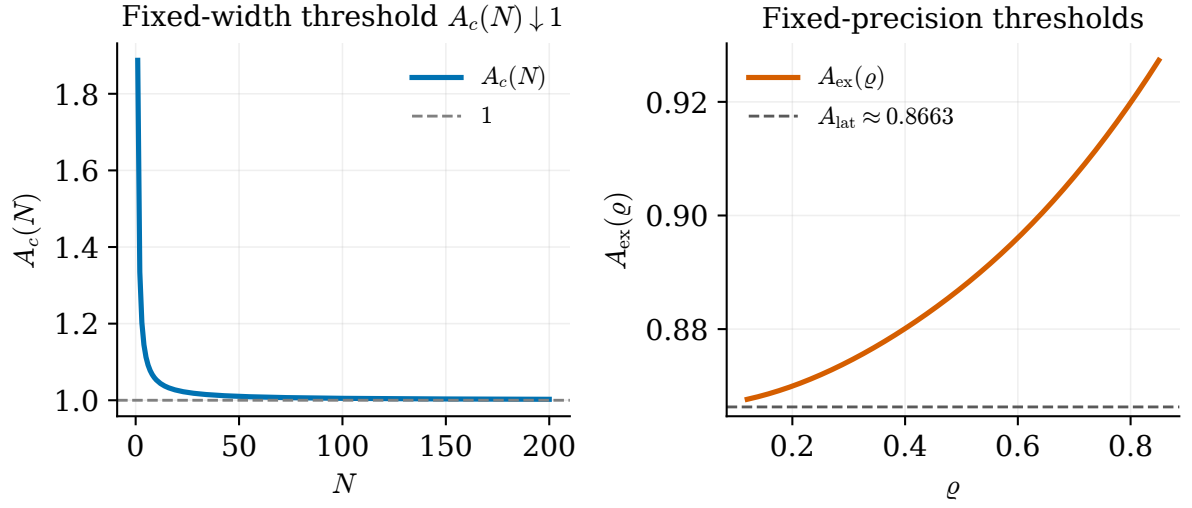


Figure 8.7: Thresholds. Left: $A_c(N)$ decreases to 1. Right: $A_{\text{ex}}(\varrho)$, evaluated numerically from its variational definition, and A_{lat} .

A. Proofs of auxiliary probabilistic results

This appendix contains proofs of Lemmas 2.2 and 2.3 and Propositions 2.5 and 2.6.

A.1. Proof of Lemma 2.2.

Proof. Put $T := T_N$ and suppress the index N on $\mathcal{R}_t$, A_t , η_t , α_t , and σ_t throughout the proof.

Step 1: The moment before τ_N . Fix $0 \leq t < T$. Since τ_N is a stopping time, $\{\tau_N > t\} \in \mathcal{F}_t$. Moreover, $\{\tau_N > t+1\} \subseteq \{\tau_N > t\}$, and on the latter event the recursion (2.33) applies. Consequently,

$$\mathcal{R}_{t+1}^2 \mathbb{1}_{\{\tau_N > t+1\}} \leq \mathbb{1}_{\{\tau_N > t\}} (A_t \mathcal{R}_t + \eta_{t+1})^2.$$

The variables $\mathbb{1}_{\{\tau_N > t\}}$, A_t , and $\mathcal{R}_t$ are $\mathcal{F}_t$ -measurable. Thus the conditional cross term vanishes by (2.34), and

$$\mathbb{E}[\mathcal{R}_{t+1}^2 \mathbb{1}_{\{\tau_N > t+1\}} \mid \mathcal{F}_t] \leq \mathbb{1}_{\{\tau_N > t\}} (A_t^2 \mathcal{R}_t^2 + \mathbb{E}[\eta_{t+1}^2 \mid \mathcal{F}_t]) \leq \mathbb{1}_{\{\tau_N > t\}} \left(\alpha_t^2 \mathcal{R}_t^2 + \frac{C\sigma_t^2}{N} \right).$$

Set $m_t := \mathbb{E}[\mathcal{R}_t^2 \mathbb{1}_{\{\tau_N > t\}}]$. Taking expectations in the preceding display and using $\mathbb{P}(\tau_N > t) \leq 1$ gives

$$m_{t+1} \leq \alpha_t^2 m_t + \frac{C\sigma_t^2}{N}.$$

Iterating this deterministic recursion, and using $m_0 \leq \mathbb{E}\mathcal{R}_0^2$, yields

$$m_t \leq \left(\prod_{u=0}^{t-1} \alpha_u^2 \right) \mathbb{E}\mathcal{R}_0^2 + \frac{C}{N} \sum_{s=0}^{t-1} \sigma_s^2 \prod_{u=s+1}^{t-1} \alpha_u^2.$$

This is (2.37).

Step 2: The moment of the stopped process. Define the deterministic backward weights

$$W_t := \prod_{u=t}^{T-1} (1 \vee \alpha_u^2), \quad 0 \leq t \leq T.$$

These weights satisfy

$$W_T = 1, \quad W_{t+1} \leq W_t, \quad \alpha_t^2 W_{t+1} \leq W_t.$$

In addition, the definition of K_N gives

$$W_0^{1/2} = \prod_{u=0}^{T-1} (1 \vee \alpha_u) \leq K_N, \quad \sigma_t W_{t+1}^{1/2} = \sigma_t \prod_{u=t+1}^{T-1} (1 \vee \alpha_u) \leq K_N.$$

For the one-step weighted estimate, note that on $\{\tau_N \leq t\}$ the stopped process is constant, whereas on $\{\tau_N > t\}$ the recursion applies. Hence

$$\overline{\mathcal{R}}_{t+1,N} = \mathbb{1}_{\{\tau_N \leq t\}} \overline{\mathcal{R}}_{t,N} + \mathbb{1}_{\{\tau_N > t\}} (A_t \mathcal{R}_t + \eta_{t+1}).$$

The two events in this decomposition are disjoint. Taking conditional expectations, using (2.34), and applying the preceding bounds therefore gives

$$\begin{aligned} & \mathbb{E} \left[W_{t+1} \overline{\mathcal{R}}_{t+1,N}^2 \mid \mathcal{F}_t \right] \\ &= \mathbb{1}_{\{\tau_N \leq t\}} W_{t+1} \overline{\mathcal{R}}_{t,N}^2 + \mathbb{1}_{\{\tau_N > t\}} W_{t+1} (A_t^2 \mathcal{R}_t^2 + \mathbb{E}[\eta_{t+1}^2 \mid \mathcal{F}_t]) \\ &\leq \mathbb{1}_{\{\tau_N \leq t\}} W_t \overline{\mathcal{R}}_{t,N}^2 + \mathbb{1}_{\{\tau_N > t\}} W_t \mathcal{R}_t^2 + \frac{CK_N^2}{N} \\ &= W_t \overline{\mathcal{R}}_{t,N}^2 + \frac{CK_N^2}{N}. \end{aligned}$$

Taking expectations and using the tower property gives, for every $0 \leq t < T$,

$$\mathbb{E}[W_{t+1}\overline{\mathcal{R}}_{t+1,N}^2] - \mathbb{E}[W_t\overline{\mathcal{R}}_{t,N}^2] \leq \frac{CK_N^2}{N}.$$

Summing from $t = 0$ to $T - 1$, the left-hand side telescopes, and we obtain

$$\mathbb{E}[W_T\overline{\mathcal{R}}_{T,N}^2] - \mathbb{E}[W_0\overline{\mathcal{R}}_{0,N}^2] \leq \frac{CK_N^2T}{N}.$$

Since $W_T = 1$, W_0 is deterministic, and $\overline{\mathcal{R}}_{0,N} = \mathcal{R}_{0,N}$, this is equivalent to

$$\mathbb{E}\overline{\mathcal{R}}_{T,N}^2 \leq W_0\mathbb{E}\mathcal{R}_{0,N}^2 + \frac{CK_N^2T}{N}.$$

Since $W_0 \leq K_N^2$, this proves (2.36).

Step 3: The probability of exit. Suppose that τ_N is the first time at which $|\mathcal{R}_{t,N}| > \delta$. On $\{\tau_N \leq T_N\}$ we have $T_N \wedge \tau_N = \tau_N$. Therefore, by the definition of the stopped process and the strict inequality in the definition of τ_N ,

$$\overline{\mathcal{R}}_{T_N,N} = \mathcal{R}_{\tau_N,N}, \quad |\mathcal{R}_{\tau_N,N}| > \delta.$$

Hence

$$\delta^2 \mathbf{1}_{\{\tau_N \leq T_N\}} \leq \overline{\mathcal{R}}_{T_N,N}^2.$$

Taking expectations and applying (2.36), we obtain

$$\begin{aligned} \delta^2 \mathbb{P}(\tau_N \leq T_N) &= \delta^2 \mathbb{E} \mathbf{1}_{\{\tau_N \leq T_N\}} \\ &\leq \mathbb{E} \overline{\mathcal{R}}_{T_N,N}^2 \\ &\leq CK_N^2 \left(\mathbb{E}\mathcal{R}_{0,N}^2 + \frac{T_N}{N} \right). \end{aligned}$$

Dividing by δ^2 proves (2.38) and completes the proof. $\square$

A.2. Proof of Lemma 2.3.

Proof. Step 1: The recursion before T . By our assumptions we can choose $\varepsilon > 0$ so small that $\mu_\varepsilon := \mathbb{E} \log(M_1 + \varepsilon) < 0$. Take $K_* \geq b/\varepsilon$. Whenever $U_t \geq K_*$,

$$U_{t+1} = M_{t+1}U_t + b \leq M_{t+1}U_t + \varepsilon K_* \leq (M_{t+1} + \varepsilon)U_t.$$

Let $T := \inf\{t : U_t \leq K_*\}$. On the event $\{T > n\}$ we have $U_t > K_*$ for every $t \leq n$, so the previous bound applies at each step and

$$\log U_n \leq \log K + \tilde{S}_n, \quad \tilde{S}_n := \sum_{j=1}^n \log(M_j + \varepsilon),$$

a random walk with increment mean $\mu_\varepsilon < 0$. Since $U_n > K_*$ on $\{T > n\}$, this forces $\tilde{S}_n > -\log(K/K_*)$.

Step 2: The tail of T . By hypothesis $\log(M_1 + \varepsilon)$ has exponential moments near the origin, so $\Lambda(s) := \log \mathbb{E}(M_1 + \varepsilon)^s$ is finite for s in a neighborhood of 0, with $\Lambda(0) = 0$ and $\Lambda'(0) = \mu_\varepsilon < 0$. Fix $s > 0$ small enough that $\Lambda(s) < 0$ and put $c_3 := -\Lambda(s) > 0$. Chernoff's inequality gives, for every integer $n \geq 1$,

$$\mathbb{P}(T > n) \leq \mathbb{P}(\tilde{S}_n > -\log(K/K_*)) \leq \left(\frac{K}{K_*}\right)^s (\mathbb{E}(M_1 + \varepsilon)^s)^n = \left(\frac{K}{K_*}\right)^s e^{-c_3 n}.$$

Since T is integer valued, $\mathbb{P}(T > c_1 \log K + r) = \mathbb{P}(T > \lfloor c_1 \log K + r \rfloor)$, and using $\lfloor c_1 \log K + r \rfloor > c_1 \log K + r - 1$ with the choice $c_1 := s/c_3$,

$$\mathbb{P}(T > c_1 \log K + r) \leq \left(\frac{K}{K_*}\right)^s e^{c_3} e^{-c_3(c_1 \log K + r)} = K_*^{-s} e^{c_3} K^{s-c_3 c_1} e^{-c_3 r} = K_*^{-s} e^{c_3} e^{-c_3 r},$$

because $s - c_3 c_1 = 0$. This is (2.39) with $c_2 := \max\{K_*^{-s} e^{c_3}, 1\}$ and completes the proof. $\square$

A.3. Proof of Proposition 2.5.

Proof. We use the Krylov–Bogolyubov averaging argument. Compare with [KB37].

Step 1: $K_{A,N}$ is Feller. The space $[0, 1]$ is compact, and hence the set of probability measures on $[0, 1]$ is weakly compact. The transition kernel is Feller: if ϕ is continuous and bounded on $[0, 1]$, then

$$K_{A,N}\phi(q) = \int_{\mathbb{R}^N} \phi(F_{A,N}(q, g)) \mathcal{G}_N(dg)$$

is continuous in q , by the continuity of $F_{A,N}$ and dominated convergence.

Step 2: A Cesàro limit is invariant. Let $\bar{\nu}_{T_j}^q \Rightarrow \bar{\nu}$. For every continuous ϕ ,

$$\bar{\nu}_{T_j}^q K_{A,N}\phi - \bar{\nu}_{T_j}^q \phi = \frac{1}{T_j} \left(K_{A,N}^{T_j} \phi(q) - \phi(q) \right).$$

The right-hand side tends to zero. Passing to the limit, using the Feller property for $K_{A,N}\phi$, gives $\bar{\nu} K_{A,N}\phi = \bar{\nu}\phi$ for every continuous ϕ . Since probability measures on $[0, 1]$ are determined by their integrals against continuous functions, $\bar{\nu}$ is invariant and satisfies (2.44).

Step 3: ν^+ is invariant on $(0, 1]$. Since $K_{A,N}(0, \cdot) = \delta_0$ and $K_{A,N}(s, \{0\}) = 0$ for $s > 0$ (all Gaussian coordinates would have to vanish), an invariant law $\bar{\nu} \neq \delta_0$ decomposes as

$$\bar{\nu} = \bar{\nu}(\{0\})\delta_0 + (1 - \bar{\nu}(\{0\}))\nu^+,$$

with ν^+ invariant on $(0, 1]$. The assertion for the vector law is then the invariant-law conclusion of Proposition 2.4. The stationary equation for $X^{(N)}$ is (2.47). $\square$

A.4. Proof of Proposition 2.6.

Proof. Step 1: Negative moments of S_N . Let $G_1, \dots, G_N$ be independent standard Gaussians and put

$$S_N := \frac{1}{N} \sum_{i=1}^N \min(G_i^2, 1).$$

If $0 < u < 1/N$, then $S_N \leq u$ implies $|G_i| \leq \sqrt{Nu}$ for every i . Hence

$$\mathbb{P}(S_N \leq u) \leq \prod_{i=1}^N \mathbb{P}(|G_i| \leq \sqrt{Nu}) \leq C_N u^{N/2}.$$

Consequently, $\mathbb{E}S_N^{-p} < \infty$ for every $p < N/2$.

Step 2: Contraction of r^{-p} near zero. Since $A > A_c(N)$,

$$\frac{d}{dp} \mathbb{E} \left[\left(A^2 \frac{\chi_N^2}{N} \right)^{-p} \right]_{p=0} = -\mathbb{E} \log \left(A^2 \frac{\chi_N^2}{N} \right) = 2\gamma_{A,N} < 0,$$

where $\ell_{A,N}$ is as in (2.25), and $\gamma_{A,N} = -\mathbb{E} \log \ell_{A,N} < 0$ in the present supercritical regime. Therefore there exists $p \in (0, N/2)$ such that

$$\mathbb{E} \left[\left(A^2 \frac{\chi_N^2}{N} \right)^{-p} \right] < 1. \tag{A.1}$$

We claim that, for some $R_0 \in (0, 1)$ and $a < 1$,

$$\sup_{0 < r \leq R_0} \mathbb{E} \left[\left(\frac{F_{A,N}(r, G)}{r} \right)^{-p} \right] \leq a. \tag{A.2}$$

Indeed, for fixed g the quotient $r^{-1} \tanh^2(A\sqrt{r}g)$ decreases in r and converges to A^2g^2 as $r \downarrow 0$. Therefore $F_{A,N}(r, G)/r$ converges almost surely to $A^2\chi_N^2/N$. Moreover, for each fixed $R > 0$ and $0 < q \leq R$, the same monotonicity gives

$$\frac{F_{A,N}(q, G)}{q} \geq \frac{F_{A,N}(R, G)}{R}.$$

The random variable on the right is bounded below by $c_{A,R,N}S_N$. The estimate from Step 1 gives $\mathbb{E}S_N^{-p} < \infty$. Dominated convergence therefore gives the convergence of the expectations, and (A.2) follows from (A.1) by taking R_0 small.

Step 3: A bound away from zero. There is $B < \infty$ such that

$$\sup_{r \in [R_0, 1]} \mathbb{E}F_{A,N}(r, G)^{-p} \leq B. \quad (\text{A.3})$$

This follows from the pointwise lower bound

$$\tanh^2(A\sqrt{r}g) \geq c_{A,R_0} \min(g^2, 1), \quad r \in [R_0, 1],$$

together with the estimate for S_N from Step 1.

Let $V(r) := r^{-p}$ for $r > 0$. Combining the two estimates gives

$$K_{A,N}V(r) \leq aV(r) + B, \quad r > 0. \quad (\text{A.4})$$

Step 4: The Cesàro limit does not charge zero. Let $Z^{(N)}$ start from $q \in (0, 1]$. Iterating (A.4) gives

$$\frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E}V(Z_t^{(N)}) \leq \frac{V(q)}{T(1-a)} + \frac{B}{1-a}.$$

Choose a weakly convergent Cesàro subsequence $\bar{\nu}_{T_j}^q \Rightarrow \bar{\nu}$. For $M < \infty$, the function $V^M(r) := \min\{r^{-p}, M\}$ for $r > 0$ and $V^M(0) := M$ is continuous on $[0, 1]$. Passing to the limit gives a bound independent of M . Since $M\bar{\nu}(\{0\}) \leq \int V^M d\bar{\nu}$, letting $M \rightarrow \infty$ first gives $\bar{\nu}(\{0\}) = 0$. Monotone convergence on $(0, 1]$ then gives

$$\int_{(0,1]} r^{-p} \bar{\nu}(dr) < \infty, \quad \bar{\nu}(\{0\}) = 0.$$

Thus $\bar{\nu} \neq \delta_0$, and Proposition 2.5 shows that $\bar{\nu} \in \mathcal{I}_{A,N}^+(q)$. The vector invariant law follows from (2.46). $\square$

B. Proofs of estimates for supercritical cutoff as $N \rightarrow \infty$

This appendix contains the proofs of the estimates used in Section 6.3, with the notation introduced there for $Z^{(N)}$ and $(q_{t,N})_{t \geq 0}$.

B.1. Proofs of Lemmas 6.7 and 6.9 and Propositions 6.8 and 6.11.

Proof of Lemma 6.7. Step 1: Negative moments of the truncated average. Let

$$U_i := \min(G_i^2, 1), \quad S_N := \frac{1}{N} \sum_{i=1}^N U_i.$$

For $\lambda \geq 1$,

$$\mathbb{E}e^{-\lambda U_1} \leq \int_{-1}^1 e^{-\lambda x^2} \frac{e^{-x^2/2}}{\sqrt{2\pi}} dx + e^{-\lambda} \leq C\lambda^{-1/2}.$$

Chernoff's bound with $\lambda = u^{-1}$ gives

$$\mathbb{P}(S_N \leq u) \leq e^N \left(\mathbb{E} e^{-U_1/u} \right)^N \leq C^N u^{N/2},$$

for $0 < u \leq 1$.

For $x, p > 0$, the Gamma integral and the change of variables $v = ux$ give

$$x^{-p} = \frac{1}{\Gamma(p)} \int_0^\infty u^{p-1} e^{-ux} du.$$

Let $p = \alpha N$ with $\alpha \in (0, 1/2)$. By Tonelli's theorem and independence,

$$\mathbb{E} S_N^{-p} = \frac{1}{\Gamma(p)} \int_0^\infty u^{p-1} \left(\mathbb{E} e^{-(u/N)U_1} \right)^N du.$$

We split the integral at $u = N$. On $[0, N]$ we use the trivial bound $\mathbb{E} e^{-(u/N)U_1} \leq 1$. On $[N, \infty)$ we use the preceding one-coordinate Laplace bound with $\lambda = u/N \geq 1$. Since $p = \alpha N < N/2$, this gives

$$\begin{aligned} \mathbb{E} S_N^{-p} &\leq \frac{N^p}{\Gamma(p+1)} + \frac{C^N N^{N/2}}{\Gamma(p)} \int_N^\infty u^{p-1-N/2} du \\ &\leq C_\alpha^N \frac{N^p}{\Gamma(p+1)}. \end{aligned}$$

Stirling's formula, after increasing the constant to handle bounded N , shows that the last expression is bounded by $e^{C_\alpha N}$. If $W_{L,i} := \min(G_i^2, L^2)$ and $L \geq 1$, then $W_{L,i} \geq U_i$. Consequently,

$$\sup_{L \geq 1} \mathbb{E} \left[\left(\frac{1}{N} \sum_{i=1}^N W_{L,i} \right)^{-\alpha N} \right] \leq e^{C_\alpha N}. \quad (\text{B.1})$$

Step 2: Choice of L , ε , and δ . Put $\mu_L := \mathbb{E} W_{L,1}$. Choose first L large, then $\varepsilon > 0$ small enough that $(1 - \varepsilon)A^2\mu_L > 1$, and finally $\delta > 0$ such that

$$a := (1 - \varepsilon)A^2(\mu_L - \delta) > 1.$$

If $\overline{W}_{L,N} := N^{-1} \sum_i W_{L,i}$, Hoeffding's inequality gives

$$\mathbb{P}(\overline{W}_{L,N} < \mu_L - \delta) \leq e^{-c_0 N}$$

with $c_0 > 0$ depending on L and δ .

Step 3: Splitting at $\overline{W}_{L,N} = \mu_L - \delta$. Fix $\alpha \in (0, 1/2)$. By (B.1), after increasing C_α to absorb the fixed factor $((1 - \varepsilon)A^2)^{-\alpha N}$,

$$\mathbb{E} \left[((1 - \varepsilon)A^2 \overline{W}_{L,N})^{-\alpha N} \right] \leq e^{C_\alpha N}.$$

On the event $\{\overline{W}_{L,N} \geq \mu_L - \delta\}$, the integrand in (6.7) is at most $a^{-\gamma N}$. On the complementary event, Hölder's inequality gives, for $0 < \gamma < \alpha$,

$$\mathbb{E} \left[((1 - \varepsilon)A^2 \overline{W}_{L,N})^{-\gamma N} \mathbf{1}_{\{\overline{W}_{L,N} < \mu_L - \delta\}} \right] \leq e^{-c_0 N(1 - \gamma/\alpha)} e^{C_\alpha N \gamma/\alpha}.$$

Choose $\gamma > 0$ so small that $\gamma < \alpha c_0/(c_0 + C_\alpha)$. Then choose $\kappa > 0$ such that

$$\kappa < \min\{\gamma \log a, c_0(1 - \gamma/\alpha) - C_\alpha \gamma/\alpha\}.$$

The good- and bad-event estimates then give (6.7) and complete the proof. $\square$

Proof of Proposition 6.8. Let $\gamma, \kappa, L, \varepsilon$ be as in Lemma 6.7, decreasing κ if necessary. For $q > 0$ define

$$h_q(g) := \frac{\tanh^2(A\sqrt{q}g)}{q} = A^2 g^2 \left(\frac{\tanh(A\sqrt{q}g)}{A\sqrt{q}g} \right)^2,$$

with the continuous interpretation at $g = 0$. Since $x \mapsto \tanh x/x$ is decreasing on $(0, \infty)$, the map $q \mapsto h_q(g)$ is decreasing for each fixed g .

Choose $R_0 \in (0, \inf I_*)$ so small that

$$\left(\frac{\tanh(A\sqrt{R_0}u)}{A\sqrt{R_0}u} \right)^2 \geq 1 - \varepsilon \quad \text{for every } |u| \leq L. \quad (\text{B.2})$$

For $|g| \leq L$, (B.2) gives

$$h_{R_0}(g) \geq (1 - \varepsilon)A^2 g^2.$$

For $|g| > L$, the map $|g| \mapsto h_{R_0}(g) = R_0^{-1} \tanh^2(A\sqrt{R_0}g)$ is increasing, and hence

$$h_{R_0}(g) \geq h_{R_0}(L) \geq (1 - \varepsilon)A^2 L^2.$$

Combining the two cases gives

$$h_{R_0}(g) \geq (1 - \varepsilon)A^2 \min(g^2, L^2) \quad \text{for every } g \in \mathbb{R}. \quad (\text{B.3})$$

Hence, for $0 < q \leq R_0$,

$$\frac{F_{A,N}(q, G)}{q} = \frac{1}{N} \sum_{i=1}^N h_q(G_i) \geq \frac{1}{N} \sum_{i=1}^N h_{R_0}(G_i) \geq (1 - \varepsilon)A^2 \frac{1}{N} \sum_{i=1}^N \min(G_i^2, L^2).$$

Since $x \mapsto x^{-\gamma N}$ is decreasing on $(0, \infty)$, applying Lemma 6.7 gives

$$\sup_{0 < q \leq R_0} \mathbb{E} \left[\left(\frac{F_{A,N}(q, G)}{q} \right)^{-\gamma N} \right] \leq e^{-\kappa N}$$

for all sufficiently large N . □

Proof of Lemma 6.9. There is $c_{A,R_0} > 0$ such that, for every $q \in [R_0, 1]$ and every $g \in \mathbb{R}$,

$$\tanh^2(A\sqrt{q}g) \geq c_{A,R_0} \min(g^2, 1). \quad (\text{B.4})$$

Indeed, for $|g| \leq 1$ this follows from the monotonicity of $\tanh x/x$ on $[0, A]$, while for $|g| \geq 1$ it follows from $\tanh^2(A\sqrt{q}|g|) \geq \tanh^2(A\sqrt{R_0})$. Thus

$$F_{A,N}(q, G) \geq c_{A,R_0} \frac{1}{N} \sum_{i=1}^N \min(G_i^2, 1).$$

The bound follows from (B.1), after absorbing $c_{A,R_0}^{-\gamma N}$ into e^{bN} . □

Proof of Proposition 6.11. Step 1: The bound for $K_{A,N}V_N$. Let R_0, γ, κ be as in Proposition 6.8, and let b be the constant in Lemma 6.9 for this choice of R_0, γ . Define

$$V_N(q) := q^{-\gamma N}, \quad q > 0.$$

For $q > 0$, the two negative-moment estimates give the Foster inequality

$$K_{A,N}V_N(q) \leq e^{-\kappa N}V_N(q) + e^{bN}. \quad (\text{B.5})$$

Indeed, if $q \leq R_0$ then

$$K_{A,N}V_N(q) = q^{-\gamma N} \mathbb{E} \left[(F_{A,N}(q, G)/q)^{-\gamma N} \right] \leq e^{-\kappa N} V_N(q),$$

while for $q \in [R_0, 1]$ the outside estimate gives $K_{A,N}V_N(q) \leq e^{bN}$.

Step 2: The bound under $\nu_{A,N}$. Let $(Z_t^{(N)})_{t \geq 0}$ start from 1. Iterating (B.5) yields

$$\mathbb{E}V_N(Z_t^{(N)}) \leq e^{-\kappa N t} + \frac{e^{bN}}{1 - e^{-\kappa N}}. \quad (\text{B.6})$$

Hence

$$\int V_N(q) \bar{\nu}_T^1(dq) \leq \frac{1}{T(1 - e^{-\kappa N})} + \frac{e^{bN}}{1 - e^{-\kappa N}}. \quad (\text{B.7})$$

Here $\bar{\nu}_T^1$ is the Cesàro average started from $q = 1$ in (2.45).

Take a weakly convergent subsequence $\bar{\nu}_{T_j}^1 \Rightarrow \bar{\nu}$ with $T_j \rightarrow \infty$. For $M < \infty$ put $V_N^M(q) := \min\{V_N(q), M\}$ for $q > 0$ and $V_N^M(0) := M$. Then V_N^M is continuous on $[0, 1]$. Applying (B.7) with $T = T_j$ and then letting $j \rightarrow \infty$, the term containing T_j^{-1} vanishes and

$$\int V_N^M(q) \bar{\nu}(dq) \leq \frac{e^{bN}}{1 - e^{-\kappa N}}.$$

Letting $M \rightarrow \infty$ and using monotone convergence, we obtain $\int_{(0,1]} q^{-\gamma N} \bar{\nu}(dq) < \infty$ and $\bar{\nu}(\{0\}) = 0$. By Proposition 2.5, $\bar{\nu}$ is a nonzero invariant law. By Proposition 6.1, $\bar{\nu} = \nu_{A,N}$. The same monotone-convergence passage gives (6.10). $\square$

B.2. Proofs of Propositions 6.12 and 6.14 and Lemmas 6.13 and 6.15.

Proof of Proposition 6.12. Step 1: $\nu_{A,N}((0, r])$ is exponentially small. Let R_0, γ be as in Proposition 6.8, and let b be the constant from Lemma 6.9 for this choice of R_0, γ . Set

$$V_N(q) := q^{-\gamma N}, \quad q > 0.$$

By Proposition 6.11, after increasing C if necessary,

$$\int_0^1 V_N(q) \nu_{A,N}(dq) \leq C e^{bN}. \quad (\text{B.8})$$

Now choose a second threshold $r \in (0, R_0)$ so small that

$$b + \gamma \log r < 0. \quad (\text{B.9})$$

Since $V_N(q) \geq r^{-\gamma N}$ on $(0, r]$,

$$\nu_{A,N}((0, r]) \leq r^{\gamma N} \int_0^1 V_N(q) \nu_{A,N}(dq) \leq C e^{-cN}. \quad (\text{B.10})$$

Step 2: Entrance into I_ after m steps.* By Lemma 2.1, every orbit of V_A started from $[r, 1]$ converges monotonically to q_* without crossing it. Hence

$$|V_A^{t+1}(q) - q_*| \leq |V_A^t(q) - q_*|, \quad q \in [r, 1].$$

Dini's theorem shows that this convergence is uniform on $[r, 1]$. Therefore there are $m \in \mathbb{N}$ and $\eta > 0$ such that

$$V_A^m([r, 1]) \subset I_{*, -\eta} := \{q \in I_* : \text{dist}(q, I_*^c) \geq \eta\}. \quad (\text{B.11})$$

Let $L < \infty$ be a Lipschitz constant for V_A on $[0, 1]$ and set

$$\eta_m := \frac{\eta}{2(1 + L + \dots + L^{m-1})}.$$

If the noise variables in (2.14) have absolute value at most η_m during the first m steps, then the stochastic radius at time m is within $\eta/2$ of its deterministic counterpart and hence belongs to I_* . Hoeffding's inequality (2.16) and a union bound over these fixed m steps give

$$\sup_{q \in [r, 1]} K_{A,N}^m(q, I_*^c) \leq C e^{-cN}. \quad (\text{B.12})$$

Step 3: Invariance gives the bound on I_^c .* Since $\nu_{A,N}$ is invariant,

$$\nu_{A,N}(I_*^c) = \int_0^1 K_{A,N}^m(q, I_*^c) \nu_{A,N}(dq) \leq \nu_{A,N}((0, r]) + \sup_{q \in [r, 1]} K_{A,N}^m(q, I_*^c).$$

Combining (B.10) and (B.12) proves (6.11), and the weaker estimate (6.12) follows immediately. $\square$

Proof of Lemma 6.13. Step 1: The score of one coordinate. Let μ_q be the law of $Y_q := \tanh^2(A\sqrt{q}G)$ on $(0, 1)$, and let f_q be the density in (6.1). Its q -score is

$$\ell_q(y) := \partial_q \log f_q(y) = -\frac{1}{2q} + \frac{\operatorname{arctanh}(\sqrt{y})^2}{2A^2 q^2}. \quad (\text{B.13})$$

Since $\operatorname{arctanh}(\sqrt{Y_q}) = |A\sqrt{q}G|$, we have

$$\ell_q(Y_q) = \frac{G^2 - 1}{2q}, \quad \mathbb{E}\ell_q(Y_q) = 0, \quad \mathbb{E}\ell_q(Y_q)^2 = \frac{1}{2q^2}. \quad (\text{B.14})$$

Fix a compact subinterval $I = [q_-, q_+] \Subset (0, 1)$. On this interval the displayed density is a C^1 map into $L^1((0, 1))$: after writing $u = \operatorname{arctanh}(\sqrt{y})$, the substitution $y = \tanh^2 u$ gives a uniform integrable Gaussian envelope for $\partial_q f_q = \ell_q f_q$. Thus $q \mapsto \mu_q$ is differentiable in total variation on I , with signed derivative measure $\partial_q \mu_q = \ell_q \mu_q$. Here and below $|\lambda|$ denotes the total variation measure of a signed measure λ . With our convention for total variation distance between probability measures, one divides the signed variation norm by two.

Step 2: The score of N coordinates. Let $Y_{1,q}, \dots, Y_{N,q}$ be independent copies of Y_q and set

$$M_{N,q} := \frac{1}{N} \sum_{i=1}^N Y_{i,q}, \quad S_{N,q} := \sum_{i=1}^N \ell_q(Y_{i,q}).$$

Differentiating the product measure gives

$$\partial_q \mu_q^{\otimes N} = S_{N,q} \mu_q^{\otimes N}. \quad (\text{B.15})$$

Equivalently, for every bounded measurable $\varphi : (0, 1)^N \rightarrow \mathbb{R}$,

$$\partial_q \mathbb{E}\varphi(Y_{1,q}, \dots, Y_{N,q}) = \mathbb{E}[\varphi(Y_{1,q}, \dots, Y_{N,q}) S_{N,q}]. \quad (\text{B.16})$$

Since $S_{N,q}$ is the density of $\partial_q \mu_q^{\otimes N}$ against $\mu_q^{\otimes N}$, its total variation norm is

$$|\partial_q \mu_q^{\otimes N}|((0, 1)^N) = \mathbb{E}|S_{N,q}| \leq (\mathbb{E}S_{N,q}^2)^{1/2} = \frac{\sqrt{N}}{\sqrt{2}q}, \quad (\text{B.17})$$

where the last identity uses (B.14) and the independence of the $Y_{i,q}$.

Step 3: Passing to $K_{A,N}$. The kernel $K_{A,N}(q, \cdot)$ is the law of $M_{N,q}$ and hence the pushforward of $\mu_q^{\otimes N}$ under the map $(y_1, \dots, y_N) \mapsto N^{-1} \sum_i y_i$. Pushforward does not increase the total variation distance. Since $q \mapsto \mu_q^{\otimes N}$ is continuously differentiable in total variation,

$$\mu_q^{\otimes N} - \mu_{q'}^{\otimes N} = \int_{q'}^q \partial_s \mu_s^{\otimes N} ds$$

as signed measures. Integrating (B.17) and dividing the signed variation by two gives

$$\|K_{A,N}(q, \cdot) - K_{A,N}(q', \cdot)\|_{\text{TV}} \leq \frac{1}{2} \left| \mu_q^{\otimes N} - \mu_{q'}^{\otimes N} \right|((0, 1)^N) \leq \frac{1}{2} \int_{q \wedge q'}^{q \vee q'} \frac{\sqrt{N}}{\sqrt{2}s} ds \leq \frac{\sqrt{N}}{2\sqrt{2}q_-} |q - q'|.$$

Since $q, q' \in I \subseteq (0, 1)$, this proves (6.13) with $C_I = (2\sqrt{2}q_-)^{-1}$. $\square$

Proof of Proposition 6.14. Put

$$V_N := \int_0^1 (q - q_*)^2 \nu_{A,N}(dq).$$

Invariance of $\nu_{A,N}$ and (2.15) give

$$\begin{aligned} V_N &= \int_0^1 \mathbb{E}[(F_{A,N}(q, G) - q_*)^2] \nu_{A,N}(dq) \\ &= \int_0^1 (V_A(q) - q_*^2) \nu_{A,N}(dq) + O(N^{-1}). \end{aligned} \quad (\text{B.18})$$

On I_* , (6.6) gives $|V_A(q) - q_*| \leq \kappa |q - q_*|$. On I_*^c , we use the trivial bound $|V_A(q) - q_*| \leq 1$ and (6.12). Hence $V_N \leq \kappa^2 V_N + \frac{C}{N}$. Since $\kappa < 1$, this gives $V_N \leq \frac{C}{(1-\kappa^2)N}$. The claim follows after enlarging C . $\square$

Proof of Lemma 6.15. Step 1: The error at time s_ .* Put $e_t^{(N)} := Z_t^{(N)} - q_{t,N}$. Since the comparison orbit starts from the same finite- N radius $q_{0,N}$, one has $e_0^{(N)} = 0$. The map V_A is Lipschitz on $[0, 1]$. Since s_* is fixed, finitely many iterations of (2.15) and (2.16) give

$$\mathbb{E}[(e_{s_*}^{(N)})^2] \leq \frac{C}{N}, \quad \mathbb{P}(|e_{s_*}^{(N)}| > \eta) \leq C \exp(-c_\eta N) \quad (\text{B.19})$$

for every fixed $\eta > 0$.

Step 2: The second moment before exit. Use the one-step noise $\zeta_{t+1}^{(N)}$ and natural filtration $\mathcal{F}_t$ from (2.14) and (2.15). Then

$$e_{t+1}^{(N)} = Z_{t+1}^{(N)} - q_{t+1,N} = V_A(Z_t^{(N)}) - V_A(q_{t,N}) + \zeta_{t+1}^{(N)}.$$

Moreover,

$$\mathbb{E}[\zeta_{t+1}^{(N)} | \mathcal{F}_t] = 0, \quad \mathbb{E}[(\zeta_{t+1}^{(N)})^2 | \mathcal{F}_t] \leq \frac{C}{N}.$$

On $\{t < \tau_*^{(N)}\}$, both $Z_t^{(N)}$ and $q_{t,N}$ lie in the stable interval I_* . Hence the mean value theorem and (6.6) give

$$|V_A(Z_t^{(N)}) - V_A(q_{t,N})| \leq \kappa |e_t^{(N)}|.$$

Therefore, using the preceding decomposition and the martingale property of $\zeta_{t+1}^{(N)}$,

$$\mathbb{E}[(e_{t+1}^{(N)})^2 | \mathcal{F}_t] = (V_A(Z_t^{(N)}) - V_A(q_{t,N}))^2 + \mathbb{E}[(\zeta_{t+1}^{(N)})^2 | \mathcal{F}_t] \leq \kappa^2 (e_t^{(N)})^2 + \frac{C}{N}$$

on $\{t < \tau_*^{(N)}\}$. Apply (2.37) to the time-shifted process $(e_{s_*+j}^{(N)})_{j \geq 0}$, with $\alpha_{j,N} = \kappa$ and $\sigma_{j,N} = C$. The geometric sum in that estimate is bounded by $(1 - \kappa^2)^{-1}$, while (B.19) supplies the $O(N^{-1})$ initial term. Hence

$$\mathbb{E}\left[(e_t^{(N)})^2 \mathbf{1}_{\{\tau_*^{(N)} > t\}}\right] \leq \frac{C}{N}. \quad (\text{B.20})$$

Step 3: Removing the stopping time. Choose $\eta > 0$ so small that

$$\kappa\delta_* + \eta < \delta_*.$$

If

$$\left|e_{s_*}^{(N)}\right| \leq \eta \quad \text{and} \quad \left|\zeta_j^{(N)}\right| \leq \eta \quad \text{for every } s_* < j \leq T_N,$$

then the recursion

$$\left|e_{t+1}^{(N)}\right| \leq \kappa \left|e_t^{(N)}\right| + \left|\zeta_{t+1}^{(N)}\right|$$

and the margin in (6.5) imply $Z_t^{(N)} \in I_*$ for every $s_* \leq t \leq T_N$. Hence (2.16) and (B.19) and a union bound give

$$\mathbb{P}(\tau_*^{(N)} \leq T_N) \leq CT_N e^{-c_0 N}.$$

Since $\left|e_t^{(N)}\right| \leq 1$, splitting according to whether $Z^{(N)}$ has exited before time t gives

$$\mathbb{E}[(e_t^{(N)})^2] = \mathbb{E}\left[(e_t^{(N)})^2 \mathbf{1}_{\{\tau_*^{(N)} > t\}}\right] + \mathbb{E}\left[(e_t^{(N)})^2 \mathbf{1}_{\{\tau_*^{(N)} \leq t\}}\right] \leq \mathbb{E}\left[(e_t^{(N)})^2 \mathbf{1}_{\{\tau_*^{(N)} > t\}}\right] + \mathbb{P}(\tau_*^{(N)} \leq t).$$

Combining this with (6.15) and (B.20) and $T_N = O(\log N)$ gives, uniformly for $t \leq T_N$,

$$\mathbb{E}[(e_t^{(N)})^2] \leq \frac{C}{N} + C(\log N)e^{-c_0 N}.$$

Multiplying by N and increasing C for finitely many small values of N proves (6.16). $\square$

B.3. Proof of Lemma 6.16.

Proof. Step 1: The last ℓ_N steps. Take C_0 as in (6.17). Fix $b > 0$ and $c \geq 0$. All constants below are independent of c . The lower bound on N and the $o(1)$ terms may depend on c , since the limit in N is taken first. By (6.18),

$$n_N(c) \leq C_0 \log N$$

for all large N . The estimates from Lemma 6.15 will be used only up to this time. For all sufficiently large N , put

$$\ell_N := \max\{0, \lfloor b \log \log N \rfloor\}, \quad s_N := \max\{0, n_N(c) - 1 - \ell_N\}.$$

Since $n_N(c) - 1 - \ell_N > 0$ eventually, $s_N + \ell_N = n_N(c) - 1$ for all large N , and $\ell_N = o(\log N)$. Because $n_N(c)$ is of order $\log N$ while $\ell_N = o(\log N)$, we have $s_N \rightarrow \infty$. Hence, after increasing N if necessary,

$$s_N \geq s_*, \quad 0 \leq s_N \leq s_N + \ell_N = n_N(c) - 1 \leq C_0 \log N.$$

The orbit estimates from Lemma 6.15 therefore apply at every time used in the last block.

Step 2: The distance at time s_N . By the deterministic asymptotic in (2.32), together with the definition of $t_N(q_0)$ and the bounded integer-time rounding in $n_N(c)$,

$$|q_{s_N, N} - q_*| \leq CN^{-1/2} \mu_A^{c - \ell_N} \quad (\text{B.21})$$

for all large N , with C independent of c . Indeed,

$$s_N - (t_N(q_0) + c - \ell_N) \in [-2, -1],$$

so the rounding contributes at most the fixed factor μ_A^{-2} . Only the lower bound on N in the locally uniform deterministic asymptotic may depend on c . Since $\tilde{Z}_0^{(N)} \sim \nu_{A,N}$ and the law $\nu_{A,N}$ is invariant, also $\tilde{Z}_{s_N}^{(N)} \sim \nu_{A,N}$. Therefore, by the triangle inequality, Cauchy's inequality, (6.16), and (6.14),

$$\begin{aligned} \mathbb{E} \left| Z_{s_N}^{(N)} - \tilde{Z}_{s_N}^{(N)} \right| &\leq \mathbb{E} \left| Z_{s_N}^{(N)} - q_{s_N,N} \right| + |q_{s_N,N} - q_*| + \mathbb{E} \left| \tilde{Z}_{s_N}^{(N)} - q_* \right| \\ &\leq CN^{-1/2} + CN^{-1/2} \mu_A^{c-\ell_N} + CN^{-1/2}. \end{aligned}$$

Hence

$$\mathbb{E} \left| Z_{s_N}^{(N)} - \tilde{Z}_{s_N}^{(N)} \right| \leq CN^{-1/2} (\mu_A^{c-\ell_N} + 1). \quad (\text{B.22})$$

Step 3: Both chains stay near q_ .* Let

$$\rho_N := (\log N)^{-2}, \quad J_N := [q_* - \rho_N, q_* + \rho_N] \cap [0, 1].$$

For every $0 \leq j \leq \ell_N$, the orbit $(q_{t,N})_{t \geq 0}$ remains closer to q_* as time increases. Hence (B.21) implies

$$q_{s_N+j,N} \in [q_* - \rho_N/2, q_* + \rho_N/2] \quad (\text{B.23})$$

for all large N , because $N^{-1/2} \mu_A^{c-\ell_N} = o(\rho_N)$. Chebyshev's inequality and (6.16) therefore give

$$\mathbb{P} \left(Z_{s_N+j}^{(N)} \notin J_N \right) \leq \mathbb{P} \left(\left| Z_{s_N+j}^{(N)} - q_{s_N+j,N} \right| > \rho_N/2 \right) \leq \frac{C}{N \rho_N^2}.$$

Similarly, stationarity of $\tilde{Z}^{(N)}$ and (6.14) imply

$$\mathbb{P} \left(\tilde{Z}_{s_N+j}^{(N)} \notin J_N \right) \leq \frac{C}{N \rho_N^2}.$$

Thus, with

$$E_j := \left\{ Z_{s_N+j}^{(N)} \in J_N, \tilde{Z}_{s_N+j}^{(N)} \in J_N \right\},$$

we have

$$\mathbb{P}(E_j^c) \leq \frac{C}{N \rho_N^2}. \quad (\text{B.24})$$

Step 4: The distance contracts during the last block. On J_N the deterministic map has derivative close to its derivative at q_* . Since $V_A'(q_*) = \mu_A$ and V_A'' is bounded in a neighborhood of q_* , the mean value theorem gives, for all large N and all $q, q' \in J_N$,

$$|V_A(q) - V_A(q')| \leq (\mu_A + C \rho_N) |q - q'|. \quad (\text{B.25})$$

We choose N large enough that $a_N := \mu_A + C \rho_N < 1$.

Let $D_j := \mathbb{E} \left| Z_{s_N+j}^{(N)} - \tilde{Z}_{s_N+j}^{(N)} \right|$. Conditional on the two radii at time $s_N + j$, the next step uses the same Gaussian vector in both chains. For fixed $g \in \mathbb{R}^N$, the map $q \mapsto F_{A,N}(q, g)$ is nondecreasing. Indeed, for $q > 0$,

$$\partial_q \tanh^2(A\sqrt{q} g_i) = \frac{A g_i}{\sqrt{q}} \tanh(A\sqrt{q} g_i) \operatorname{sech}^2(A\sqrt{q} g_i) \geq 0,$$

and the endpoint $q = 0$ follows by continuity. Monotonicity therefore removes the absolute value inside the expectation, and for every $q, q' \in [0, 1]$,

$$\mathbb{E} |F_{A,N}(q, G) - F_{A,N}(q', G)| = |V_A(q) - V_A(q')|.$$

Applying this identity conditionally gives

$$\mathbb{E}\left[\left|Z_{s_N+j+1}^{(N)} - \tilde{Z}_{s_N+j+1}^{(N)}\right| \middle| Z_{s_N+j}^{(N)}, \tilde{Z}_{s_N+j}^{(N)}\right] = \left|V_A(Z_{s_N+j}^{(N)}) - V_A(\tilde{Z}_{s_N+j}^{(N)})\right|.$$

On E_j , (B.25) bounds this by $a_N |Z_{s_N+j}^{(N)} - \tilde{Z}_{s_N+j}^{(N)}|$. On E_j^c we use only the trivial bound by one. Taking expectations and using (B.24) gives

$$D_{j+1} \leq a_N D_j + \frac{C}{N\rho_N^2} \quad 0 \leq j < \ell_N. \quad (\text{B.26})$$

Iterating the recursion and using $a_N < 1$ gives

$$D_{\ell_N} \leq a_N^{\ell_N} D_0 + \frac{C}{N\rho_N^2} \sum_{k=0}^{\ell_N-1} a_N^k \leq a_N^{\ell_N} D_0 + \frac{C}{N\rho_N^2}.$$

Since $\rho_N = (\log N)^{-2}$,

$$\frac{1}{N\rho_N^2} = \frac{(\log N)^4}{N} = o(N^{-1/2}).$$

Hence

$$D_{\ell_N} \leq a_N^{\ell_N} D_0 + o(N^{-1/2}). \quad (\text{B.27})$$

Step 5: The distance at time $n_N(c) - 1$. The block length recovers the factor $\mu_A^{\ell_N}$, while the perturbation $C\rho_N$ has no asymptotic effect. Indeed,

$$a_N^{\ell_N} = \mu_A^{\ell_N} \left(1 + \frac{C\rho_N}{\mu_A}\right)^{\ell_N},$$

and $\ell_N \rho_N \rightarrow 0$. Therefore

$$a_N^{\ell_N} = \mu_A^{\ell_N} (1 + o(1)). \quad (\text{B.28})$$

Combining (B.22), (B.27) and (B.28) gives

$$\sqrt{N} D_{\ell_N} \leq C \mu_A^{\ell_N} (\mu_A^{c-\ell_N} + 1) + o(1) \leq C \mu_A^c + C \mu_A^{\ell_N} + o(1) = C \mu_A^c + o(1).$$

Here $\mu_A^{\ell_N} \rightarrow 0$ because $\mu_A \in (0, 1)$ and $\ell_N \rightarrow \infty$. The constants in the preceding estimate are independent of c , while the $o(1)$ term may depend on its fixed value. Since D_{ℓ_N} is the expectation in (6.19), the claimed bound follows. $\square$

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